Hello, Adele:
Tate's Thesis and Automorphicity

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Contents

Chapter

Introduction

Overview

[Incomplete]

Adeles have gone by various names over the course of mathematical history. A term first introduced by Chevalley as a wordplay on ideals [Che40], they have been referred to as the "ring of valuation vectors" by John Tate in his seminal thesis [Tat50], "repartitions" by Jean-Pierre Serre [insert Serre Alg Grps book citation here], before eventually Weil threatened everyone at gunpoint to settle on Chevalley's original nomenclature. Whatever one may choose to call them privately, they have now become a staple of the language of the modern number theorist. This thesis is an exposition of some of the basic results concerning them, and their most standard use cases in computations with Hecke characters and in the theory of automorphic representations.

Chapter I covers the motivations behind and basic properties of the restricted product to get the reader familiar with the adelic language.
Chapter II recounts the substance of Tate's famous PhD thesis, which proves the classical Hecke functional equation between zeta functions of global fields using an adelic formulation. We follow primarily the exposition of [RV98] and [Wei74].
Chapter III will be completed later

None of the material covered in this thesis is novel, recent, or even niche mathematics, and there are a plethora of sources on this; the References section of this thesis likely contains but a tiny fraction of them. To me this thesis is two things primarily: (1) a process by which I teach and familiarize myself with this field of mathematics via written exposition, and (2) a celebration of this strange yet elegant algebraic object that has become so ubiquitous in number theory. If for some inexplicable reason you, the reader, are someone distinct from myself or my advisor, I hope you find beauty and fascination galore from reading about this stuff. I know I sure did writing it.

Format, Background, and Conventions

Format

In the days of old, mathematics was written densely and in black and white due to cost and typesetting considerations. With the wonders of modern technology I have chosen to make use of a variety of boxes and colors throughout this text because it makes it far less painful to read, review, and skim my own work. The three main classes of boxed environments used will be as follows:

> **Definition 0.0.1.** A bright, almost-blaring red-backgrounded, red-striped box to house a vital Definition™.

> **Example 0.0.2.** An amber-striped environment to indicate an important or prototypical Example™ of a concept is contained in its confines.

> **Theorem 0.0.3.** A neutrally-toned gray-shaded, box to house an important result, whether that be a Theorem™ (such as this particular box), a Proposition™, etc.

As you might have deduced, Definitions, Examples and Results follow a [Subsection.Number] numbering scheme, all using a common counter. Equations throughout will follow a (Section-Number) numbering scheme.

As is standard in mathematical writing I've also filled this text with technically unnecessary but hopefully-enriching Remarks™. Less-standard variants of asides I've used here are Sanity Checks™ (statements that should be intuitively obvious and verifiable in some epsilon time to an attentive reader) and Warnings™ (cautionary interventions about potentially confusing or unintuitive notation, terminology or concepts).

Assumed Background and Conventions

A writer of mathematics always challenged with balancing mathematical acessibility with the limits of time and space. As such, we assume some familiarity with measure theory, functional analysis, algebraic number theory and algebraic geometry, all at an undergraduate level. As a litmus test we suggest the reader go through the following subsection on Conventions and Notations. The goal of this text is that it should be broadly understandable to a reader conceptually comfortable with some dense subset of the conventions described below.

Miscellaneous Notation Conventions

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ will denote the set of Naturals, Integers, Rationals, Reals and Complex Numbers respectively. Naturals include 0.
- S^1 denotes the circle, i.e the set of norm-one complex numbers.
- $\mathbb{F}_q$ denotes the (unique!) finite field with q elements.

- When we have an infinite product or a vector with infinitely many values we say **"almost everywhere"** to mean "in all but finitely many components". If the components of the infinite product are indexed by $i \in I$ then we write $(x_i)_i$ to mean the "vector" of a single element in the infinite product, and x_i to mean the i-th coordinate.
- For A a subset of a set X, we denote by A^c or $X \setminus A$ the complement of A in X.

Algebra and Number Theory

- Given subgroups H, K of a group G with operation denoted by addition, we denote by $H + K$ the group of all elements of the form $h + k$ with $h \in H$ and $k \in K$.
- All rings are assumed to be unital. $R^\times$ will denote the set of invertible elements of a ring R.
- We use $H \leq G$ and $I \trianglelefteq R$ to say "H is a subgroup of the group G" and "I is an ideal of the ring R" respectively. If $r_1, r_2, \ldots$ is a set of elements in a group G (resp. a ring R), we denote by $\langle r_1, r_2, \ldots \rangle$ the subgroup (resp ideal) generated by the r_i.
- The symbol $\mathfrak{p}$ ALWAYS refers to a prime ideal, and in the case that we are working in a discrete valuation ring we denote its uniformizer by π. When necessary we add subscripts to indicate the relevant field we are working over.
- By a **local field** we mean a valued field whose additive group is locally compact with respect to the topology induced by the valuation. Note this convention includes $\mathbb{R}$ and $\mathbb{C}$, though they are sometimes excluded from the definition in other texts. By a **global field** we mean a finite extension of $\mathbb{Q}$ or $\mathbb{F}_p(t)$. We assume the reader is familiar with the standard fact that every local field arises as the completion of a global field.
- For K a field equipped with an absolute value $|\cdot|$ we denote by $\mathcal{O}_K$ its **valuation ring** (i.e the set of elements with absolute value ≤ 1). Moreover for any valuation v on the field we denote by K_v the **completion** of K with respect to v, and $\mathcal{O}_v$ the valuation ring of K_v. Note this convention particularly in the case of $K = \mathbb{Q}$; when we write $\mathbb{Z}_p$ we are ALWAYS referring to the p-adic integers!
- We denote by $\mathscr{P}_K$ the set of all **places** (i.e equivalence classes of absolute values) of a global field K. We say a place is **finite** if it arises as a $\mathfrak{p}$-adic absolute value for $\mathfrak{p}$ a prime of $\mathcal{O}_K$, and **infinite** otherwise. We write $v \nmid \infty$ to mean the place v is finite and $v \mid \infty$ to say it is infinite. For an absolute value corresponding to a place $v \in \mathscr{P}_K$ we write $|\cdot|_v$.
- We say a place is **non-Archimedean** if any valuation corresponding to it is ultrametric, and **Archimedean** otherwise. In particular every place of a global field of positive characteristic is non-Archimedean. In characteristic zero, on the other hand, the notion of Archimedean and infinite places exactly coincide. We also recall that completing a global field at any Archimedean place yields either $\mathbb{R}$ or $\mathbb{C}$.

Functions

- We denote by $\mathscr{C}(X)$ the set of continuous functions on a topological space X, and by $\mathscr{C}_0(X)$ the set of continuous functions that vanish at infinity (recall a function **vanishes at infinity** if $f(x) \to 0$ as $x \to \infty$[1].)

[1] More precisely, the absolute value of x.

- $\mathrm{Supp}(f)$ denotes the **support** (i.e closure of pre-image of non-zero values) of a function. We write $\mathscr{C}_c(X)$ for the set of continuous compactly supported functions on X.

- Given a function f defined on a space X and a subset $Y \subseteq X$, we write $f|_Y$ for the restriction of f to Y.

Measures and Topological Groups

- The phrase **"almost everywhere"** means we are considering a set whose complement has measure zero. We use terms like **comeasure/covolume**/etc, to mean "measure/volume/etc of the complement".

- By a **locally compact group** we mean G is Hausdorff and every point has a compact neighborhood. Note we will ALWAYS assume the Hausdorff hypothesis. In many cases the locally compact groups we work with are also assumed to be abelian. We use the abreviation "**LCA group**" in such cases.

- Recall that every locally compact group admits a (bi-invariant) **Haar measure**, which we often denote by $\mathrm{d}x$ or $\mathrm{d}y$ or $\mathrm{d}z$, and so on. This will be a slight abuse of notation since we may change the variable depending on what is being integrated, but before any such integration we explicitly describe the Haar measure. Given a Borel (i.e measurable) set $A \subseteq G$, we denote by $\mathrm{Vol}(A, \mathrm{d}x)$ the measure of A with respect to $\mathrm{d}x$.

- For G a topological group, we denote by $\widehat{G}$ the **Pontryagin dual group** of continuous characters $G \to S^1$. See Appendix **??** for a detailed discussion.

- Given x a point of a metric space, we note by $B_\epsilon(x)$ the OPEN ball of radius ϵ centered at x. Putting a bar above it denotes its closure, i.e the closed ball of the same radius.

Acknowledgements

[TODO]

Chapter I

Hello, Adele

"*In retrospect, we see now that the real numbers appear there as ... neither more nor less interesting to the arithmetician than its p-adic companions, and that there is at least one language and one technique, that of the* ***adeles****, for bringing them all together under one roof and making them cooperate for a common purpose.*"

Preface to *Basic Number Theory*
André Weil

The adelic language gives a beautiful solution to an age-old problem: that of doing Fourier analysis on a global field K. The classical approach was to pass to a completion of K, but with the discovery of (infinitely many) non-trivial absolute values, it was not too far a stretch to imagine a structure that captured the information of all the completions at once, and "bringing them all together", in the words of Weil.

Instead of taking just the direct product of all the completions, however, we take the *restricted direct product*, which imposes an "almost-everywhere integrality" condition on the elements. The reason for this is twofold: (1) to make the natural embedding of K discrete and cocompact, and (2) to give the group a locally compact Hausdorff structure, without which we could not do the appropriate harmonic analysis that will come up later.

This chapter proceeds in a somewhat reverse historical order. Section 1 recounts the basic results of the adeles and their unit group the ideles in somewhat reverse historical order (or rather, one might say history discovered the ideles in the reverse order). We build the theory in this section with pedantry and care, to allow for a full appreciation for the nuances of the adelic formulation.

Remark. Some texts (e.g [Ber+13]) use the accented terms "adéles" and "idéles", but Chevalley introduced these terms more as a wordplay on "ideals," and so they don't necessarily exactly correspond to French words. Besides, I've never seen accents used for these terms by a French writer other than Chevalley's original paper(see [Wei74] and [Lan94]), so I have chosen not to accent them in this text either.

Section 1

Adeles and Ideles

§1.1 Introduction: Meeting the Adeles and Ideles

Recall the ring of profinite integers;

$$\widehat{\mathbb{Z}} := \varprojlim_{n\in\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = \prod_{p\in\mathscr{P}_{\mathbb{Q}}} \mathbb{Z}_p,$$

where the inverse limit is taken over the natural maps of quotient rings ordered by divisibility. Imbued with the natural profinite topology, this is a compact Hausdorff space, and thus has all the nice properties that come along with that.

Warning (Overloaded Notation). Throughout this thesis for a locally compact Hausdorff topological group G we use $\widehat{G}$ to denote the Pontryagin dual of G. This is visibly not what we mean by $\widehat{\mathbb{Z}}$ above. Unfortunately the notation $\widehat{R}$ is commonly to denote completions of rings, so distinguishing the two is simply a matter of context.

Suppose we wanted to do a similar thing with $\mathbb{Q}$ and all of its completions. We run into a slight issue here, since $\prod \mathbb{Q}_p$ is not locally compact. Local compactness (and Hausdorfness) is rather necessary if one desires to do any sort of analysis over arbitrary topological groups, since we would like to have a natural Haar measure on our group (see Appendix A for more details). Amending this issue is precisely the motivation behind the construction of the *restricted product*, which we dive into now.

Let J be an indexing set such that for every $v \in J$ we have a locally compact Hausdorff group G_v indexed by v. We fix a finite subset J_∞ of J (which will later represent the infinite places), and suppose that for all $v \notin J_\infty$ we have a compact open (and thus closed) subgroup $H_v \leq G_v$.

Definition 1.1.1. The **restricted direct product** (also just called the **restricted product**) of the G_v with respect to the H_v is defined as

$$\coprod\!\!\!\!\prod_{v\in J} G_v := \left\{ (x_v) \in \prod_{v\in J} G_v : x_v \in H_v \text{ for all but finitely many } v \right\}.$$

Remark (Notation). The notation I've used for the restricted product (product and coproduct symbol overlayed on one another) is non-standard, but more distinct than the usual $\prod'$. I first saw this notation in [Sut15].

Let us denote this restricted product as G. Note $G \subseteq \prod G_v$ (the usual direct product of groups). We want to topologize G, but the subspace topology induced by the product topology on $\prod G_v$ is insufficient. For reasons that will become clear later, we'll want subsets like $\prod H_v$ to be open, which won't be true in the subspace topology unless $H_v = G_v$ for all but finitely many $v \in J$.

Definition 1.1.2. The **restricted product topology** on $\coprod\!\!\!\!\prod G_v$ is given by the following neighborhood basis of $g = (g_v)_v$:

$$\left\{\prod U_v : U_v \subseteq_{\text{open}} G_v,\ g_v \in U_v \text{ and } U_v = H_v \text{ for all but finitely many } v\right\},$$

i.e. the (non-empty) open sets around g are exactly the ones that contain H_v in cofinitely many components, and arbitrary open neighborhoods of g_v in the rest.

Observation. Suppose we have a family of compact open subgroups $H'_v \subseteq G_v$ such that $H'_v = H_v$ for all but finitely many $v \in J$. Then we note that $\coprod G_v$ wrt to the H_v is the same (both set-theoretically and topologically) as $\coprod G_v$ wrt to the H'_v. It thus suffices to specify the H_v for all but finitely many v. A parallel observation we make is that for any finite positive integer n we have the topological isomorphism:

$$\begin{aligned} \textstyle\coprod_v G_v^n &\cong \prod^n \left(\textstyle\coprod_v G_v\right) \\ (\ldots, (g_{v,1}, \ldots, g_{v,n}), \ldots)_v &\mapsto ((g_{v,1})_v, \ldots, (g_{v,n})_v), \end{aligned} \tag{1.1-1}$$

where, naturally, G_v^n is the n-fold direct-product of G_v with itself.

We introduce a handy piece of notation that will come in useful later. Let S be any finite subset of the index set J containing J_∞. Then we define

$$G_S := \prod_{v \in S} G_v \times \prod_{v \notin S} H_v, \tag{1.1-2}$$

endowed with the usual product topology. We note, for any S, that this is by definition an open neighborhood of the identity $(e_v)_v$ of G. Moreover, for any $T \supseteq S$ consider the inclusion map $i_{S,T} : G_S \hookrightarrow G_T$, which is an open embedding. Then we see that th restricted product G is in fact the directed union of the G_S's as open subspaces, i.e.

$$G = \varinjlim_S G_S. \tag{1.1-3}$$

We leave it to the reader to verify this abstract-nonsense (but ultimately useful and simpler!) characterization, which ultimately comes down to checking the relevant universal property is satisfied.

Proposition 1.1.3. Let G_v, H_v be as defined above and G be the restricted product. Then

(a) G is a locally compact group.

(b) A subset $Y \subseteq G$ has compact closure if and only if $Y \subseteq \prod K_v$ for some family of compact subsets $K_v \subseteq G_v$ such that $K_v = H_v$ for all but finitely many v.

Proof. **(a)**
We first verify that it is indeed a topological group. The group structure is clear; for continuity of the group maps, recall any basic open set is of the form

$$\mathcal{U} := \prod_{v \in S} U_v \times \prod_{v \notin S} H_v$$

for some finite indexing set S and opens $U_v \subseteq G_v$. Since each H_v is a subgroup, the preimage of $\mathcal{U}$ under the inversion map is:

$$\prod_{v \in S} (U_v)^{-1} \times \prod_{v \notin S} H_v,$$

which is also open since each G_v is a topological group, thereby giving us continuity of inversion.
For multiplication, the topological isomorphism in Equation 1.1-1 gives us $G \times G \cong \prod_v (G_v \times G_v)$, allowing us to similarly look at preimages component by component. In particular, consider any element

$$\Big((k_v, k'_v)_{v\in S}, (k_v, k'_v)_{v\notin S} \Big)$$

in the preimage of $\mathcal{U}$ under the addition map. We note that by definition, away from some finite index set $T \supseteq S$, we must have that $(k_v, k'_v) \in H_v \times H_v$ for all $v \notin T$, and clearly $H_v \times H_v$ is in the pre-image of $\mathcal{U}$. For any $v \in T$, we know from the fact that G_v is a topological group that (k_v, k'_v) has some open neighborhood $W_v \times W'_v$ contained in the pre-image of $\mathcal{U}$. Thus:

$$\Big((k_v, k'_v)_{v\in S}, (k_v, k'_v)_{v\notin S} \Big) \in \prod_{v\in T} (W_v \times W'_v) \times \prod_{v\notin T} (H_v \times H_v) \subseteq \mathrm{add}^{-1}(\mathcal{U}),$$

and so multiplication is continuous. Thus G is a topological group.

Warning. Our work above seems (and may very well be) a tad pedantic with the details, and indeed several texts leave the fact of the restricted product being a topological group as a trivial observation. However it is worth being careful about constructions such as this; one might be tempted to think that if we have *any* directed system of topological groups with inclusion maps and define a colimit as in 1.1-3, that the product must surely be a topological group. The editors of MIT's second-edition *Encyclopedic Dictionary of Mathematics* thought so too. However this turns out NOT the case: the colimit of the directed system of $G_i := \mathbb{Q} \times \mathbb{R}^i$ does not have continuous addition; [TSH98] shows that we do get a nice output in this colimit construction when the groups are all locally compact.

We now show local compactness. Since each H_v is compact, Tychonoff tells us $\prod_{v\notin S} H_v$ is also compact. Thus G_S is the product of finitely many locally compact spaces with a compact space, and is thus itself locally compact. Moreover, the product topology on G_S is the SAME as the subspace topology induced by G, and since any $(x_v) \in G$ is contained in such a G_S, we conclude G must be locally compact as well.

(b)
If Y is contained in some product of compact sets $K = \prod K_v$ (which is compact by Tychonoff) its closure $\overline{Y}$ is a closed subset of a compact subset of a Hausdorff space G, and thus $\overline{Y}$ is compact too.
For the other direction, suppose $\overline{Y}$ is a compact subset of G. Since the sets G_S defined above cover G, a finite number of G_S must cover $\overline{Y}$. Note a union of a finite number of G_S is just a set of the form $G_{\mathscr{S}}$, for $\mathscr{S}$ being the union of all the S such that G_S is in the cover of $\overline{Y}$.
Consider the (continuous!) projection maps $\rho_v : G \to G_v$. Thus each $\rho_v(\overline{Y}) \subseteq G_v$ is compact. And since all but finitely many components of $G_{\mathscr{S}}$ are equal to H_v, the image $\rho_v(\overline{Y}) \subseteq H_v$ for all but finitely many v.
Thus $\overline{Y}$, and therefore Y, is contained in a set of the desired form. □

Observation. For each v we have a natural topological embedding $G_v \hookrightarrow G$ into the v-th component of the restricted product. Thus we naturally identify G_v with a closed subgroup of G.

We are now ready to define the adeles. Let K be a global field, and let K_v denote its completion at a place v. With respect to addition, note K_v is a locally compact additive group. For all finite places v, recall K_v admits its local ring of integers $\mathcal{O}_v$ as an open compact subgroup.

DEFINITION 1.1.4. The **adele group** $\mathbb{A}_K$ of a global field K is the restricted direct product of K_v with respect to the open subgroups $\mathcal{O}_v$. Since this admits a natural ring structure as well, it is also referred to as the **adele ring**.

Note we have an embedding

$$\begin{aligned} K &\hookrightarrow \mathbb{A}_K \\ x &\mapsto (x, x, \dots) \end{aligned}$$

This is well-defined since K always has a natural embedding in K_v for any place v, and by unique factorization, every element is a local integer for all but finitely many v.

Similarly, we can conisder $K^\times$, the locally compact group of units of K wrt to multiplication. Then the local units $\mathcal{O}_v^\times$ are an open compact subgroup of each $K_v^\times$.

DEFINITION 1.1.5. The **idele group** $\mathbb{I}_K$ of a global field K is the restricted direct product of the $K_v^\times$ with respect to $\mathcal{O}_v^\times$.

We have a similar embedding

$$\begin{aligned} K^\times &\hookrightarrow \mathbb{I}_K \\ x &\mapsto (x, x, \dots) \end{aligned}$$

which is again well-defined by a similar argument as with the adeles. This motivates the following definition.

DEFINITION 1.1.6. The elements in the image of the diagonal embedding of K (resp. $K^\times$) into $\mathbb{A}_K$ (resp. $\mathbb{I}_K$) are called the **principal adeles** (resp. **principal ideles**).

WARNING. The notation $\mathbb{I}_K$ and $\mathbb{A}_K^\times$ is used interchangeably in some literature, but one should be careful; although the group $\mathbb{I}_K$ is *algebraically equivalent* to the group of units $\mathbb{A}_K^\times$, this is NOT a topological homeomorphism if we take the naive subspace topology. In general, recall for any topological ring R, we topologize the unit group group $R^\times$ by the subspace topology of its image under the injection

$$\begin{aligned} R &\hookrightarrow R \times R \\ r &\mapsto (r, r), \end{aligned}$$

where of course the codomain is given the natural product topology. We do this because taking the subspace topology may fail to make $R^\times$ a topological group, as in the Example below.

EXAMPLE 1.1.7. Take $K = \mathbb{Q}$, and consider an open neighborhood, in the *subspace topology*, of the multiplicative identity $1 \in \mathbb{A}_\mathbb{Q}^\times$. Every such neighborhood has the form

$$U := \left(\mathbb{R} \times \prod_{p \in S} U_p \times \prod_{p \notin S} \mathbb{Z}_p \right) \cap \mathbb{A}_\mathbb{Q}^\times,$$

where S is some finite collection of primes, and U_p is a neighborhood of $1 \in \mathbb{Q}_p$ for each $p \in S$.

However, consider the following set:

$$V := \mathbb{R}^\times \times \prod_{p<\infty} \mathbb{Z}_p^\times,$$

which is visibly open in the idelic topology. We claim $U \nsubseteq V$ for any U described above. The reason for the non-inclusion is that for any U we can select some $(x_p)_p \in \mathbb{I}_\mathbb{Q}$ such that $x_p \in \mathbb{Z}_p \setminus \mathbb{Z}_p^\times$ for some $p \notin S$. This thus means $(x_p)_p \notin V$. It follows that the relative open sets are too coarse, i.e. the idelic topology is *finer* than the induced subspace topology.

The natural question, then, is whether the topology on $\mathbb{I}_K$ as the unit group of $\mathbb{A}_K$ is equivalent to the restricted product topology. The answer, as it neatly turns out, is yes!

Proposition 1.1.8. The subspace topology on $\mathbb{I}_K \subseteq \mathbb{A}_K \times \mathbb{A}_K$ (the image under the diagonal embedding) is exactly the restricted product topology.

Proof. Suppose $\mathbb{A}_K^\times$ is given the idelic topology. It is relatively clear that ϕ is an open map; indeed, for any basic open set in the idelic topology

$$U = \prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_V^\times,$$

we have that the image of U is:

$$\phi(U) = \left(\prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v^\times\right) \times \left(\prod_{v\in S} U_v^{-1} \times \prod_{v\notin S} \mathcal{O}_V^\times\right),$$

which is visibly open in the topology of $\mathbb{A}_K \times \mathbb{A}_K$.
Thus it remains to show ϕ is continuous if $\mathbb{A}_K^\times$ is given the idelic topology. Take an open neighborhood U of the multiplicative identity of $\mathbb{A}_K \times \mathbb{A}_K$. Any such neighborhood is of the form

$$U = \left(\prod_{v\in S} U_{1,v} \times \prod_{v\notin S} \mathcal{O}_v\right) \times \left(\prod_{v\in T} U_{2,v} \times \prod_{v\notin T} \mathcal{O}_v\right),$$

where S, T are finite indexing sets. In the following expressions, for any set set $S_v \subseteq K_v$ we let $S^\times$ be shorthand for $S_v \cap K_v^\times$. Taking intersection of U with the image of ϕ we get:

$$\begin{aligned} U \cap \phi(\mathbb{A}_K^\times) = &\prod_{v\in S\cap T} \left(U_{v,1}^\times \cap \left(U_{v,2}^\times\right)^{-1}\right) \times \left(\left(U_{v,1}^\times\right)^{-1} \cap U_{v,2}^\times\right) \\ &\times \prod_{v\in S\setminus(S\cap T)} \left(U_{v,1} \cap \mathcal{O}_v^\times\right) \times \left(\left(U_{v,1}^\times\right)^{-1} \cap \mathcal{O}_v^\times\right) \\ &\times \prod_{v\in T\setminus(S\cap T)} \left(\mathcal{O}_v^\times \cap \left(U_{v,2}^\times\right)^{-1}\right) \times \left(\mathcal{O}_v^\times \cap U_{v,2}^\times\right) \\ &\times \prod_{v\notin S\cup T} \mathcal{O}_v^\times \times \mathcal{O}_v^\times \end{aligned}$$

Note in the above (rather heinous) expression we used our observation in Eqn 1.1-1 to "absorb" the direct product of $\mathbb{A}_K$ into the restricted product itself. In any case, pulling back to $\mathbb{A}_k^\times$ corresponds to just looking at the first coordinate in each of the products. We note that for all the $v \in S \cup T$ that the set in the v-th component is visibly an open neigborhood of the identity, and since this is still a finite set the pullback is indeed equal to $\mathcal{O}_v^\times$ on cofinitely many components. We deduce it is an open set of the idelic topology, thereby proving ϕ is continuous and thus a homeomorphism onto its image. □

It will be useful to explore the behavior of the adeles under base-change - in particular, extending scalars by to finite separable extensions. This type of technique will allow us to show results for any $\mathbb{A}_K$ by just considering $K = \mathbb{Q}$ or $K = \mathbb{F}_q(t)$.

Proposition 1.1.9. Take E/K a finite separable extension of global fields, and let $\{u_1, \ldots u_n\}$ be a K-basis of E. Then, recalling we have a canonical diagonal embedding $K_v \hookrightarrow \prod_{w|v} E_w$, the natural map

$$\begin{aligned} \alpha : \prod_{j=1}^{n} \mathbb{A}_K &\to \mathbb{A}_E \\ ((x_{v,j})_v)_j &\mapsto \sum_j u_j (x_{v,j})_v \end{aligned} \tag{1.1-4}$$

is an isomorphism of topological groups.

Proof. For every place v of K, we define:

$$E_v := \prod_{w|v} E_w = E \otimes_K K_v.$$

where the product is taken over all places w of E lying above v, of which there are at most n. This is manifestly a topological algebra over K_v, into which we have a natural diagonal embedding. By B.0.8, we know E_v admits $\{u_1, \ldots, u_n\}$ as a K_v-basis, and thus the map

$$\begin{aligned} \alpha_v : K_v^{\oplus n} &\xrightarrow{\sim} E_v \\ (x_j) &\mapsto \sum_j u_j x_j \end{aligned}$$

is an algebraic and topological isomorphism.

In a similar vein, (for $v \nmid \infty$) we can define

$$\mathcal{O}_{E_v} = \prod_{w|v} \mathcal{O}_{E_w},$$

and again by B.0.8, there exists a topological and algebraic isomorphism

$$\alpha_v : \mathcal{O}_{K_v}^{\oplus n} \xrightarrow{\sim} \mathcal{O}_{E_v}$$

for all v away from a finite set S_0 of places containing the Archimedean ones. For any $S \supseteq S_0$, consider the products $\mathbb{A}_{K,S}$ and $\mathbb{A}_{E,S}$ as defined in 1.1-2. We note then that:

$$(\mathbb{A}_{K,S})^{\oplus n} = \bigoplus_{i=1}^{n} \left(\prod_{v \in S} K_{v,i} \times \prod_{v \notin S} \mathcal{O}_{K_v,i} \right) \cong \prod_{v \in S} E_v \times \prod_{v \notin S} \mathcal{O}_{E_v} =: \mathbb{A}_{E,S}.$$

Thus for any such S, by appropriately restricting the collection $\{\alpha_v\}$ induces a map

$$\alpha_S := (\mathbb{A}_{K,S})^{\oplus n} \xrightarrow{\sim} \mathbb{A}_{E,S}.$$

Recalling our abstract-nonsense equivalent definition of the adeles in Eqn 1.1-3, we deduce that the map α, as the colimit of the directed system of isomorphisms under inclusions, is a topological and algebraic isomorphism. $\square$

§1.2 Adelic Characters and Measures

We first recall the basic objects of Pontryagin duals.

DEFINITION 1.2.1. Let G be a locally compact topological group.

- A **quasi-character** of G is an element $\chi \in \mathrm{Hom}_{cont}(G, \mathbb{C}^\times)$.
- A **character** of G is an element $\chi \in \mathrm{Hom}_{cont}(G, S^1)$.

REMARK. Some texts refer to "quasi-characters" as "characters" and refer to "characters" as "unitary characters". We use the other convention because it is consistent with Tate's formulation in his original thesis [Tat50] and because it gives us an excuse to say and write the prefix "quasi".

We first begin with the construction of the so-called *standard characters* of local fields, as these are the building blocks of the adelic characters. Throughout we denote by F a local field, $\mathcal{O}_F$ its (local!) valuation ring with prime ideal $\mathfrak{p}$ and uniformizer π.

- For $F = \mathbb{R}$, we define
$$\psi(x) := e^{-2\pi i x}. \tag{1.2-1}$$
- For $F = \mathbb{Q}_p$, we define our character by the composition of the following maps:
$$\psi : \mathbb{Q}_p \twoheadrightarrow \mathbb{Q}_p/\mathbb{Z}_p \xrightarrow{\sim} \mathbb{Z}[1/p]/\mathbb{Z} \hookrightarrow S^1,$$
where the first map is the canonical quotient map, the second is the inverse of the map $\alpha + \mathbb{Z} \mapsto \alpha + \mathbb{Z}_p$, and the last is the restriction of the canonical isomorphism $\mathbb{R}/\mathbb{Z} \xrightarrow{\sim} S^1$. Calculating this chain of maps it is not too hard to verify that
$$\psi(x) := e^{2\pi i r/p^n}, \tag{1.2-2}$$
where n is the smallest non-negative integer such that $\mathfrak{p}^n \in \mathbb{Z}_p$, and $r \in [0, p^n - 1]$ is the integer equivalent to $p^n x$ modulo p^n.
- For $F = \mathbb{F}_p((t))$, recall that every $x \in F$ has a unique Laurent series representation (either by definition or by B.0.7, depending on your choice of constructing the local field), so we define
$$\psi\left(x = \sum_{i \geq n}^{\infty} a_i t^i\right) := e^{2\pi i a_{-1}/p}, \tag{1.2-3}$$
where naturally we lift $a_{-1} \in \mathbb{F}_p$ to $\mathbb{Z}$ to make sense of the exponent. One must prove, of course, that this coefficient a_{-1} is well-defined regardless of choice of uniformizer, which we accept on faith here. Section II-11 in [Ser12] gives a good proof of this for the interested reader.
- All the remainder of the cases fall under when F'/F a finite separable extension of any of the above. In characteristic 0, define
$$\psi_{F'}(x) := \psi_F\left(\mathrm{Tr}_{F'/F}(x)\right). \tag{1.2-4}$$
In the case of positive characteristic we pick a choice of isomorphism $F' \cong \mathbb{F}_q((t))$ and we pick the coefficient

of $2\pi i/p$ in the exponent to be $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p(a_{-1})}$. Again, we take on faith here that this is independent of choice of isomorphism.

Definition 1.2.2. We will call the characters described in Eqns 1.2-1, 1.2-2, 1.2-3 and 1.2-4 as the **standard characters** of their respective local fields.

Remark. In the function field case, the standard character has the following geometric interpretation: for X a smooth, irreducible complete geometrically connected algebraic curve over $\mathbb{F}_q$. Then $K := K(X)$ – the curve's function field – is a global field of positive characteristic in which $\mathbb{F}_q$ is algebraically closed; equivalently, we can define K as the stalk of the structure sheaf at the generic point. Consider the (coherent) sheaf Ω^1 of 1-forms on X, and let Ω^1_K be its stalk at the generic point, which can be interpreted as the one-dimensional K-module of "meromorphic" (in the sense that they are generically non-vanishing) 1-forms on X. Thus $\omega \in \Omega^1_K$ can be written as $f\,\mathrm{d}t$ for some $f \in K$, with some choice of separating transcendence basis $\{t\}$. Pick some $v \in \mathscr{P}_K$, corresponding to a point of X. Then the coefficient a_{-1} in the expansion of f seen as an element of K_v $(= \mathbb{F}_{q_v}((t_v))$, for some uniformizing element t_v at v), corresponds to the residue of ω at v, written $\mathrm{Res}_v(\omega)$. We refer the reader to Chapter II of [Ser12] for a more detailed exposition.

We now build the notion of a conductor of a (quasi-)character, which will be an invaluable tool in computations later.

Definition 1.2.3. Let ψ be a nontrivial additive quasi-character on a nonarchimedean local field F. Then the **conductor** of ψ is the ideal $\mathfrak{p}^m$, where

$$m := \min\left\{r \in \mathbb{Z} : \psi|_{\mathfrak{p}^r} = 1\right\},$$

i.e the conductor is the largest fractional ideal of $\mathcal{O}_F$ on which ψ is trivial.
Now suppose χ is a multiplicative quasi-character on $F^\times$. Then its **conductor** is $\mathfrak{p}^n$, where

$$n := \min\left\{r \geq 0 : \chi|_{1+\mathfrak{p}^r} = 1\right\},$$

i.e the conductor is the largest possible unit congruence subgroup of $\mathcal{O}_F$ on which χ is trivial. By convention, if χ is trivial on $\mathcal{O}_F^\times$ then its conductor is $\mathcal{O}_F^\times$ itself

Remark (Terminology). The origin of the term "conductor" comes from Dedekind, who denoted by "*Führer*" (whose meaning was closer to "guide", "attendant" or "conductor" in the original German, rather than its modern assocations) a nonzero ideal of an order in a number field, which is why classical texts denote conductors using $\mathfrak{f}$. Sometimes the conductors are defined as the integers m and n themselves rather than the ideals, but we avoid this convention.

Proposition 1.2.4. For all but finitely many places v, the standard character ψ_v on K_v has conductor equal to $\mathcal{O}_v$.

Proof. Since the trace of a finite-separable field extension K/F carries valuation rings into valuation rings, it suffices to show this for the DVRs $\mathbb{Z}_p$ and $\mathbb{F}_q[[t]]$.

In the characteristic zero case we note that our character in Eqn 1.2-2 factors through the quotient $\mathbb{Q}_p/\mathbb{Z}_p$, and so by construction must have conductor $\mathbb{Z}_p$.
In the positive characteristic case, again we notice by our explicit definition that for element of $\mathbb{F}_p[[t]]$ we have all the exponents of t are non-negative, and so $a_{-1} = 0$, and thus looking at Eqn 1.2-3 we see its conductor must be exactly the valuation ring. □

We now proceed to discuss the theory of characters on the adelic groups, with the key result being that of Pontryagin duality commuting with the restricted product. This result in fact holds for any restricted product, so the following analysis is done in greater generality so we can hit the adeles and ideles with one stone, as it were.

LEMMA 1.2.5. As with the notation of Definition 1.1.1, let G_v be a countable set of locally-compact abelian Hausdorff groups with compact open respective subgroups H_v, and let G be their restricted product.

(a) (Breaking Down) Any quasi-character $\chi \in \mathrm{Hom}_{cont}(G, \mathbb{C}^\times)$ is trivial on almost all H_v.

(b) (Building Up) Fix quasi-characters $\chi_v \in \mathrm{Hom}_{cont}(G_v, \mathbb{C}^\times)$ such that $\chi_v|_{H_v} = 1$ for almost all v. Then $\chi := \prod \chi_v$ lies in $\mathrm{Hom}_{cont}(G, \mathbb{C}^\times)$.

The same results hold if we replace every instance of "quasi-character" with "character".

Proof. Breaking Down
Since $\mathbb{C}^\times$ has "no small subgroups", we may fix a neighborhood $V \ni 1$ that does not contain any non-trivial subgroup. By considering pre-images we know there must exist some open neighborhod $U = \prod U_v$ of the identity of G such that $\chi(U) \subseteq V$. Let S denote the (finite!) index set such that $U_v \neq H_v$ exactly when $v \in S$. Consider the subgroup of G

$$H := \prod_{v \in S} 1 \times \prod_{v \notin S} H_v \subseteq U.$$

Note that $\chi(H) \subseteq U$, but since its image must be a group it must be $\{1\}$, giving us what we want. In particular, for any $x = (x_v)_v \in G$, if we define

$$\chi(x_v) := \chi(\ldots, 1, 1, x_v, 1, 1, \ldots),$$

we deduce for any x that $\chi(x_v) = 1$ almost everywhere.

Building Up
By checking definitions one confirms that χ is a well-defined homomorphism $G \to \mathbb{C}^\times$, so it remains to verify continuity. Let S denote the the (finite!) set of indices where $\chi_v|_{H_v} \neq 1$, and let $s := |S|$. For any subset $X \subseteq G$, we can define

$$X^{(s)} := \left\{ \prod_{i=1}^{s} g_i : g_i \in X \right\},$$

i.e. the set of all elements of G that are formed as products of exactly s elements of X. Note that for any neighborhood $V \ni 1$ in $\mathbb{C}^\times$, there must exist some other neighborhood W such that $W^{(s)} \subseteq V$ (one could achieve this by taking an arbitrarily small ball around 1, for example). By a standard result of character theory (see (a) of A.2.3), for every $v \in S$ we must have some neighborhood U_v of the identity of G_v such that $\chi(U_v) \subseteq W$. It follows that:

$$\prod_{v \in S} U_v \times \prod_{v \notin S} H_v \subseteq \chi^{-1}(W^{(s)}) \subseteq \chi^{-1}(V),$$

from which it follows (again, by (a) of A.2.3) that χ is continuous. □

We now have the tools needed to construct the dual group of the restricted product. For each component G_v, by local compactness we have the dual group $\widehat{G_v}$ of continuous characters that satisfies Pontryagin duality. Let J_∞ be the finite index set corresponding to infinite places, then for every $v \notin J_\infty$ we define

$$K(G_v, H_v) := \left\{ \chi_v \in \widehat{G_v} : \chi_v|_{H_v} = 1 \right\},$$

i.e characters with trivial restriction to H_v. By selecting a neighborhood $V \ni 1$ in S^1 small enough to not contain any non-trivial subgroups of S^1, we note that $K(G_v, H_v)$ is a basis element of the compact-open topology of $\widehat{G_v}$ (see A.2.1), and so is open in $\widehat{G_v}$.
Since $\chi_v \in K(G_v, H_v)$ is trivial on H_v, it factors through the quotient G_v/H_v, inducing a topological group isomorphism

$$K(G_v, H_v) \xrightarrow{\sim} \widehat{G_v/H_v}.$$

By openness of $H_v \subseteq G_v$, we get that the quotient G_v/H_v is a discrete group, and standard character theory (see (c) of A.2.3), the dual $\widehat{G_v/H_v} \cong K(G_v, H_v)$ must be compact. As such, with respect to the compact open subgroups $K(G_v, H_v)$, it makes sense to define the restricted product of the dual groups $\widehat{G_v}$.

Theorem 1.2.6 (Restricted Product Commutes with Dual Groups). With the notation above, we have an isomorphism of topological groups

$$\widehat{G} \xrightarrow{\sim} \coprod \widehat{G_v}.$$

Proof. Consider the map

$$\begin{aligned} \phi : \coprod \widehat{G_v} &\to \widehat{G} \\ (\chi_v)_v &\mapsto \prod_v \chi_v. \end{aligned}$$

By the construction of the restricted product and the 1.2.5, this is a well-defined group isomorphism. It remains to show ϕ is a homeomorphism.
By translation invariance of open sets in topological groups it suffices to prove bicontinuity at the identity in the domain and codomain. From part (b) of 1.1.3, we recall any compact neighborhood of the identity $K \subseteq G$ is of the form $\prod K_v$ for compact K_v that are equal to H_v almost everywhere. For U an open neighborhood of $1 \in S^1$, consider the open neighborhood of the identity

$$W(K, U) := \left\{ \chi \in \widehat{G} : \chi(K) \subseteq U \right\},$$

i.e a basis element of the compact-open topology on $\widehat{G}$ (see A.2.1). We want to show any such $W(K, U)$ is open. For any $\chi \in W(K, U)$, let S be the (finite!) set of indices for which χ is non-trivial on K_v, and let $s = |S|$. As in the proof of 1.2.5, we can select some neighborhood $V \ni 1$ in $\mathbb{C}^\times$ such that $V^{(s)} \subseteq U$, and consider the following open neighborhood of the identity in $\coprod \widehat{G_v}$:

$$X := \prod_{v \in S} W(K_v, V) \times \prod_{v \notin S} K(G_v, H_v) \subseteq_{\text{open}} \coprod \widehat{G_v}.$$

We note that $\phi^{-1}(\chi) \in X$ by construction, so in particular we have $\phi(X) \subseteq W(K, U)$, and so ϕ is continuous. For continuity in the other direction, let $X = \prod W(K_v, U)$ be an open neighborhood of the identity in $\coprod \widehat{G_v}$. Then, note

$$\widehat{G} \supseteq W(\coprod K_v, U) \subseteq \phi(X),$$

proving continuity in the other direction. $\square$

The above result tells us the following neat isomorphism for the dual adele and idele groups, which will allow us to work component-wise without worry.

COROLLARY 1.2.7. For a global field K,

$$\widehat{\mathbb{A}_K} \cong \coprod \widehat{K_v}, \quad \widehat{\mathbb{I}_K} \cong \coprod \widehat{K_v^\times},$$

where the restricted products are taken with respect to the various $\widehat{K_v / \mathcal{O}_v}$ and $\widehat{K_v^\times / \mathcal{O}_v^\times}$ respectively.

PROPOSITION 1.2.8. For any global field K, there exists a non-trivial quasicharacter ψ on $\mathbb{A}_K$ that is trivial on the image of K under the diagonal embedding.

Proof. By our Breaking-Down-Building-Up Lemma (1.2.5), we may take a "product" of characters ψ_v that are trivial on almost all the completions $\mathcal{O}_v$. On each component we may choose the character defined in Definition 1.2.2; then Proposition 1.2.4 tells us that these are trivial on $\mathcal{O}_v$ at almost all the places v, so the global character

$$\begin{aligned} \psi : \mathbb{A}_k &\to S^1 \\ (x_v) &\mapsto \prod_v \psi_v(x_v). \end{aligned}$$

is in fact well-defined. This character is non-trivial, since ψ_v are all non-trivial characters, so we can see that the $\psi(\ldots, 1, x_v, 1, \ldots) \neq 1$ for any x_v such that $\psi_v(x_v) \neq 1$.

For triviality on K, since the trace map carries the image of K into the image of its base subfield, it suffices to check triviality on $\mathbb{Q}$ or $\mathbb{F}_p(t)$. For any $x \in K = \mathbb{Q}$, recall we have a unique factorization

$$x = b + \sum_{p \text{ prime}} \frac{a_p}{p^{\mathrm{ord}_p(x)}}, \; a_p, b \in \mathbb{Z},$$

where a_p is nonzero for only finitely many primes. Recalling the expression for our ψ_v defined in Eqns 1.2-1 and 1.2-2, we have:

$$\begin{aligned} \psi_{\mathbb{Q}}(x) &= \prod_{v \in \mathscr{P}_{\mathbb{Q}}} \psi_v(x) \\ &= e^{-2\pi i x} \prod_{p \text{ prime}} e^{2\pi i a_p / p^{\mathrm{ord}_p(x)}} \\ &= e^{-2\pi i b}, \end{aligned}$$

which we deduce is equal to 1 since $b \in \mathbb{Z}$.

For $K = \mathbb{F}_p(t)$, we make use of the result known as the so-called *residue formula.* In the notation and terminology of the remark following Definition 1.2.2, the residue formula (see Sections II-12 and II-13 in [Ser12]) says that the sum of residues for any $\omega \in \Omega^1_K$ is well defined and equal to 0. Using this fact and identifying $\omega = x\,dt$ we get:

$$\begin{aligned}\psi_K(x) = \prod_{v \in \mathscr{P}_K} \psi_v(x) &= \prod_v e^{2\pi i \operatorname{Res}_v(\omega)/p} \\ &= \exp(2\pi i/p(\sum \operatorname{Res}_v(\omega))) \\ &= e^0 = 1,\end{aligned}$$

as desired. □

WARNING. In the positive characteristic case, our construction above relied on a *choice* of separating transcendence basis $\{t\}$. In future we will refer to the "standard" nontrivial adelic quasicharacter as defined above, and the reader should mentally add a "up to choice of $\{t\}$" after every such mention henceforth.

THEOREM 1.2.9 (Local-Fields are Self-Dual). Let F be a local field equipped with an absolute value $|\cdot|$, and fix a non-trivial continuous additive character $\psi \in \widehat{F}$. Then the map

$$\begin{aligned}\Psi : F &\mapsto \widehat{F} \\ a &\mapsto (x \mapsto \psi(ax))\end{aligned}$$

is an isomorphic homeomorphism of topological groups.

Proof. Since left-multiplication gives a continuous automorphism of the additive group of a field, we immediately see the map Ψ is a well-defined group homomorphism. We also see it is injective by the fact that ψ was assumed to be non-trivial, so $\ker(\Psi) = 0$. So it remains to show Ψ is a surjective homeomorphism.
We begin by showing Ψ is a continuous map. For $\lambda, \epsilon \in \mathbb{R}_{>0}$, note sets of the form

$$W(\lambda,\epsilon) := \left\{\chi \in \widehat{F} : |1-\chi(x)| \leq \epsilon,\ \forall x \in \overline{B_\lambda(0)} \subseteq F\right\}$$

form a neighborhood base for the trivial character in the compact-open topology (see A.2.1 for a refresher). Fix such an open set for choice of ϵ, λ, and now for continuity it suffices to show there exists some open neighborhood U of $0 \in F$ carried into $W(\lambda,\epsilon)$ via Ψ. However, note by continuity of our original character ψ that there exists some $\delta > 0$ such that

$$x \in B_\delta(0) \implies |1-\psi(x)| < \epsilon.$$

Then note the open set $U := B_{\delta/\lambda}$ suffices, since for any $y \in U$ and $x \in \overline{B_\lambda(0)}$ we have $|xy| < \delta$, and so $|1-\Psi(y)(x)| < \epsilon$. Thus Ψ is continuous.
Running the construction in reverse, one can also verify that Ψ is in fact an open map, and thus homeomorphic to its image.
Finally we need to show surjectivity. Let $H := \Psi(F) \leq \widehat{F}$. Since we showed Ψ was an open map, we deduce H is an open – and hence closed – subgroup of $\widehat{F}$, and thus both H and $\widehat{F}/H$ are LCA groups. By Pontryagin duality considerations we can identify

$$\widehat{\widehat{F}/H} \cong \left\{x \in \widehat{\widehat{F}} : x(h) = 1\ \forall h \in H\right\},$$

where of course by $x(h)$ we mean the evaluation map. But by nontriviality of our original ψ, there exists some $y \in F$ for which $\psi(y) \neq 1$, so for any $x \in F \setminus \{0\}$ we note that $\psi_{yx^{-1}} \in H$ evaluates to $\psi(y) \neq 1$ on x. It follows that $\widehat{\widehat{F}/H}$ is trivial, from which we deduce $H = \widehat{F}$, giving us surjectivity. □

Corollary 1.2.10. Let K be a global field, and ψ the standard non-trivial character of $\mathbb{A}_K$, as defined in 1.2.8. Then the map

$$\begin{aligned}\Psi : \mathbb{A}_K &\to \widehat{\mathbb{A}_K} \\ a &\mapsto (x \mapsto \psi(ax)),\end{aligned}$$

is an isomorphic homeomorphism of topological groups.

Proof. We know that each local component is self-dual by 1.2.9, and that the dual of the restricted product is the restricted product of the duals of each component in 1.2.7. □

Proposition 1.2.11. The map

$$\begin{aligned}\Psi' : K &\to \widehat{\mathbb{A}_K/K} \\ a &\mapsto (x \mapsto \psi(ax))\end{aligned}$$

is an isomorphic homeomorphism of topological groups. In particular, ψ_a as defined in 1.2.10 is trivial on K if and only if $a \in K \subseteq \mathbb{A}_K$.

Proof. As we will see later in Theorem 1.3.5, the quotient $\mathbb{A}_K/K$ is compact. By a standard result of Pontryagin duality (see A.2.3), the dual $\widehat{\mathbb{A}_K/K}$ must be discrete, and in particular homeomorphically embed as a discrete closed subgroup of $\widehat{\mathbb{A}_K}$. By 1.2.10 we've seen that $\widehat{\mathbb{A}_K} \cong \mathbb{A}_K$, so we have that $\widehat{\mathbb{A}_K/K}/K$ is a closed, discrete K-subvector space of the compact $\mathbb{A}_K/K$, from which we deduce the cardinality of $\widehat{\mathbb{A}_K/K}/K$ must be finite. However, any global field has infinite cardinality, so we must have $\widehat{\mathbb{A}_K/K} = K$, completing our proof. □

The above results motivate the following definition.

Definition 1.2.12. For a self-dual LCA group, a Haar measure $\,\mathrm{d}x\,$ is **self-dual** if it is equal to its own dual measure, in the sense of Definition A.3.4.

Our goal next will be to find an explicit measure $\,\mathrm{d}x_v\,$ on K_v that is self-dual for each $v \in \mathscr{P}_K$. We begin by fleshing out the theory of measures on the general restricted product; specifically, we want a Haar measure that will be equivalent to a "restricted product" of Haar measures on each of the components. This will require a bit of care, since the restricted product is not actually a product space.

Proposition 1.2.13. For all but finitely many v, let $\,\mathrm{d}g_v\,$ be a (left) Haar measure on G_v normalized so that $\mathrm{Vol}(H_v, \mathrm{d}g_v\,) = 1$. Then there is a unique Haar measure $\,\mathrm{d}g\,$ on G such that the restriction $\,\mathrm{d}g_S\,$ of $\,\mathrm{d}g\,$ to any G_S is exactly the product measure.

Proof. Choose such a set S satisfying the conditions of finiteness and containin J_∞, and define $\,\mathrm{d}g_S\,$ as the product of the measures $\,\mathrm{d}g_v\,$. Note the measure of the compact group $\prod_{v \notin S} H_v$ is indeed finite, by how we've normalized things. It is a verification of axioms to deduce the rest that $\,\mathrm{d}g_S\,$ is indeed a Haar measure on G_S.
Suppose we have another finite set of indices $T \supseteq S$. Then, by construction, it follows that $\,\mathrm{d}g_S\,$ coincides with the restriction of $\,\mathrm{d}g_T\,$ to $G_S \subseteq G_T$.

Since G is locally compact, we know it has a (unique up to scaling!) Haar measure, which restructs to a Haar measure on any (clopen!) G_S. Fixing any S, we can scale our Haar measure on G so that when it restricts to G_S it exactly coincides with dg_S. $\square$

The above result motivates the following.

Definition 1.2.14. Given G the restricted product of LCA groups G_v, we define the **restricted product measure**

$$dg = \prod_v dg_v \tag{1.2-5}$$

as the unique Haar measure induced by Proposition 1.2.13.

In other words, to define a (self-dual) Haar measure on $\mathbb{A}_K$ it suffices to define a (self-dual) Haar measure on each local component K_v that gives measure 1 to $\mathcal{O}_v$ for almost all v.
Since a Haar measure is unique up to scaling, it suffices to specify the finite (positive) volume of any pre-compact set with nonempty interior to determine the measure.

- For $F = \mathbb{R}$, let dx be the standard Lebesgue measure.
- For $F = \mathbb{C}$, let dx be twice the standard Lebesgue measure, i.e. $|dz \wedge dz|$.
- For F non-Archimedean, we set $\mathrm{Vol}(\mathcal{O}_F, dx) = (N\mathfrak{D})^{-1/2}$, where $\mathfrak{D}$ is the different of F. Note that this value is equal to $q^{-d/2}$, where q is the order of the residue field and $d = \mathrm{ord}_{\mathfrak{p}}(\mathfrak{D})$.

Warning. For F non-Archimedean of positive characteristic, note the definition of the different is dependent on some *choice* of isomorphism $F \cong \mathbb{F}_q((t))$, and thus is non-canonical. Nevertheless, it will be clear the ensuing computations that will use the norm of the different are independent of this choice, which is what we care about. Thus we may make such definitions without much worry.

Proposition 1.2.15. The measures defined above are self-dual with respect to the standard characters defined in 1.2.2.

Proof. The results for when $F = \mathbb{R}$ or $\mathbb{C}$ are standard in the theory of classical Fourier analysis, so we omit them here. When F is nonarchimedean, recall that the Fourier inversion theorem tells us that it suffices to check for any nonzero Schwartz-Bruhat function. To that end, let f be the characteristic function of $\mathcal{O}_F$. We thus get

$$\widehat{f}(y) = \int_F f(x)\overline{\psi(xy)}\, dx = \int_{\mathcal{O}_F} \overline{\psi(xy)}\, dx .$$

We are (slightly) abusing notation here since $\widehat{f}$ should be a function on $\widehat{F}$, but Theorem 1.2.9 allows us to do this with no worries.
For any $x \in \mathcal{O}_F$, by definition of the codifferent, we note that $\psi(xy) = 1$ if and only if $y \in \mathfrak{D}_F^{-1}$. Thus,

$$\widehat{f}(y) = \begin{cases} \mathrm{Vol}(\mathcal{O}_F, dx) = (N\mathfrak{D})^{-1/2}, & y \in \mathfrak{D}_F^{-1} \\ 0, & \text{otherwise,} \end{cases}$$

where the second equality follows from character orthogonality, since we are integrating over a group. Taking the Fourier transform again we get:

$$\hat{\hat{f}}(x) = \int_F \hat{f}(y)\overline{\psi(xy)}\,\mathrm{d}y = \int_{\mathfrak{D}^{-1}} (N\mathfrak{D})^{-1/2}\overline{\psi(xy)}\,\mathrm{d}y\,,$$

where again we slightly abuse notation here using the isomorphism of F with its dual. By similar logic to the previous step, for any $y \in \mathfrak{D}^{-1}$, we note that $\psi(xy)$ restricted to the codifferent is trivial if and only if $x \in \mathcal{O}_F$. Thus, we have:

$$\hat{\hat{f}}(x) = \begin{cases} (N\mathfrak{D})^{-1/2}\operatorname{Vol}(\mathfrak{D}^{-1}, \mathrm{d}x\,), & x \in \mathcal{O}_F \\ 0, & \text{otherwise.} \end{cases}$$

Finally, we see that

$$\operatorname{Vol}(\mathfrak{D}^{-1}, \mathrm{d}x\,) = N\mathfrak{D}\operatorname{Vol}(\mathcal{O}_F, \mathrm{d}x\,) = (N\mathfrak{D})^{1/2},$$

and thus $\hat{\hat{f}}$ is the characteristic function of $\mathcal{O}_F$. Thus $\hat{\hat{f}} = f$, and in particular $\hat{\hat{f}}(x) = f(-x)$ since $\mathcal{O}_F$ is an additively closed subgroup, so the Fourier inversion formula holds, proving self-duality of the measure. □

We are now ready to define the standard measure on the adeles.

Definition 1.2.16. For every place $v \in \mathscr{P}_K$, let $\mathrm{d}x_v$ denote the self-dual Haar measure of K_v with respect to the standard characters defined in 1.2.2. Then the **Tamagawa measure** on $\mathbb{A}_K$ is given by

$$\mathrm{d}x := \prod_v \mathrm{d}x_v\,.$$

Sanity Check. It's good to verify that the above is actually well defined. By Proposition 1.2.13, it suffices to show that the self-dual Haar measures satisfy $\operatorname{Vol}(\mathcal{O}_v, \mathrm{d}x_v\,) = 1$ for almost all finite places v. This is true by the fact the different is trivial (i.e equal to $\mathcal{O}_v$) at all the unramified – hence cofinitely many – places. We take this as fact here, but refer the interested reader to Chapter II of [Ser79] for a more detailed discussion on the different and discriminant ideals.

If we impose the counting measure on K, that combined with the Tamagawa measure on $\mathbb{A}_K$ induce a unqique Haar measure on the quotient $\mathbb{A}_K/K$ compatible with the exact sequence of topological groups

$$0 \to K \to \mathbb{A}_K \to \mathbb{A}_K/K \to 0.$$

Theorem 1.2.17. The volume of $\mathbb{A}_K/K$ (with respect to the uniquely defined measure discussed above) is 1.

Remark. We won't discuss the proof of Theorem 1.2.17 here, but we mention it as it is the starting point of rather deep theory on a currently open problem. Handwaving a lot of details, for any global field K, if we normalize measures as done above so the quotient $\mathbb{A}_K/K$ has volume 1, a conjecture of Weil says the the volume of the certain Haar measure of $G(K)\backslash G(\mathbb{A}_K)$ is 1, for any simpy connected (as an algebraic group, NOT a topological space!) semisimple algebraic group G.

§1.3 Adelic Approximations and Quotients

DEFINITION 1.3.1. We denote by $\mathbb{A}_\infty$ (resp. $\mathbb{I}_\infty$) the open subgroup $\mathbb{A}_{K,\mathscr{P}_\infty} \subseteq \mathbb{A}_K$ (resp. $\mathbb{I}_{K,\mathscr{P}_\infty} \subseteq \mathbb{I}_K$).

EXAMPLE 1.3.2 (Idelic form of $\mathrm{Cl}(K)$). By definition, for any idele $(x_v)_v \in \mathbb{I}_K$, we must have that $\mathrm{ord}_v(x_v) = 0$ for cofinitely many places. Thus, if we denote by $\mathrm{Div}(K)$ the set of all fractional ideals of K, we note the map

$$\begin{aligned} \mathfrak{i} : \mathbb{I}_K &\to \mathrm{Div}(K) \\ (x_v)_v &\mapsto \prod_{v \text{ finite}} \mathfrak{p}_v^{\mathrm{ord}_v(x_v)} \end{aligned}$$

is a well defined surjective group homomorphism with kernel $\mathbb{I}_\infty$. Moreover, for any principal fractional ideal $(x) \trianglelefteq K$, we note that it arises from the diagonally embedded $x \in \mathbb{I}_K$. Thus, we get the idelic expression of the ideal class group:

$$\mathrm{Cl}(K) \cong \mathbb{I}_K / K^\times \mathbb{I}_\infty$$

THEOREM 1.3.3. For every global field K,

$$\mathbb{A}_K = K + \mathbb{A}_\infty$$

where K is identified with its natural diagonal embedding.

Proof. Given some $x = (x_v) \in \mathbb{A}_K$, we want to find some $y \in K$ such that $x_v - y \in \mathcal{O}_v$ for all $v \nmid \infty$.
By definition, $x_v \in \mathcal{O}_v$ for all but finitely many finite places v, each corresponding to a prime $\mathfrak{p}_v \trianglelefteq \mathcal{O}_v$. Letting $p_v \in \mathcal{O}_K$ be an element of v-adic valuation 1 in each of these ideals, note that there must exist some $n_v \in \mathbb{N}$ such that $p_v^{n_v} x_v \in \mathcal{O}_v^\times$. Set

$$m := \prod_{v \in T} p_v^{n_v}$$

where T is the (finite!) set of finite places where $x_v \notin \mathcal{O}_v$. By the Chinese Remainder Theorem (note we use comaximality of primes in Dedekind domains here!), we can find some $\lambda \in \mathcal{O}_K$ such that $mx_v - \lambda \in \mathfrak{p}_v^{n_v}$ for all $v \in T$. Letting $y = \lambda / m$, note that:

$$\mathrm{ord}_{p_v}(x_v - y) = \mathrm{ord}_{p_v}(m^{-1}(mx_v - \lambda)) = n_v - \mathrm{ord}_{p_v}(m) = 0 \ \forall v \in T,$$

so by construction $x_v - y \in \mathcal{O}_v$ for all $v \nmid \infty$, as desired. □

COROLLARY 1.3.4. With the notation above, $K \cap \mathbb{A}_\infty = \mathcal{O}_K$

Proof. This follows from the standard fact that for a global field K we have

$$\mathcal{O}_K = \bigcap_{\text{finite } v \in \mathscr{P}_K} \mathcal{O}_v.$$

□

REMARK. Naturally, in the above corollary the notion of $\mathcal{O}_K$ and finite places of K when it has positive characteristic is dependent on a *choice* of isomorphism $K \cong \mathbb{F}_q(t)$.

Theorem 1.3.5. Any global field K is a discrete, cocompact subgroup of its adele ring, i.e. $\mathbb{A}_k/K$ is compact.

Proof. Let K_0 denote the smallest global subfield of K (i.e $\mathbb{Q}$ in characteristic 0 or $\mathbb{F}_p(t)$, after some *choice* of separating transcendence bases $\{t\}$ in positive characterstic), and let $n := [K : K_0]$. Then by 1.1.9, we have the following commutative diagram:

$$\begin{tikzcd} \prod_{j=1}^n \mathbb{A}_{K_0} \arrow[r] & \mathbb{A}_K \\ \prod_{j=1}^n K_0 \arrow[r] \arrow[u, hook] & K \arrow[u, hook] \end{tikzcd}$$

where the horizontal maps are topological isomorphisms. Thus $\mathbb{A}_K/K$ is compact if and only if $(\mathbb{A}_{K_0}/K_0)^n$ is compact, which is true if and only if $\mathbb{A}_{K_0}/K_0$ is compact.

Thus it suffices to consider $K = K_0$, in which case we have exactly one infinite place. Define a subset C of the adele group by

$$C := \left\{ x \in \mathbb{A}_K : |x_\infty|_\infty \leq \frac{1}{2} \text{ and } |x_v|_v \leq 1,\ \forall v \neq \infty \right\}$$

Note the condition on the finite places v simply amounts to saying that $x_v \in \mathcal{O}_v$. If we show that $C \cap K = \{0\}$ and $\mathbb{A}_K = C + K$, we will be done, since C is visibly compact and isomorphic to $\cong \mathbb{A}_K/K$.

If $x \in K \cap C$, then we must have $x \in \mathcal{O}_K$, since $x \in \mathcal{O}_v$ for every $v \nmid \infty$. If $K = \mathbb{Q}$ then this condition ($x \in \mathbb{Z}$) combined with the condition that $|x_\infty|_\infty \leq 1/2$ forces. $x = 0$. If $K = \mathbb{F}_p(t)$, then let the infinite place be the one defined by $1/t$. Then $|x|_\infty = p^{-\operatorname{ord}_{1/t}(x)}$, which is ≥ 1 for anything in $\mathcal{O}_K = \mathbb{F}_p[t]$. Thus $C \cap K = \{0\}$. This also proves discreteness since we can strengthen the defining inequalities of C to be strict, turning it into an open set of $\mathbb{A}_K$ meeting K only in the image of 0.

It remains to show any $x \in \mathbb{A}_K$ can be written as $y + c$, for y a principal adele and $c \in C$. Let S be the set of finite places where $x_v \notin \mathcal{O}_v$, so for each such v we want to construct some $\mu(v)$, integral at all finite places distinct from v, such that $x_v - \mu(v) \in \mathcal{O}_v$. Writing x_v in its (unique!) Laurent expansion form, we get

$$x_v = \sum_{i=-N}^{\infty} a_i \pi_v^i,$$

where each coefficient a_i is in $\mathbb{Z}$ or $\mathbb{F}_q$, depending on whether $K = \mathbb{Q}$ or $\mathbb{F}_p(t)$. Then we can let $\mu(v) := \sum_{i=-N}^{0} a_i p^i \in K$, so we immediately get $x_v - \mu(v) \in \mathcal{O}_v$, and moreover by construction $\mu(v)$ is integral at all places away from v. Therefore, for Then note that considering

$$y' := \sum_{v \in S} \mu(v) \in K,$$

we have that $x_v - y'_v \in \mathcal{O}_v$ for all finite places v.

For the infinite place, we claim we can now choose $y'' \in \mathcal{O}_K$ such that $|x_\infty - y' - y''|_\infty \leq 1/2$. For $K = \mathbb{Q}$, we can simply select y'' to be the closest integer to $x_\infty - y'$. If $K = \mathbb{F}_q(t)$, we may select y'' to be the non-negative degree component of $x_\infty - y'$, since subtracting this leaves us with a polynomial in t^{-1} with no constant term, and thus has norm $\leq q^{-1} \leq 1/2$, as we want. In either case, setting $y := y' + y''$, our constructions give us that $x - y \in C$, completing the proof. $\square$

Corollary 1.3.6. The image of the diagonal embedding of $K^\times$ is discrete in $\mathbb{I}_K$.

Proof. By 1.3.5 we know that K is discrete in $\mathbb{A}_K$, and so in particular the image of K under the map ϕ in Eqn **??** is also discrete in the product topology, and hence in the subspace topology of $\phi(\mathbb{A}_K^\times)$, which we have shown is the idelic topology. □

Remark. The discreteness of the image of the embedding is a more delicate fact than one might think on first inspection. We will see this when discussing strong approximation (Corollary 1.3.17) later on.

Example 1.3.7 (Volume of Quotient in non-Tamagawa Measure). For various purposes we may not always want to use the Tamagawa measure on $\mathbb{A}_K$, as we will see in more detail in §**??**, for example. Here we delve into an example of the types of explicit computations done.

Let K be a number field, and let $\mathrm{d}x$ be the measure induced on $\mathbb{A}_K$ by setting $\mathrm{Vol}(\mathcal{O}_v, \mathrm{d}x_v) = 1$ for all finite places v, and otherwise $\mathrm{d}x_v$ is the self-dual measure as defined in 1.2.15. Then:

$$\mathrm{Vol}(\mathbb{A}_K/K, \mathrm{d}x) = |\mathrm{disc}(\mathcal{O}_K)|^{1/2}.$$

Warning (Abuse of Notation). The reader will likely have noticed the measure $\mathrm{d}x$ is not actually a measure of the quotient group. What we technically mean is the Haar measure on the quotient induced by the counting measure on (the discrete normal subgroup!) K.

Proof Sketch. An examination of the proof of Theorem 1.3.5 shows that the set

$$C := [-1/2, 1/2] \times \prod_{p \text{ prime}} \mathbb{Z}_p$$

is a fundamental domain (i.e set of representatives for the orbits of the addition action by K on $\mathbb{A}_K$) for the discrete subgroup $\mathbb{Q} \subseteq \mathbb{A}_\mathbb{Q}$; that is to say $\mathrm{Vol}(\mathbb{A}_\mathbb{Q}/\mathbb{Q}, \mathrm{d}x) = \mathrm{Vol}(C, \mathrm{d}x)$. In the case of $K = \mathbb{Q}$ one immediately sees C has measure 1 = $\mathrm{disc}(\mathbb{Z})$.

To generalize the above construction, for $K/\mathbb{Q}$ a finite separable extension of degree n we may pick some isomorphism

$$\eta : \mathbb{R}^n \xrightarrow{\sim} K \otimes_\mathbb{Q} \mathbb{R} \left(\cong \prod_{v \in \mathscr{P}_{K,\infty}} K_v \right)$$

by sending the canonical basis to a $\mathbb{Q}$-basis of K $\{\xi_1, \ldots, \xi_n\}$ that is also a $\mathbb{Z}$-basis of $\mathcal{O}_K$. It is not too hard to show via similar methods that

$$C := \eta\left([-1/2, 1/2]^n\right) \times \prod_{v \notin \mathscr{P}_{K,\infty}} \mathcal{O}_v$$

is a fundamental set for K in $\mathbb{A}_K$. If $\mathrm{d}x_\infty$ denotes the product measure over the infinite places of K, then we note the proof comes down to showing

$$\mathrm{Vol}(\eta\left([0,1]^n\right), \mathrm{d}x_\infty) = |\mathrm{disc}(\mathcal{O}_K)|^{1/2}.$$

The key trick is to consider the embeddings

$$\sigma_h : K \otimes_{\mathbb{Q}} \mathbb{R} \hookrightarrow \mathbb{C},\ h \in [1, n]$$

that arise as ($\mathbb{R}$-linear) extensions of the canonical n distinct embeddings $K \hookrightarrow \mathbb{C}$. The problem then reduces to computing the pushforward measure $\prod_{h\in[n]} \mathrm{d}\sigma_h(\theta(u))$, which one can check to be $\pm\det(N)\prod_{h\in[n]} \mathrm{d}x_h$, where N is the matrix given by $[(\sigma_h(\xi_i))_i]$, i.e rows are given by the images of $u \in \mathbb{R}^n$ under composition by η followed by σ_h. One can also readily check the determinant of this matrix is the square of the discriminant of K. We refer the reader to Chapter 5 Section 4 of [Wei74] for a more detailed exposition. □

Since we have an analogous result for discrete embeddings of the base field into the adeles and ideles, one might also hope for a similar cocompactness analogue; i.e that $K^\times$ is also cocompact in $\mathbb{I}_K$. Unfortunately, this is not the case. To analyze this phenomenon, we first standardize the absolute value functions we're using.

Definition 1.3.8. For K a global field, the **normalized absolute value** on its completion K_v at a place v is defined as follows:

(a) For $K_v = \mathbb{R}$ we set $|\cdot|_v$ to be the usual absolute value function.

(b) For $K_v = \mathbb{C}$ we set $|z|_v := z\bar{z}$, which is the square of the usual absolute value function on $\mathbb{C}$.

(c) For K_v non-Archimedean with uniformizer π and residue field of order q, we set $|\pi|_v := q^{-1}$, which extends uniquely to K_v.

At first glance these normalized absolute values may seem arbitary and even a bit odd - why set it to be the square of the standard absolute value on $\mathbb{C}$, for example? The answer is these absolute values are precisely the ones for which $|x|_v$ is the modulus of the automorphism $y \mapsto xy$. That is to say,

$$|x|_v = \frac{\mu(x\mathcal{O}_v)}{\mu(\mathcal{O}_v)},$$

for μ a Haar measure on K_v. Note we may replace $\mathcal{O}_v$ in the above expression with any measurable set of finite positive measure. We leave the verification of these facts to the reader; they are not hard to check.

Definition 1.3.9. Let K be a global field, and let $|\cdot|_v$ denote the normalized absolute value on the completion K_v. Then the **idelic norm** (sometimes called the **idelic absolute value**) is defined by:

$$\begin{aligned} |\cdot|_{\mathbb{I}_K} : \mathbb{I}_K &\to \mathbb{R}_{\geq 0} \\ x = (x_v)_v &\mapsto \prod_v |x_v|_v \end{aligned}$$

Sanity Check. Note that $|\cdot|_{\mathbb{I}_K}$ does indeed converge, since by definition any $x = (x_v) \in \mathbb{I}_K$ is integral (i.e has local norm ≤ 1) for cofinitely many v.

Remark. Often we will care about this absolute value's relationship and values on the ideles, hence it is often also known as the **idelic norm**. Whenever we say this we simply mean the restriction of the above to $\mathbb{I}_K$.

Theorem 1.3.10 (Product Formula). Let K be a global field. Then for every $x \in K^\times$ we have $|x|_{\mathbb{I}_K} = 1$

Proof. Let E/K be finite separable. Then

$$\begin{aligned} |x|_{\mathbb{I}_E} &= \prod_{u \in \mathscr{P}_K} \prod_{v \in \mathscr{P}_E,\, v|u} |x|_v \\ &= \prod_{u \in \mathscr{P}_K} \prod_{v \in \mathscr{P}_E,\, v|u} |N_{E_v/K_u}(x)|_u \\ &= \prod_{u \in \mathscr{P}_K} |N_{E/K}(x)|_u, \end{aligned}$$

and since the norm carries $E^\times$ into $K^\times$ showing the result for K proves it for E. In particular, it suffices to prove this for $\mathbb{Q}$ or $\mathbb{F}_q(t)$. Furthermore, mutiplicativity of the adelic norm allows us to verify the result for just the irreducibles of $\mathcal{O}_K$.

- For $K = \mathbb{Q}$, since $\mathcal{O}_\mathbb{Q} = \mathbb{Z}$ we can check it for positive primes p. By definition $|p|_q = 1$ for all primes $q \neq p$. Thus we only have to multiply the p-adic and infinite place, giving us:

$$|p|_{\mathbb{I}_K} = |p|_\infty \cdot |p|_p = p \cdot \frac{1}{p} = 1.$$

- For $K = \mathbb{F}_q(t)$, we only have to consider irreducible polynomials $p(t) \in \mathbb{F}_q[t]$. Again this has trivial absolute value at all but two places. This gives us:

$$|p(t)|_{\mathbb{I}_K} = |p(t)|_\infty \cdot |p(t)|_{p(t)} = q^{\deg p} \cdot \left(|\mathbb{F}_q[t]/\langle p(t)\rangle|\right)^{-1} = 1.$$

This completes our proof. $\square$

Remark. Some texts and notes (including [RV98] and the "Adele ring" Wikipedia page) refer to 1.3.10 as the *Artin product formula*. A less trivial result, also due to Artin, is the converse: that any field K for which the "product formula" holds must be a global field. For a more precise statement with proof, we direct the reader to the 1945 paper of Artin and Whaples [AW45].

Example 1.3.11 (Decomposition of $\mathbb{I}_\mathbb{Q}$). We claim we can describe the idele group over $\mathbb{Q}$ as follows:

$$\mathbb{I}_\mathbb{Q} = \mathbb{Q}^\times \times \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times,$$

where $\widehat{\mathbb{Z}} := \varprojlim_{n \in \mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$, where the inverse limit is taken via the natural maps induced by divisibility. The notation is unfortunate here; the hat here signifies completion, NOT Pontryagin dual.
To see this, consider the map

$$\begin{aligned} \phi : \mathbb{I}_\mathbb{Q} &\to \mathbb{Q}^\times \\ x = (x_v)_v &\mapsto \operatorname{sgn}(x_\infty) \prod_p |x_p|_p^{-1}. \end{aligned}$$

Recalling our normalized absolute values defined in 1.3.8, it is immediate from its construction that $\phi(x) = x$ for any $x \in \mathbb{Q}^\times \subseteq \mathbb{I}_\mathbb{Q}$. Thus, $\mathbb{I}_\mathbb{Q} = \mathbb{Q}^\times \times \ker(\phi)$. It is also immedaite by construction that $\ker(\phi) = \mathbb{R}_{\geq 0} \times \prod_p \mathbb{Z}_p^\times$, and the identity $\widehat{\mathbb{Z}}^\times \cong \prod_p \mathbb{Z}_p^\times$ completes our proof.

Observation. The product formula allows us to generalize the above example to see why $\mathbb{I}_K/K^\times$ is *never* compact.
In the case of K being a number field, consider the idele

$$x := (t, 1, 1, \ldots),$$

which is 1 everywhere except a single infinite place. Note then that $|x|_{\mathbb{A}_K} = t$ or t^2, depending on whether the place is real or complex. Regardless, this observation tells us that $\mathbb{I}_K$ continuously surjects onto $\mathbb{R}_{\geq 0}$ (a non-compact image!) under the idelic norm map.
When K has positive characteristic, pick a choice of isomorphism $K \cong \mathbb{F}_q(t)$. Then at any completion, the norm of any element must be a power of q. The product formula then tells us that the image of the idelic norm must be some finite-index subgroup of $q^{\mathbb{Z}}$, thus noncompact (as an infinite discrete group).

Example 1.3.11 gave an explicit description of the non-compactness of $\mathbb{I}_K/K$, but it fortunately also provides a direction in which we may amend this.

Definition 1.3.12. For K a global field, recall we have the idelic norm $|\cdot|_{\mathbb{I}_K} : \mathbb{I}_K \to \mathbb{R}_{>0}$. Given this, the **ideles of norm 1** are:

$$\mathbb{I}_K^1 := \ker(|\cdot|_{\mathbb{I}_K})$$

and the **norm-1 idele class group** is the quotient:

$$C_K^1 := \mathbb{I}_K^1/K^\times$$

We have the short-exact sequence

$$1 \to C_K^1 \to C_K \to |\mathbb{I}_K|_{\mathbb{I}_K} \to 1$$

which is in fact split-exact in the number field case, by our previous reasoning for non-compactness of C^K. We should guess that this C_K^1 should be the compact quotient we're looking for. Before getting into that, we make an interesting observation about the norm-1 ideles that will come in useful later.

Lemma 1.3.13. The subspace topologies of $\mathbb{I}_K^1$ taken with respect to $\mathbb{I}_K$ or $\mathbb{A}_K$ are the same.

Proof. Suppose we give $\mathbb{I}_K^1$ the idelic subspace topology by default; it then suffices to show that the injection $\phi : \mathbb{I}_K^1 \hookrightarrow \mathbb{A}_K$ is a homeomorphism onto its image. We have seen in Example 1.1.7 that the subspace adelic topology is coarser than the idelic topology, so ϕ is automatically continuous.
To show it is an open map onto its image, it suffices to show any basic open set of $\mathbb{I}_k^1$ contains a subset open in the adelic subspace topology. To this end, take some basic open neighborhood of the multiplicative identity $1 \in \mathbb{I}_K^1$ in the idelic subspace topology. This has the form:

$$U = \prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v^\times \cap (\mathbb{I}_K^1)$$

where each U_v is some open ball around $1 \in K_v^\times$, and S is some finite index set containing the infinite places. Now consider the set

$$W := \prod_{v \in S} W_v \times \prod_{v \notin S} \mathcal{O}_v,$$

where the index set S is the same as in the description of U, each $W_v \subseteq U_v$, shrunk appropriately so that for every $\gamma \in W$ satisfies the (strict!) inequality

$$\prod_{v \in S} |\gamma_v|_v < 2.$$

We see that W an open neighborhood of 1 in the adelic topology. We claim that $W \cap \mathbb{I}_K^1 \subseteq U$. To see this, note that, for $\gamma \in W \cap \mathbb{I}_K^1$, by definition we have:

$$|\gamma|_{\mathbb{A}_K} = 1 = \prod_{v \in S} |\gamma_v|_v \prod_{v \notin S} |\gamma_v|_v < 2 \prod_{v \notin S} |\gamma_v|_v,$$

where the final inequality follows by our construction of W. Since for every $v \notin S$, we have $\gamma_v \in \mathcal{O}_V$, we have $|\gamma_v|_v \leq 1$, which tells us in particular that $1 < 2|\gamma_v|_v$. This, in turn, implies $|\gamma_v|_v > 1/2$ which then also implies $\gamma_v \in \mathcal{O}_v^\times$ for all $v \notin S$, and so we have our desired containment in U. □

The main result we wish to show is the following.

> THEOREM 1.3.14. The norm-one idele class-group C_K^1 is always compact.

To prove this result we will need an adelic version of a classic result in the so-called geometry of numbers. In particular, recall the Blichfeldt-Minkowski lemma, which states that any compact, convex, origin-symmetric set of $\mathbb{R}^n$ with volume greater than 2^n must contain at least one nonzero integer lattice point.
Using the (alternative, depending on your experience) characterization of a lattice as a discrete, co-compact subgroup of an $\mathbb{R}$-vector space, in light of 1.3.5 we are led to the following analogous statement about the diagonal embedding $K \hookrightarrow \mathbb{A}_K$.

> LEMMA 1.3.15 (Adelic Blichfeldt-Minkowski). For any $\gamma = (\gamma_v) \in \mathbb{I}_K$, consider the (compact, convex and origin-symmetric!) subset
>
> $$X_\gamma := \{x \in \mathbb{A}_K : |x_v|_v \leq |\gamma_v|_v \ \forall v \in \mathscr{P}_K\}.$$
>
> Then there exists some constant $C > 0$ such that if $|\gamma|_{\mathbb{A}_K} > C$ then $X_\gamma \cap K$ meets $K^\times$.

Proof. Consider the following compact subset:

$$D := \{(d_v) \in \mathbb{A}_K : |d_v|_v \leq 1/4,\ v \mid \infty,\ |d_v|_v \leq 1,\ v < \infty\}.$$

This closed subset is the product of the valuation rings at all the finite places and the balls of radius 1/4 (resp. 1/2) at all the real (resp. complex) places. Thus, under our normalized measure dx on $\mathbb{A}_K$, this has finite and positive measure, since there are only a finite number of infinite places. Moreover, for any $d, d' \in D$ our construction tells us $|d'_v - d_v|_v \leq 1$ for all places v.
Let D_0 be a fundamental domain for K in $\mathbb{A}_K$ (e.g we may choose our construction in the proof of 1.3.5), which must also have finite positive measure by compactness of the quotient. We now define our constant $C := dx\,(D_0)/\,dx\,(D)$.
Since the idelic absolute value is the module of the multiplication automorphism, we get:

$$|\gamma|_{\mathbb{A}_K} > C \implies dx\,(\gamma D) > dx\,(D_0).$$

Thus, the set γD cannot lie within a fundamental domain for K, and thus must contain two distinct elements

$\gamma d_1, \gamma d_2$ that project to the same nonzero value in $\mathbb{A}_K/K$, i.e such that $d := \gamma d_1 - \gamma d_2 \in K^\times$. Thus, we have:

$$|d_v|_v \leq |\gamma_v|_v |d_{1,v} - d_{2,v}|_v \leq |\gamma_v|,$$

and thus $d \in X_\gamma$ by definition. Thus, $X_\gamma \cap K^\times$ is non-empty, completing our proof. □

We are now ready to show compactness of C_K^1.

Proof of 1.3.14. By Lemma 1.3.13, it suffices to find some $W \subseteq \mathbb{A}_K$, compact in the adelic topology, such that the continuous map

$$W \cap \mathbb{I}_K^1 \to C_K^1$$

is a surjection. To this end, consider a positive real $C > 0$, and pick any $\gamma \in \mathbb{I}_K$ with $|\gamma|_{\mathbb{A}_K} > C$, and any $\beta \in \mathbb{I}_K^1$ and define

$$W_C := \left\{x \in \mathbb{A}_K : |x_v|_v \leq \left|\beta_v^{-1}\gamma_v\right|_v \ \forall v \in \mathscr{P}_K\right\},$$

which is visibly compact in $\mathbb{A}_K$. By the adelic Blichfeldt-Minkowski lemma (1.3.15), we may pick C large enough so that W_C contains a non-zero principal adele $\alpha \in K^\times \subseteq \mathbb{I}_K^1$.
We now set

$$W := \{x \in \mathbb{A}_K : |x_v|_v \leq |\gamma_v|_v \ \forall v \in \mathscr{P}_K\},$$

which is again visiby compact in $\mathbb{A}_K$, and note by construction we have $\alpha\beta \in W$. We deduce $\alpha\beta \in W \cap \mathbb{I}_K^1$, with image $\overline{\beta}$ in C_K^1. Since $\beta \in \mathbb{I}_K^1$ was arbitrary, we deduce the quotient map is indeed surjective, and thus C_K^1 is compact. □

Remark. The above result is important for two primary reasons. Firstly, it is akin to the finiteness of the class number of a number field, in the sense that it implies "finiteness" of generalized ideal class groups of abelian number field extensions. Secondly, as we will see in Chapter II, compactness limits the variety of so-called Hecke characters that can arise when one is doing Fourier anaylysis following the method of Tate.

We close with perhaps the most powerful approximation result on the adeles: strong approximation.

> **Theorem 1.3.16.** For K a global field, let w be a fixed finite place and let S be a finite subset of $\mathscr{P}_K \setminus \{w\}$. For every v in S, select arbitrary $a_v \in K$ and $\epsilon_v \in \mathbb{R}_{>0}$. Then there exists some $x \in K$ such that
>
> $$|x - a_v|_v \leq \epsilon_v, \forall v \in S$$
> $$|x|_v \leq 1, \forall v \notin S \cup \{w\},$$
>
> with no restriction on $|x|_w$.

Proof. Pick some $\gamma \in \mathbb{A}_K$ satisfying the following conditions:

1) $|\gamma_v|_v \leq \epsilon_v$, for every $v \in S$.

2) $|\gamma_v|_v = 1$, for every $v \in T := \mathscr{P}_K \setminus (\{w\} \cup S)$.

3) $|\gamma_w|_w > C \prod_{v \in S} |\gamma_v|_v^{-1}$ for the place w, where C is the "minimal volume" constant associated to $\mathbb{A}_K$ by the adelic Blichfeldt-Minkowski lemma (1.3.15).

Picking such a γ is possible since T is a cofinite index set, so the conditions imposed satisfy being cofinitely integral everywhere. Lemma 1.3.15 then tells us that $X_\gamma \cap K$ contains a non-zero principal adele z.
Now consider the set

$$W := \{w \in \mathbb{A}_K : |w_v|_v \leq 1, \ \forall v \in \mathscr{P}_K\}.$$

Recalling the proof of Theorem 1.3.5, we note our construction there generalizes to show there exists some $y \in K$ such that $x - y \in W$. Using the notation of that proof, to construct $\mu(v)$ for all the finite non-integral places, the process is identical, so it remains to check the correction at the infinite places. In particular we want some y' so that $|x_w - \sum_{v \in S} \mu(v) - y'|_w \leq 1$ for all infinite places w.

Moreover, for any $b \in K^\times$ we may write $\mathbb{A}_K = K + bW$, since multiplication by b is an automorphism of K. Thus for an arbitary $x = (x_v)_v \in \mathbb{A}_K$, we have

$$x = a + y,$$

for some $a \in K$ and $y \in zW$, where z is the principal adele in $X_\gamma \cap K$ discussed earlier in this proof. Thus we have:

$$|x_v - a_v|_v = |y_v|_v \leq |z_v|_v \leq |\gamma_v|_v,$$

which, by construction, gives us our desired inequality for every place $v \in S \cup T$. □

Corollary 1.3.17 (Strong Approximation). For any place $w \in \mathscr{P}_K$, the diagonal embedding

$$K \hookrightarrow \coprod_{v \in \mathscr{P}_K \setminus \{w\}} K_v := \mathbb{A}_K^w$$

is dense. The corresponding statement also holds for $K^\times$ and $\mathbb{I}_K^w := \mathbb{I}_K^1 \cap \mathbb{A}_K^w$.

Proof. This is immediate from the above theorem and the definition of the adelic norm; barring a restriction on the place w, we may make the norm of the difference between any given adele and a principal adele arbitrary small. The second statement is a consequence of the fact that $K^\times \subseteq \mathbb{I}_K^1$ and that $\mathbb{I}_K^1$ inherits the subspace topology from $\mathbb{A}_K$ (Lemma 1.3.13). □

Comparing the above result to Theorem 1.3.5 gives us a better appreciation for the volatilty of the embedding $K \hookrightarrow \mathbb{A}_K$. In the full adele ring, the image is so sparse it is discrete, but if we remove even a single place from the restricted product the image is dense!

Example 1.3.18 (Discrete and Dense). Let K be a number field and let S_∞ be the set of Archimedean places. Ignoring them, we get the so-called **finite adeles**, given by

$$\mathbb{A}_K^{fin} = \coprod_{v \nmid \infty} K_v.$$

A similar construction gives us $\mathbb{I}_K^{fin}$, the **finite ideles**. Consider the intersection of $K^\times$ with the compact open $\prod_{v \nmid \infty} \mathcal{O}_v$, which we note corresponds exactly with the unit group $\mathcal{O}_K^\times$. When $K = \mathbb{Q}$ or is imaginary quadratic, this is finite.

The statement of strong approximation also generalizes to simply connected (in teh sense of algebraic groups, not topological spaces!) algebraic groups. In general such an algebraic group G is said to have the *strong approximation* property if the embedding $G(K) \hookrightarrow G(\mathbb{A}_{K,S})$ has dense image for some finite set $S \subsetneq \mathscr{P}_K$.

§1.4 The S-Groups

Recall that if S denotes a finite set of places of a global field K, then the adele ring can be described as the direct limit

$$\mathbb{A}_K = \varinjlim_S \mathbb{A}_{K,S}.$$

It is useful to have "S-analogues" of certain compactness results and volume calculations, as we'll see in the chapter of Tate's thesis. Often (but not always!) we'll be interested in the cases when $S \supseteq S_\infty = \mathscr{P}_{K,\infty}$, i.e when S contains all the Archimedean places of K.

Warning. The notation used here is conventional but non-ideal if one wishes to generalize the theory of adeles beyond number fields, as it conflates the Archimedean and infinite places, which is not the case in positive characteristic. In particular S_∞ is EMPTY if $char(K) > 0$.

Definition 1.4.1. Take S any finite set of places of a global field K. Then:

(a) The S-**adeles** of K are defined as:

$$\mathbb{A}_{K,S} := \prod_{v \in S} K_v \times \prod_{v \notin S} \mathcal{O}_v$$

(b) The **ring of S-integers** is defined as:

$$\mathcal{O}_{K,S} := K \cap \mathbb{A}_{K,S}$$

i.e the elements of K that are v-integral for all $v \notin S$.

(c) The S-**ideles** are defined as:

$$\mathbb{I}_{K,S} := \prod_{v \in S} K_v^\times \times \prod_{v \notin S} \mathcal{O}_v^\times$$

(d) The S-**ideles of norm 1** defined as:

$$\mathbb{I}^1_{K,S} := \mathbb{I}^1_K \cap \mathbb{I}_{K,S}$$

Sanity Check. The S-ideles are algebraically the units of the S-adeles, and $\mathcal{O}^\times_{K,S} = K^\times \cap \mathbb{I}_{K,S}$.

The primary result of this subsecton will be the following theorem, which characterizes the unit group of $\mathcal{O}_{K,S}$ defined above.

Theorem 1.4.2 (S-unit theorem). For K a global field any nonempty finite set of places S containing the infinite ones, we have an isomorphism

$$\mathcal{O}_{K,S}^{\times} \cong \mu(K) \times \mathbb{Z}^{|S|-1},$$

where $\mu(K)$ is the (finite!) group of roots of unity in K.

Example 1.4.3 (Dirichlet's Unit Theorem). When K is a number field with r_1 real embeddings and r_2 complex embeddings, then setting $S = S_\infty = \mathscr{P}_{K,\infty}$, 1.4.2 tells us that $\mathcal{O}_K^\times = \mathcal{O}_{K,S_\infty}^\times$ is finitely generated of rank $r_1 + r_2 - 1$, which is precisely the classical theorem of Dirichlet. Thus we may see 1.4.2 as a vast generalization of this result, of which we will proceed to give a neat proof using the adelic formulation.

To prove this S-unit theorem we will need a few intermediary results, the firsst of which is the following lemma.

Lemma 1.4.4. For K a global field, $x \in K$ is a root of unity (i.e has finite multiplicative order) if and only if $|x|_v = 1$ for all $v \in \mathscr{P}_K$.

Proof. One direction is clear, so for the other one suppose x has norm 1 at all places.
When K is a number field, this is a classical result of Kronecker, but we recount a detailed proof here. We note x must be an algebraic integer, all of whose complex embeddings have norm 1. Its minimal polynomial $f(t)$ is integral and has the form = $\prod_i (t - x_i)$ taken over all Galois conjugates of x. Consider the coefficient of t^k. It is the sum of $\binom{n}{k}$ terms, where n is the degree of f and where each term is a product of x_i's, and thus must have complex norm 1. Thus the coefficient, since it lies in $\mathbb{Z}$, must be bounded in absolute value by $\binom{n}{k}$. Since $x \in \mathcal{O}_K$, we know $n \leq [K : \mathbb{Q}]$. Moreover, for any $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$, we note that $|\sigma(x^m)| = |\sigma(x)|^m = 1$ for all $m \geq 1$. Thus the minimal polynomials of all such x^m have coefficients bounded by $\binom{[K:\mathbb{Q}]}{\lfloor [K:\mathbb{Q}]/2 \rfloor}$ (i.e the largest binomial coefficient), and their degree is also bounded by $[K : \mathbb{Q}]$. Thus the powers of x are roots of a finite set of polynomials, so the cyclic group generated by x is finite, making it a root of unity.
When K has positive characteristic, making a choice of isomorphism $K \cong \mathbb{F}_q(t)$, the condition tells us that x is a rational function with no zeroes or poles, hence lies in the field of constants, and thus has finite order. $\square$

Next, we need a generalization of the compactness of the norm-1 idele class group C_K^1 to the S-ideles.

Lemma 1.4.5. The quotient $\mathbb{I}_{K,S}^1 / \mathcal{O}_{K,S}^\times$ is compact, for any set S.

Proof. Firstly $\mathcal{O}_{K,S}^\times$ is visibly a subgroup of $\mathbb{I}_{K,S}^1$, using the fact that $K^\times$ embeds in $\mathbb{I}_K$ and the product formula (1.3.10), thus the quotient makes sense.
We also see that $\mathbb{I}_{K,S}$ is by definition an open subgroup of $\mathbb{I}_K$, and thus $\mathbb{I}_{K,S}^1$ is open (hence closed!) in $\mathbb{I}_K^1$ under the subspace topology. Since C_K^1 is a compact (Hausdorff) quotient, it follows $\mathbb{I}_{K,S}^1 / \mathcal{O}_{K,S}^\times$ is a closed subgroup, and thus also compact. $\square$

We now are ready to prove the main result.

Proof of 1.4.2. Consider the "adelic circle", defined as the product of all the norm-1 elements of each complemetion, i.e.

$$C_{\mathbb{A}} := \prod_{v \in \mathscr{P}_K} C_v^1,$$

where $C_v^1 := \{x_v : |x_v|_v = 1\} \subseteq K_v^\times$. Note when v is non-Archimedean then $C_v^1 = \mathcal{O}_v^\times$. By Tychonoff, this is a compact subgroup of $\mathbb{I}_{K,S}$ for any S. Thus we have the exact sequence:

$$1 \to C_{\mathbb{A}} \to \mathbb{I}_{K,S} \to \prod_{v \in S} (K_v^\times / C_v^1) \to 1. \tag{1.4-1}$$

Note that when v is non-Archimedean then

$$K_v^\times / C_v^1 = K_v^\times / \mathcal{O}_v^\times \cong \mathbb{Z},$$

and when v is Archimedean we have

$$K_v^\times / C_v^1 \cong \mathbb{R}_{>0} \cong \mathbb{R}.$$

Thus, if we have n non-Archimedean places and m Archimedean places in S, Eqn. 1.4-1 becomes

$$1 \to C_{\mathbb{A}} \to \mathbb{I}_{K,S} \to \mathbb{Z}^n \times \mathbb{R}^m \to 1. \tag{1.4-2}$$

Now we consider the (canonical) image of $K^\times$ under the diagonal embedding in each of the objects in the sequence 1.4-2. Firstly, we see $K^\times \cap C_{\mathbb{A}}$ consists of all elements that have norm 1 at every place, which Lemma 1.4.4 tells us is precisely $\mu(K)$. For the middle object, we recover exactly $\mathcal{O}_{K,S}^\times$. For the final object, we may identify the image of $K^\times$ with some arbitrary algebraic object under the embedding and quotient maps, which we denote by L. This transforms 1.4-2 to the following short exact sequence:

$$1 \to \mu(K) \to \mathcal{O}_{K,S}^\times \to L \to 1. \tag{1.4-3}$$

It remains to show that $L \cong \mathbb{Z}^{n+m-1}$. Consider the natural continuous map

$$\begin{aligned} \phi : \prod_{v \in S} (K_v^\times / C_v^1) &\to |\mathbb{I}_K|_{\mathbb{I}_K} \\ (x_v)_{v \in S} &\mapsto \prod_{v \in S} |x_v|_v . \end{aligned}$$

When K is a number field, this can abstractly be seen as a continuous homomorphism $\mathbb{Z}^n \times \mathbb{R}^m \to \mathbb{R}_{>0}$ that is nontrivial on each copy of $\mathbb{R}$, whose kernel is thus isomorphic to $\mathbb{Z}^n \times \mathbb{R}^{m-1}$. Since the image of $K^\times$ under its canonical embedding (i.e. L) lies within this kernel, and is in fact a discrete and cocompact subgroup. Any subgroup of $\mathbb{Z}$ is isomorphic to $\mathbb{Z}$, and any discrete subgroup of $\mathbb{R}^{m-1}$ must be finite free abelian, and the cocompactness conditions enforces a rank $m-1$ condition as well. Thus, $L \cong \mathbb{Z}^{n+m-1}$, as desired.
When K has positive characteristic, recalling the image of the idelic norm is some finite index subgroup of $q^{\mathbb{Z}}$ (hence isomorphic to $\mathbb{Z}$), we can view ϕ as a surjective map $\mathbb{Z}^n \to \mathbb{Z}$ (recall a function field has no Archimedean places!) whose kernel thus must be isomorphic to $\mathbb{Z}^{n-1}$, as desired. □

The above result motivates the following definition.

Definition 1.4.6. For S any nonempty finite set of places of a global field K, the S-**class group** of K is

$$C_{K,S} := \mathbb{I}_K / (K^\times \cdot \mathbb{I}_{K,S})$$

Observation. We have the following isomorphism:

$$\mathbb{I}_K^1 / (K^\times \cdot \mathbb{I}_{K,S}^1) \xrightarrow{\sim} \mathbb{I}_K / (K^\times \cdot \mathbb{I}_{K,S}) =: C_{K,S}.$$

We always have a canonical inclusion from the left side to the right side, and for any element of the right side we may always represent it by an idele of norm 1 by adjusting the value in K_v for any $v \in S$. Putting these observations, along with previous results, together, we get the following result.

Proposition 1.4.7. Let S be any nonempty finite set of places of a global field K. Then $C_{K,S}$ is a finite group.

Proof. Follows from the above observation and the proof of 1.4.2. □

The above result will be crucial in volume and residue calculations of L-functions over number fields, which will be covered in the subsequent chapter.

Example 1.4.8 (The Ideal Class Group). When $S = S_\infty$, we get precisely $\mathbb{I}_\infty$ defined in 1.3.1. Examining the Example following that definition, we get

$$C_{K,S_\infty} = \mathrm{Cl}(K),$$

which, from Proposition 1.4.7, is another way to show the ideal class group of a number field is finite.

Chapter II

The Thesis of Tate (a.k.a, GL_1)

"In [his thesis], Tate provides an elegant and unified treatment of the analytic continuation of the L-functions attached by Hecke to his Größencharaktere... The power of the methods of abelian harmonic analysis in the setting of Chevalley's adeles/ideles provided a remarkable advance over the classical techniques used by Hecke."

Ch. 6 of *An Introduction to the Langlands Program*
Stephen S. Kudla

One can generalize Dirichlet characters to quasi-characters of ray class groups of number fields, and then one can generalize even further to continuous characters of the idele class group C_K for K a global field. It is a famous result of Hecke that $L(s,\chi)$ has analytic continuation and satisfies an appropriate functional equation for any number field K and any idele class character χ. The original formulation of this result involved a complicated computation with theta functions and Mellin transforms. Moreover in the functional equation a so-called *root number* $W(\chi)$ associated to each quasi-character χ kept popping up, with no good explanation for its value.

Under the guidance of Emil Artin, in his 1950 thesis John Tate made use of Fourier analysis on adele groups to reprove the results of Hecke, first by establishing local functional equations at all the completions of a global field and then using the adelic formulation to give an elegant description of a global functional equation on quasi-characters of the idele group.

Because of the limits of space and time we do not go into Hecke's original approach via theta functions here, but we recommend the interested reader to read Chapter VII of [NS10] for a detailed exposition of that method, in order to gain a better appreciation of the relative elegance of Tate's packaging.

The novice reader may also question why the moniker of "GL_1" for this chapter. The answer lies in deeper theory, and the reader should (after reading this chapter, of course) continue to the subsequent chapter to find out the reason why.

Section 2

The Local Theory

§2.1 The Local Functional Equations

In his thesis, Tate worked with some ad-hoc function spaces over local and global fields. We standardize our workings over the so-called *Schwartz-Bruhat functions* on local fields. Over an Archimedean field (i.e $\mathbb{R}$ or $\mathbb{C}$), recall a **Schwartz** function (more commonly known as a *rapidly-decreasing function*) is a smooth $\mathbb{C}$-valued function f such that

$$\lim_{|x|\to\infty} p(x)D^k f(x) = 0$$

for all polynomials $p(x)$ and k-th derivative operators D^k (corresponding to $\mathrm{d}^k / \mathrm{d}x^k$ for $F = \mathbb{R}$ and $\partial^k/(\partial x^i \partial y^{k-i})$ for $F = \mathbb{C}$). These functions classicaly form a useful space on which to do Fourier analysis, and so we define a similarly useful generalization for our adelic purposes.

Definition 2.1.1. A **Schwartz-Bruhat** function on a local field F is a Schwartz function if F is Archimedean and a smooth[a], compactly-supported function in F is non-Archimedean. We denote by $\mathscr{S}(F)$ the complex vector space of Schwartz-Bruhat functions on F.

[a] A function on a non-Archimedean local field is smooth if it is locally constant.

This gives us the correct analogue of the space of tempered distributions in classical Fourier theory, as we'll see in §3.1.

Before analyzing the global zeta function of a function on an adele group of a global field, we first understand the induced zeta functions on each of the local components. The goal of this subsection will be to describe a functional equation of zeta functions on a local field by introducing so-called *local L-factors* associated to a character χ (and its dual under the Fourier transform $\chi^\vee$) of a local field. In particular, these local L-factors will be realized as the gcd of some local zeta functions. We start with a slew of notations and definitions, so buckle up.

When F is non-Archimedean, let $\mathcal{O}_F$ denote its ring of integers, $\mathfrak{p}$ its maximal ideal with uniformizer π and residue field $\mathbb{F}_q$. Recall that we can write $F^\times \cong \mathcal{O}_F^\times \times \mathcal{S}_F$, where $\mathcal{S}_F = |F^\times|$, i.e. the image of $F^\times$ in the valuation map, up to choice of uniformizer π and homomorphic retraction $F^\times \to \mathcal{O}_F^\times$.
We make the observation that any quasi-character χ of $F^\times$ factors into a product:

$$\chi = \mu|\cdot|^s,$$

where $s \in \mathbb{C}$ and μ is the pullback of a character on $\mathcal{O}_F^\times$ (defined uniquely by restriction of χ!). This is because compactness of $\mathcal{O}_F^\times$ forces any quasi-character on it to have image in S^1, and the only quasi-characters on $\mathcal{S}_F$ are maps $t \mapsto t^s$.

Definition 2.1.2. For any quasi-character $\chi = \mu|\cdot|^s$ of $(F^\times)$, we call $\mathfrak{Re}(s)$ the **exponent** of χ. Furthermore, we say χ is **unramified** if $\mu = 1$. We denote by $\mathscr{X}(F^\times)$ the space of equivalence classes of quasi-characters on $F^\times$, where two are considered equivalent if their quotient is unramified.

WARNING. Recall that for any $q \in \mathbb{R}_{>0}$ we have the formula:

$$q^i = \cos(\log(q)) + i\sin(\log(q)).$$

In particular, in the non-Archimedean case the value of s is NOT uniquely determined (e.g $s = i$ and $s = (1+2\pi)i$ would correspond to the same map), which is why we take the exponent to be the real part, which IS uniquely determined.

Note that the group of unramified quasi-characters has the structure of a Riemann surface, since it is determined by the complex power of the absolute value map. In particular it is isomorphic to $\mathbb{C}$ in the Archimedean case and $\mathbb{C}/((2\pi i/q)\mathbb{Z})$ in the non-Archimedean case. Thus the total space of quasi-characters is the disjoint union of at most countably many Riemann surfaces, and our quotient space $\mathscr{X}(F^\times)$ is a Riemann surface.

We now state certain L-functions that will show up in the functional equation involving the (local!) zeta functions, which will be described later.

(1) In the non-Archimedean case, we set:

$$L(\chi) := \begin{cases} (1-\chi(\pi_F))^{-1}, & \text{if } \chi \text{ is unramified,} \\ 1, & \text{otherwise.} \end{cases} \tag{2.1-1}$$

(2) In the Archimedean case we recall by Gelfand-Mazur that $F = \mathbb{C}$ or $\mathbb{R}$.

(i) If $F = \mathbb{C}$, note any χ is of the form:

$$\chi_{s,n} : z \mapsto |z|^s \left(\frac{z}{|z|}\right)^n$$

for $s \in \mathbb{C}$ and $n \in \mathbb{Z}$ uniquely defined. Here we use our result in A.1.2 to characterize the characters of $U_\mathbb{C} = S^1$, and by $|z|$ we mean the usual absolute value on $\mathbb{C}$, not our normalized one. We then set:

$$L(\chi_{s,n}) := \Gamma_\mathbb{C}(s + \frac{|n|}{2}) := (2\pi)^{-\left(s+\frac{|n|}{2}\right)}\Gamma(s + \frac{|n|}{2}). \tag{2.1-2}$$

(ii) If $F = \mathbb{R}$ then $U_F = \{\pm 1\}$. Here any quasi-character is of the form $\chi = \mu|\cdot|^s$, where μ and s are uniquely defined,with μ either trivial or the sign homomorphism $\mathrm{sgn} : x \mapsto x/|x|$. We then set:

$$L(\chi) := \begin{cases} \Gamma_\mathbb{R}(s) := \pi^{-s/2}\Gamma(s/2), & \mu = 1, \\ \Gamma_\mathbb{R}(s+1), & \mu = \mathrm{sgn}. \end{cases} \tag{2.1-3}$$

DEFINITION 2.1.3. The functions described in Eqns 2.1-3, 2.1-2 and 2.1-1 are the **local L-factors** of their corresponding local fields. We also introduced the following convenient notation:

$$L(s,\chi) := L(\chi|\cdot|^s).$$

Moreover we define the **shifted dual** (or sometimes **twisted dual**) of χ by:

$$\chi^\vee := \chi^{-1}|\cdot|,$$

so that
$$L\left((\chi\,|\cdot|^{s})^{\vee}\right) = L\left(1-s,\chi^{-1}\right).$$

In Tate's original thesis (which only treated number fields) he normalizes his Haar measurs on the local field to be self-dual relative toa choice of additive character ψ, from which he obtained the nice classical Fourier identity $f(x) = \widehat{\widehat{f}}(x)$. Here instead we explore the functional equation's dependency on choice of additive character and Haar measure.

Given $\mathrm{d}x$ a Haar measure on F, define
$$\mathrm{d}^{\times}x := c \cdot \frac{\mathrm{d}x}{|x|} \tag{2.1-4}$$
for some $c \in \mathbb{R}_{>0}$, which we always set to 1 in the Archimedean case. Then $\mathrm{d}^{\times}x$ is a Haar measure on $F^{\times}$. One may ask why we don't always set $c = 1$, and the reason for allowing ourselves this flexibility will become clear when we get to the computations of the global zeta functions in 3.2.

Fixing a nontrivial additive character $\psi \in \widehat{F}$, for any $f \in \mathscr{S}(F)$ we can define its Fourier transform by:
$$\widehat{f}(y) := \int_F f(x)\psi(xy)\,\mathrm{d}x\,. \tag{2.1-5}$$
Note this almost the same formulation as in A.3.1, though we drop the conjugation in the second factor of the integrand for notational convenience. The function $\widehat{f}$ is well-defined and in-fact again lies in $\mathscr{S}(F)$, though it depends on our choice of ψ and $\mathrm{d}x$.

Definition 2.1.4. Given $f \in \mathscr{S}(F)$ and $\chi \in \mathscr{X}(F^{\times})$, we define the associated **local zeta function** to be:
$$Z(f,\chi) := \int_{F^{\times}} f(x)\chi(x)\,\mathrm{d}^{\times}x\,.$$

The convenience of defining the space $\mathscr{X}(F^{\times})$ in 2.1.2 is hopefully clear with the L-factors and zeta functions defined, as it allows us to write zeta functions, a priori taking quasicharacters χ as inputs, as functions on $\mathbb{C}$, which is frequently used in literature for ease of notation. The following is the main result of this subsection.

Theorem 2.1.5 (The Local Functional Equation). Let F be a local field, and take $f \in \mathscr{S}(F)$ and $\chi = \mu\,|\cdot|^{s} \in \mathscr{X}(F^{\times})$ with exponent $\sigma = \mathfrak{Re}(s)$. Then:

(a) If $\sigma > 0$ then the local zeta function $Z(f,\chi)$ is absolutely convergent.

(b) If $\sigma \in (0,1)$ then the following functional equation holds:
$$Z(\widehat{f},\chi^{\vee}) = \gamma(\chi,\psi,\mathrm{d}x)Z(f,\chi)$$
for some γ independent of f, a priori defined on the strip $\sigma \in (0,1)$, and over all of $\mathscr{X}(F^{\times})$ via analytic continuation.

(c) There exists $\varepsilon(\chi,\psi,\mathrm{d}x) \in \mathbb{C}^{\times}$ satisfying:
$$\gamma(\chi,\psi,\mathrm{d}x) = \varepsilon(\chi,\psi,\mathrm{d}x)\frac{L(\chi^{\vee})}{L(\chi)}$$
for all $\chi \in \mathscr{X}(F^{\times})$.

Observation. Note that part (a) tells us that $Z(f,\chi)$ is absolutely convergent to the right of 0 on the complex plane, from which it follows by construction that $Z(\widehat{f},\chi^\vee)$ is absolutely convergent to the left of 1. Using this domain overlap along with part (b), we see that this theorem automatically yields a meromorphic continuation of $Z(,f,\chi)$ to all of $\mathscr{X}(F^\times)$.

proof of (a) in 2.1.5. To show $Z(f,\chi)$ is absolutely convergent when $\sigma > 0$, using our relationship between our Haar measures in Equation 2.1-4 and the fact that μ is unitary, it suffices to show that:

$$I(f,\sigma) = c\int_{F\setminus\{0\}} |f(x)|\,|x|^{\sigma-1}\,\mathrm{d}x \;<\infty.$$

In the Archimedean case, finiteness of $I(f,\sigma)$ follows from the fact that f is a Schwartz function, and thus rapidly decaying as $|x|$ approaches infinity, and that $|x|^{\sigma-1}$ is integrable near 0[1], so we are happy.

In the non-Archimedean case, f is locally constant with compact support, so it must factor through some finite group quotient of the form $\mathfrak{p}^n/\mathfrak{p}^m$, for some integers $m \leq n$ (here $\mathfrak{p}^k = \langle \pi^k \rangle$ represents the fractional ideal of $\mathcal{O}_F$ generated by the k-th power of its uniformizer). By linearity and translation-invariance of the Haar measure, it suffices to check finiteness of the integral for functions f that are just the characteristic functions of the various ideals $\mathfrak{p}^j$.

Using unique factorizaition in any fractional ideal, we can decompose it into a disjoint union:

$$\mathfrak{p}^j\setminus\{0\} = \bigsqcup_{k=j}^{\infty} \pi^k\mathcal{O}_F^\times.$$

Thus our integral becomes:

$$\begin{aligned} I(f,\sigma) &= c\int_{F\setminus\{0\}} |f(x)|\,|x|^{\sigma-1}\,\mathrm{d}x \\ &= \int_{F^\times} |f(x)|\,|x|^{\sigma}\,\mathrm{d}^\times x \\ &= \sum_{k\geq j}\int_{\pi^k\mathcal{O}_F^\times} |x|^{\sigma}\,\mathrm{d}^\times x \\ &= \sum_{k\geq j}\mathrm{Vol}(\mathcal{O}_F^\times,\mathrm{d}^\times x)\,q^{-k\sigma} \\ &= \mathrm{Vol}(\mathcal{O}_F^\times,\mathrm{d}^\times x)\frac{q^{-j\sigma}}{1-q^{-\sigma}} \end{aligned}$$

which is finite when $\sigma > 0$, as desired. □

To prove the next part of our main theorem, we use a lemma that is key in Tate's thesis.

Lemma 2.1.6. For $g \in \mathscr{S}(F)$ and any quasi-character χ with exponent $\sigma \in (0,1)$, we have:

$$Z(f,\chi)Z(\widehat{g},\chi^\vee) = Z(\widehat{f},\chi^\vee)Z(g,\chi).$$

Proof. Expanded out, the left side of the Lemma is the following product of integrals:

$$Z(f,\chi)Z(\widehat{g},\chi^\vee) = \int_{F^\times} f(x)\chi(x)\,\mathrm{d}^\times x\cdot\int_{F^\times}\widehat{g}(y)\chi^{-1}(y)\,|y|\,\mathrm{d}^\times y\,,$$

[1]For $F = \mathbb{C}$, we should be a bit careful here since the Haar measure and absolute value here are not the usual ones, but we note that we have $\frac{|\mathrm{d}z\wedge \mathrm{d}\bar{z}|}{|z|^{2(\sigma-1)}} = \frac{|\mathrm{d}z|}{|z|^{\sigma-1}}\times\frac{\mathrm{d}\bar{z}}{|z|^{\sigma-1}}$, so this integral remains well-defined.

both of which are absolutely convergent by part (a) of 2.1.5. We can turn this into a double integral over the product space by:

$$\iint_{F^\times \times F^\times} f(x)\widehat{g}(y)\chi(xy^{-1})\,|y|\;\mathrm{d}^\times x\;\mathrm{d}^\times y\,.$$

Using invariance of the product measure under the "shearing" automorphism $(x,y) \mapsto (x,xy)$, we obtain:

$$\iint_{F^\times \times F^\times} f(x)\widehat{g}(xy)\chi(y^{-1})\,|xy|\;\mathrm{d}^\times x\;\mathrm{d}^\times y\,,$$

which, by Fubini's theorem, is equal to:

$$\int_{F^\times}\left(\int_{F^\times} f(x)\widehat{g}(xy)\,|x|\;\mathrm{d}^\times x\right)\chi(y^{-1})\,|y|\;\mathrm{d}^\times y\,.$$

To prove the lemma it remains to show that the inner integral is symmetric in f and g. Expanding out the Fourier transform of g (recalling our convention in Equation 2.1-5 and transferring it to an integral over F using Equation 2.1-4), we get that the inner integral is equal to:

$$c\iint_{F\times F} f(x)g(z)\psi(xyz)\,\mathrm{d}z\;\mathrm{d}x\,.$$

Note we replaced $F^\times$ with F, which is of no issue since $\{0\} \subseteq F$ is a set of measure zero. We switch the order of integration via Fubini to obtain:

$$c\iint_{F\times F} f(x)g(z)\psi(xyz)\,\mathrm{d}x\;\mathrm{d}z\; = c\int_F g(z)\widehat{f}(zy)\,\mathrm{d}z\; = \int_{F^\times} g(z)\widehat{f}(zy)\,|z|\;\mathrm{d}^\times z$$

which gives us the desired result. □

Proof of (b) in 2.1.5. First suppose $Z(f,\chi) \neq 0$. Then our functional equation in Lemma 2.1.6 tells us that:

$$\gamma(\chi,\psi,\mathrm{d}x\,) := \frac{Z(\widehat{f},\chi^\vee)}{Z(f,\chi)}$$

is independent of our choice of $f \in \mathscr{S}(F)$. Note that:

$$\chi^\vee = \left(\mu\,|\cdot|^s\right)^\vee = \mu^{-1}\,|\cdot|^{1-s}$$

By (the already proved!) part (a) of Theorem 2.1.5, we know $Z(\widehat{f},\chi^\vee)$ is absolutely convergent when $\sigma = \mathfrak{Re}\,(s) < 1$, and we have already seen that the denominator of γ is absolutely convergent for all positive σ, so the functional equation in terms of γ holds for all $\sigma \in (0,1)$ as desired.
If $Z(f,\chi) = 0$, we want to show that $Z(\widehat{f},\chi^\vee) = 0$ as well, which will follow from Lemma 2.1.6 so long as we exhibit, for any χ, the existence of a $g \in \mathscr{S}(F)$ so that $Z(g,\chi)$ is not identically zero. Morover, we also want to show γ is meromorphic as a function of all $\chi \in \mathscr{X}(F^\times)$. For both these we turn to the proof of part (c) of the theorem, where we will compute explicit expressions for the zeta function and for the γ-factor which will directly imply these results. □

proof of (c) in 2.1.5 in the Archimedean case. As shown in the proof of part (b) of the theorem, the value of the γ function is independent of our choice of $f \in S(F)$. Thus we may choose a special function, or family of functions, for each of the possible cases of F to make our computations convenient. We will select a standard measure $\mathrm{d}x$ for each case, which is self-dual for an appropriate choice of ψ, and then analyze dependency on arbitrary $\mathrm{d}x$ and ψ afterward, in §2.2.

Case 1: $F = \mathbb{R}$

We take $\mathrm{d}x$ to be the usual Lebesgue measure, and set our additive character to be:

$$\psi(x) = e^{-2\pi i x}.$$

As we noted in our discussion of the local L-factors, any $\chi \in X(\mathbb{R}^\times)$ is of the form $|\cdot|^s$ or $\operatorname{sgn}|\cdot|^s$.

Sub-Case (i): $\chi = |\cdot|^s$

Take $f(x) = e^{-\pi x^2}$, which is visibly in $S(\mathbb{R})$. Then recalling that we set the scaling constant in the measure transformation of Equation 2.1-4 to 1, we get:

$$Z(f,\chi) = \int_{\mathbb{R}^\times} e^{-\pi x^2} |x|^s \, \mathrm{d}^\times x = 2\int_0^\infty e^{-\pi x^2} x^{s-1} \, \mathrm{d}x .$$

Using a substitution $u = \pi x^2$ (hence $\mathrm{d}u = 2\pi x \, \mathrm{d}x$), the integral becomes:

$$\pi^{-s/2} \int_0^\infty e^{-u} u^{s/2-1} \, \mathrm{d}u = \pi^{-s/2}\Gamma(s/2).$$

and we see from Equation 2.1-3 that this is exactly our local L-factor, so in fact $Z(f,\chi) = L(\chi)$ for all characters of this form. Classical Fourier analysis also tells us f is self-dual via the Fourier transform, i.e.

$$\widehat{f}(y) = \int_{\mathbb{R}} e^{-\pi x^2} e^{-2\pi i x y} \, \mathrm{d}x = f(x).$$

Thus, we have:

$$Z(\widehat{f},\chi^\vee) = \int_{\mathbb{R}^\times} \widehat{f}(x)\chi^\vee(x) \, \mathrm{d}^\times x = \int_{\mathbb{R}^\times} f(x)\chi^\vee(x) \, \mathrm{d}^\times x = Z(f,\chi^\vee)$$

which is equal to $L(\chi^\vee)$ by our previous calculation in this proof. Thus if $\chi = |\cdot|^s$, we have:

$$\gamma(\chi,\psi,\mathrm{d}x) = \frac{L(\chi^\vee)}{L(\chi)}$$
$$\implies \varepsilon(\chi,\psi,\mathrm{d}x) = 1. \tag{2.1-6}$$

Sub-Case (ii): $\chi = \operatorname{sgn}|\cdot|^s$

In this case we choose $f(x) = xe^{-\pi x^2}$, which is again immediately verifiable as rapidly decaying. Expanding the local zeta function we get:

$$\begin{aligned} Z(f,\chi) &= \int_{\mathbb{R}^\times} xe^{-\pi x^2} \frac{x}{|x|} |x|^s \, \mathrm{d}^\times x \\ &= \int_{\mathbb{R}^\times} e^{-\pi x^2} |x|^{s+1} \, \mathrm{d}^\times x \\ &= \pi^{-(s+1)/2}\Gamma\left(\frac{s+1}{2}\right). \\ &= \Gamma_{\mathbb{R}}(s+1) \end{aligned}$$

which we again recall from Equation 2.1-3 as by definition our real local L-factor $L(\chi)$.
Classical Fourier duality on the Schwartz space tells us that multiplication by i corresponds to multiplication by x, so we get the identity

$$\widehat{f}(y) = iye^{-\pi y^2} = if(y),$$

from which it follows

$$Z(\widehat{f},\chi^\vee) = \int_{\mathbb{R}^\times} if(y)\chi^\vee \, \mathrm{d}^\times x = iL(\chi^\vee).$$

Thus, by a similar calculation to the last sub-case we deduce:

$$\varepsilon(\chi, \psi, \mathrm{d}x\ ) = i. \tag{2.1-7}$$

Case 2: $F = \mathbb{C}$

Here we let our measure on $\mathbb{C}$ be $|\,\mathrm{d}z\ \wedge\ \mathrm{d}\bar{z}\,| = 2\,|\,\mathrm{d}x\ \wedge\ \mathrm{d}y\,|$, double the standard Lebesgue measure. In polar coordinates this is $2r\,|\,\mathrm{d}r\ \wedge\ \mathrm{d}\theta\,|$, which also implies the Haar measure on $F^{\times}$ is $\mathrm{d}^{\times}z\ = (2/r)\,\mathrm{d}r\ \mathrm{d}\theta$ (we divide by r^2 since our normalized absolute value on $\mathbb{C}$ is the square of the usual one). We choose this as it is self-dual with respect to the additive quasi-character

$$\psi(z) := e^{-2\pi i(z+\bar{z})}.$$

We also adjust the standard norm on $\mathbb{C}$ to agree with the module, i.e

$$|\,z\,| := z\bar{z}.$$

Since $\mathbb{C}^{\times} \cong \mathbb{R}_{>0} \times S^1$, any multiplicative quasi-character must be of the form:

$$\chi_{s,n} := z \mapsto |\,z\,|^s \left(\frac{z}{|\,z\,|}\right)^n$$

for $s \in \mathbb{C}$ and $n \in \mathbb{Z}$ uniquely defined. We choose our class of functions to be:

$$f_n(z) := \begin{cases} \frac{1}{2\pi}\,(\bar{z})^n\, e^{-2\pi z\bar{z}}, & n \geq 0, \\ \frac{1}{2\pi} z^{-n} e^{-2\pi z\bar{z}}, & n < 0. \end{cases}$$

We note that f is visibly a Schwartz function for any n. Computing the local zeta function for non-negative n, we get:

$$\begin{aligned}
Z(f_n, \chi_{s,n}) &= \int_{\mathbb{C}^{\times}} f_n(z)\chi_{s,n}(z)\,\mathrm{d}^{\times}z \\
&= \frac{1}{2\pi}\int_{\mathbb{C}^{\times}} (\bar{z})^n\, e^{-2\pi z\bar{z}}(z\bar{z})^s e^{in\arg(z)}\,\mathrm{d}^{\times}z \\
&= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\infty} (re^{-i\theta})^n e^{-2\pi r^2}(r^2)^s e^{in\theta}(2/r)\,\mathrm{d}r\ \mathrm{d}\theta \\
&= \frac{1}{\pi}\int_0^{2\pi}\int_0^{\infty} r^{2s+n-1}e^{-2\pi r^2}\,\mathrm{d}r\ \mathrm{d}\theta \\
&= (2\pi)^{-\left(s+\frac{n}{2}\right)}\int_0^{\infty} e^{-2\pi r^2}(2\pi r)^{s+\frac{n}{2}-1}4\pi r\,\mathrm{d}r \\
&= (2\pi)^{-\left(s+\frac{n}{2}\right)}\int_0^{\infty} e^{-t}t^{s+\frac{n}{2}-1}\,\mathrm{d}t \\
&= (2\pi)^{-\left(s+\frac{n}{2}\right)}\Gamma(s+\frac{n}{2}) = L(\chi_{s,n})
\end{aligned}$$

where we recall our definition of the local L-factor in 2.1-2. A similar calculation also shows that $Z(f_n, \chi_{s,n}) = L(\chi_{s,n})$ when $n < 0$.

It remains to compute the "dual" local zeta function. We do this via a mixture of handwaving and clever observations. First, classical Fourier duality tells us that we have

$$\widehat{f_n}(z) = \frac{1}{2\pi}i^{|\,n\,|}f_{-n}(z)$$

for all n. Next, we notice that $\chi_{n,s}^{\vee} = \chi_{1-s,-n}$ for any s and n. Using the fact that our computation was linear in

the exponents s and n, we deduce that
$$Z(\widehat{f}_n, \chi_{s,n}^\vee) = i^{|n|} L(\chi_{s,n}^\vee).$$

As a consequence we get that:
$$\begin{aligned} \gamma(\chi_{s,n}, \psi, \mathrm{d}x) &= i^{|n|} \frac{L(\chi_{s,n}^\vee)}{\chi_{s,n}} \\ \implies \varepsilon(\chi_{s,n}, \psi, \mathrm{d}x) &= i^{|n|}, \end{aligned} \tag{2.1-8}$$

completing the complex case.

To conclude our primary analysis on the Archimedean local fields, we note by looking at all the (nowhere-vanishing!) local epsilon factors computed in 2.1-6, 2.1-7 and 2.1-8, and their corresponding local L-factors that $\gamma(\chi, \psi, \mathrm{d}x)$ is indeed meromorphic as a function of χ, and that the ϵ-factor has the desired properties. Moreover, we exhibited, for each $\chi \in \mathscr{X}(F^\times)$, a function $f \in \mathscr{S}(F)$ for which $Z(f, \chi)$ was visibly not identically zero, which was also required for the last part of the proof of part (b) of Theorem 2.1.5. □

Before we proceed complete the proof of the final part of the theorem for the non-Archimedean case, we need to build some more tools and lemmas to use.

Definition 2.1.7. Given a topological ring R with additive character $\psi : R \to S^1$, multiplicative character $\chi : R^\times \to S^1$, and Haar measure $\mathrm{d}^\times u$ on the unit group, we define the associated **Gauss sum** to be:
$$g(\chi, \psi) := \int_{R^\times} \chi(u)\psi(u)\, \mathrm{d}^\times u\ .$$

Remark. Gauss sums turn out to be a hugely convenient notation that show up in a host of different applications, from theta-functions to cyclotomic fields. In fact, Weil made use of these Gauss-sums as a key step in his diagonal variety point-counting argument to motivate the Weil Conjectures in his original 1936 paper.

We will use this language and notation of conductors and Gauss sums to simplify the integrands of the local zeta functions via following key lemma.

Lemma 2.1.8. Let χ be a multiplicative character on $\mathcal{O}_F^\times$ and ψ be an additive character, and suppose they have conductors $\mathfrak{p}^n$ and $\mathfrak{p}^m$ respectively. Then:

(a) If $m < n$ then $g(\chi, \psi) = 0$.

(b) If $m = n$ then $|g(\chi, \psi)|^2 = c\,\mathrm{Vol}(\mathcal{O}_F, \mathrm{d}x)\,\mathrm{Vol}(1 + \mathfrak{p}^n, \mathrm{d}^\times x)$.

(c) If $m > n$ then $|g(\chi, \psi)|^2 = c\,\mathrm{Vol}(\mathcal{O}_F, \mathrm{d}x)\,\big(\mathrm{Vol}(1 + \mathfrak{p}^n, \mathrm{d}^\times x) - \mathrm{Vol}(1 + \mathfrak{p}^{m-1}, \mathrm{d}^\times x)\big)$.

Proof. Case 1: $m < n$

We decompose $\mathcal{O}_F^\times$ into the disjoint union of (finitely many!) cosets of the unit subgroups $U_m := 1 + \mathfrak{p}^m$. Note for any $a, b \in \mathcal{O}_F$ we have
$$\psi(a(1 + \pi^m b)) = \psi(a)\psi(a\pi^m b) = \psi(a),$$

by definition of the conductor $\mathfrak{p}^m = \langle \pi^m \rangle$. Since $n \geq m$, we note χ is constant on cosets of U_m in $\mathcal{O}_F^\times$, the Gauss sum thus becomes:

$$g(\chi, \psi) = \sum_{\mathcal{O}_F^\times / U_m} \left(\psi(a)\chi(a) \int_{U_m} \chi(u)\, \mathrm{d}^\times u \right).$$

However since $m < n$, the subgroup U_m is a strict superset of U_n, and so, by definition of the conductor, the character χ is non-trivial on U_m. Recall the integral of a nontrivial character over a compact subgroup is zero, which in turn makes the entire Gauss sum zero, as desired. This proves (i).

Case 2: $m \geq n$

Expanding out the expression and making use of Fubini, we get:

$$\begin{aligned}
|g(\chi,\psi)|^2 &= g(\chi,\psi)\overline{g(\chi,\psi)} \\
&= \int_{\mathcal{O}_F^\times} \chi(x)\psi(x)\, \mathrm{d}^\times x \cdot \overline{\int_{\mathcal{O}_F^\times} \chi(y)\psi(y)\, \mathrm{d}^\times y} \\
&= \iint_{\mathcal{O}_F^\times} \chi(xy^{-1})\psi(x-y)\, \mathrm{d}^\times x \ \mathrm{d}^\times y\,.
\end{aligned}$$

Making the substitution $xy^{-1} \mapsto z$ the above integral becomes

$$\int_{\mathcal{O}_F^\times} \chi(z)h(z)\, \mathrm{d}^\times z\,,$$

where

$$h(z) := \int_{\mathcal{O}_F^\times} \psi(yz - y)\, \mathrm{d}^\times y \ = c\int_{\mathcal{O}_F^\times} \psi(y(z-1))\, \mathrm{d}y\,.$$

Note we can switch the Haar measure in the second equality using Equation 2.1-4 and the fact that $|y| = 1$ for all $y \in \mathcal{O}_F^\times$. Making use of the fact that $\mathcal{O}_F$ is a local ring, we get:

$$h(z) = c\left(\int_{\mathcal{O}_F} \psi(y(z-1))\, \mathrm{d}y \ - \int_{\mathfrak{p}} \psi(y(z-1))\, \mathrm{d}y \right).$$

Note that if $\nu_{\mathfrak{p}}(z-1) < m-1$ then in the above integrands $y(z-1)$ takes values outside of the conductor of ψ, and thus the integrands are non-trivial characters. If $\nu_{\mathfrak{p}}(z-1) = m-1$ then the integrand is trivial over $\mathfrak{p}$, but not $\mathcal{O}_F$, and otherwise it is trivial over all of $\mathcal{O}_F$. These observations give us:

$$h(z) = \begin{cases} 0, & \nu_{\mathfrak{p}}(z-1) < m-1 \\ -c\dfrac{\mathrm{Vol}(\mathcal{O}_F, \mathrm{d}x\,)}{q}, & \nu_{\mathfrak{p}}(z-1) = m-1 \\ c\,\mathrm{Vol}(\mathcal{O}_F, \,\mathrm{d}x\,)\left(1 - \frac{1}{q}\right), & \nu_{\mathfrak{p}}(z-1) \geq m. \end{cases}$$

All that remains is to substitute these cases back into the integral $\int \chi(z)h(z)\, \mathrm{d}^\times z$ to show the desired equalities in part (ii) and part (iii), which follows from direct computation, as the reader may choose to verify. □

Now we have the necessary tools and results to prove the final part of Theorem 2.1.5 in the non-Archimedean case. We first make a quick observation to classify the multiplicative characters we are working with. Recall from our discussion in the prelude to Definition 2.1.2 that any such character of $F^\times$ is of the form:

$$\chi_{s,n} := \left(x \mapsto |x|^s\, \omega(\frac{x}{|x|}) \right),$$

where $x \mapsto x/|x|$ is homomorphic retraction to $\mathcal{O}_F^\times \hookrightarrow F^\times$ and ω is a unitary character with conductor $\mathfrak{p}^n$. Note the indices s, n do not completely determine the character $\chi_{s,n}$, as there is some choice of ω to be made, but this notation will end up sufficing for the computations ahead. Note also that in the unramified case (i.e $n = 0$), the unitary character ω is trivial. This fact will be used in the proof.

proof of (c) in 2.1.5 in the non-Archimedean case. Supposing our additive character ψ has conductor $\mathfrak{p}^m$, we define:

$$f(x) := \begin{cases} \psi(x), & x \in \mathfrak{p}^{m-n} \\ 0, & \text{otherwise.} \end{cases} \tag{2.1-9}$$

This is a Schwartz-Bruhat function as it is compactly-supported by definition, and also locally constant, since $\psi = \psi(0) = 1$ near 0, as $\mathbb{C}^\times$ has "no small subgroups". We now compute the local zeta functions $Z(f, \chi_{s,n})$ in two cases: n zero, and n positive.

Case 1: $n = 0$

This means that ω is trivial, i.e $\chi_{s,0}$ is an unramified character. We recall ψ is trivial on $\mathfrak{p}^m$, and therefore f is zero outside of the conductor of ψ. Using similar methods as in the proof of 2.1.8, we compute:

$$\begin{aligned} Z(f, \chi_{s,0}) &= \int_{F^\times} f(x)\chi_{s,0}(x)\, \mathrm{d}^\times x \\ &= \int_{\mathfrak{p}^m \setminus \{0\}} \psi(x)\, |x|^s\, \omega(x/\,|x|)\, \mathrm{d}^\times x \\ &= \int_{\mathfrak{p}^m \setminus \{0\}} |x|^s\, \mathrm{d}^\times x \\ &= \sum_{k \geq m} \left(\int_{\pi^k \mathcal{O}_F^\times} |x|^s\, \mathrm{d}^\times x \right) \\ &= \mathrm{Vol}(\mathcal{O}_F^\times, \mathrm{d}^\times x\,) \sum_{k \geq m} q^{-ks} \\ &= q^{-ms}\, \mathrm{Vol}(\mathcal{O}_F^\times, \mathrm{d}^\times x\,) L(\chi_{s,0}), \end{aligned} \tag{2.1-10}$$

where we recall our definition of the local L-factor in the unramified case in Equation 2.1-1.

Case 2: $n > 0$

Recalling that f is zero outside of $\mathfrak{p}^{m-n}$, by similar logic as in the $n = 0$ case we get:

$$\begin{aligned} Z(f, \chi_{s,n}) &= \sum_{k \geq m-n} \left(\int_{\pi^k \mathcal{O}_F^\times} \psi(x)\, |x|^s\, \omega(x/\,|x|)\, \mathrm{d}^\times x \right) \\ &= \sum_{k \geq m-n} \left(\int_{\mathcal{O}_F^\times} \psi(\pi^k u) \left|\pi^k u\right|^s \omega(u)/\omega(\pi^k)\, \mathrm{d}^\times u \right) \\ &= \sum_{k \geq m-n} q^{-ks} \left(\int_{\mathcal{O}_F^\times} \psi(\pi^k u)\omega(u)\, \mathrm{d}^\times u \right) \\ &= \sum_{k \geq m-n} q^{-ks} g(\omega, \psi_{\pi^k}). \end{aligned}$$

In the last line the Gauss sum notation comes as a typographical godsave for us, where $\psi_{\pi^k}(x) := \psi(x\pi^k)$. We note that the conductor of ψ_{π^k} is $m - k$, so in particular if $k > m - n$ then it contains the conductor of ω. We thus get:

$$Z(f,\chi_{s,n}) = q^{(n-m)s} g(\omega,\psi_{\pi^{m-n}}) = q^{(n-m)s} c \operatorname{Vol}(\mathcal{O}_F, \mathrm{d}x\,) \operatorname{Vol}(1+\mathfrak{p}^n, \mathrm{d}^\times x\,)(\overline{g(\omega,\psi_{\pi^{m-n}})})^{-1}, \tag{2.1-11}$$

where the first equality follows from part (a) of Lemma 2.1.8 and the second from part(b) of the same lemma. Thus the zeta function, as a function of $s \in \mathbb{C}$, is a visibly holomorphic function with no zeroes or poles.

Now we need to compute the zeta function of the dual, where here we define the Fourier transform also with respect to our additive character ψ of conductor $\mathfrak{p}^m$, i.e.

$$\widehat{f}(y) := \int_F f(x)\psi(xy)\,\mathrm{d}x \; = \int_{\mathfrak{p}^{m-n}} \psi(x(y+1))\,\mathrm{d}x\,. \tag{2.1-12}$$

We proceed with similar casework to before.

Case 1: $n = 0$

Since the conductor of ψ is $\mathfrak{p}^m$, we observe from Equation 2.1-12 that $\widehat{f}(y)$ is the integral of $\psi(x(y+1))$ taken over $\mathfrak{p}^m$. The character is non-trivial on $\mathfrak{p}^m$, and thus $\widehat{f}(y)$ is zero, exactly when $y \notin \mathcal{O}_F$. Thus in this case we have:

$$\widehat{f}(y) = \begin{cases} \operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,), & y \in \mathcal{O}_F \\ 0, & \text{otherwise,} \end{cases}$$

i.e we have a scaled characteristic function of the ring of integers. Using this, we compute:

$$\begin{aligned} Z(\widehat{f},\chi_{s,0}^\vee) &= \int_{\mathcal{O}_F\setminus\{0\}} \widehat{f}(x)\chi_{s,0}^\vee(x)\,\mathrm{d}^\times x \\ &= \operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,)\sum_{k\geq 0}\left(\int_{\mathcal{O}_F^\times} \left|\pi^k x\right|^{1-s}\,\mathrm{d}^\times x\right) \\ &= \operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,)\operatorname{Vol}(\mathcal{O}_F^\times, \mathrm{d}^\times x\,)\frac{1}{1-q^{(1-s)}} \\ &= \operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,)\operatorname{Vol}(\mathcal{O}_F^\times, \mathrm{d}^\times x\,)L(\chi_{s,0}^\vee). \end{aligned} \tag{2.1-13}$$

Combing 2.1-10 and 2.1-13, we get that for $\chi = \chi_{s,0}$ we have:

$$\begin{aligned} \gamma(\chi,\psi,\mathrm{d}x\,) &= q^{ms}\operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,)\frac{L(\chi^\vee)}{L(\chi)} \\ \implies \varepsilon(\chi,\psi,\mathrm{d}x\,) &= q^{ms}\operatorname{Vol}(\mathfrak{p}^m, \mathrm{d}x\,). \end{aligned} \tag{2.1-14}$$

Case 2: $n > 0$

Examining 2.1-5 again, we see that the integrand $\psi(x(y+1))$ is trivial for all $x \in \mathfrak{p}^{m-n}$ exactly when $y+1 \in \mathfrak{p}^n$, and otherwise it is a nontrivial character of x and thus $\widehat{f}(y) = 0$. We deduce then that $\widehat{f}$ is the characteristic function of $\mathfrak{p}^n - 1$ scaled by the factor $\operatorname{Vol}(\mathfrak{p}^{m-n}, \mathrm{d}x\,)$. Noting that $\mathfrak{p}^n - 1 \subseteq \mathcal{O}_F^\times$ (since $n > 0$), we compute the dual local zeta function:

$$\begin{aligned} Z(\widehat{f},\chi_{s,n}^\vee) &= \int_{\mathfrak{p}^n-1} \widehat{f}(x)\chi_{s,n}^\vee(x)\,\mathrm{d}^\times x \\ &= \operatorname{Vol}(\mathfrak{p}^{m-n}, \mathrm{d}x\,)\int_{\mathfrak{p}^n-1} |x|^{1-s}\,\overline{\omega}(x)\,\mathrm{d}^\times x \\ &= q^{n-m}\operatorname{Vol}(\mathcal{O}_F, \mathrm{d}x\,)\int_{1+\mathfrak{p}^n} \overline{\omega}(-1)\overline{\omega}(x)\,\mathrm{d}^\times x \\ &= q^{n-m}\operatorname{Vol}(\mathcal{O}_F, \mathrm{d}x\,)\operatorname{Vol}(1+\mathfrak{p}^n, \mathrm{d}^\times x\,)\omega(-1), \end{aligned} \tag{2.1-15}$$

where the last line follows from the fact that the conductor of ω is equal to the conductor of its conjugate, so it is trivial on all of $1+\mathfrak{p}^n$. Combining 2.1-11 and 2.1-15 for $\chi = \chi_{s,n}$, we get:

$$\varepsilon(\chi, \psi, \mathrm{d}x) = \gamma(\chi, \psi, \mathrm{d}x) = \frac{1}{c}q^{(m-n)(s-1)\omega(-1)}\overline{g(\omega, \psi_{\pi^{m-n}})} = \frac{1}{c}q^{(m-n)(s-1)}g(\overline{\omega}, \psi_{\pi^{m-n}}). \tag{2.1-16}$$

The final equality above made use of the following identity (the validity of which we leave the reader to verify) that, for any Gauss sum:

$$\overline{g(\chi, \psi)} = \chi(-1)g(\overline{\chi}, \psi).$$

We now conclude our analysis. The equality involving the epsilon factor and relevant local L-factors have been shown in 2.1-14 and 2.1-16, from all of which $\gamma(\chi, \psi, \mathrm{d}x)$ is visibly meromorphic in χ, completing our proof. □

Our work is not yet complete. We have implicitly shown the necessary conditions to allow the proofs of parts (b) and (c) to hold in our computations, in that we have demonstrated nontrivial zeta functions and holomorphicity of the ϵ-factors on the strip $\mathfrak{Re}(s) = (0,1)$ to allow γ to be meromorphically continued to all of $\mathscr{X}(F^\times)$. This will be discussed this more explicitly in the following subsection. There we also analyze dependence of the γ-factors on choice of additive character and Haar measure, thus allowing the results to hold outside of our specific choices made in our computations.

§2.2 The Epsilon and Gamma Factors

In the previous subsection we proved a functional equation of the zeta function on local fields, which involved the so-called gamma and epsilon factors, which were functions of our mulitplicative character χ, our fixed additive character ψ, and our choice of Haar measure $\mathrm{d}x$ on our local field. In here we analyze the dependency of these factors on all three inputs, and prove some results about these ϵ-factors in general, and their connection to the so-called root number.

For convenience of reference here we summarize all our computations of the previous subsection. First, for each local field F, for each multiplicative character $\chi \in \mathscr{X}(F^\times)$, we record the standard additive character, Haar measure, and Schwartz-Bruhat function f with respect to which we made our computations.

	Multiplicative character χ	Additive character ψ	Haar measure $\mathrm{d}x$	Standard function f
$F=\mathbb{R}$	$\chi = \lvert\cdot\rvert^s$	$e^{-2\pi i x}$	Lebesgue measure	$e^{-\pi x^2}$
	$\chi = \operatorname{sgn}\lvert\cdot\rvert^s$			$xe^{-\pi x^2}$
$F=\mathbb{C}$	$\chi_{s,n} : z \mapsto \lvert z\rvert^s \left(\frac{z}{\lvert z\rvert}\right)^n,\ n\in\mathbb{Z}$	$e^{-2\pi i(z+\overline{z})}$	$2\lvert\mathrm{d}z \wedge \mathrm{d}\overline{z}\rvert$	$\frac{1}{2\pi}(\overline{z})^n e^{-2\pi z\overline{z}},\ n\geq 0$
				$\frac{1}{2\pi}(z)^{-n} e^{-2\pi z\overline{z}},\ n< 0$
F non-Arch	$\chi_{s,n} : x \mapsto \lvert x\rvert^s \omega\left(\frac{x}{\lvert x\rvert}\right), n = \text{conductor}(\omega)\in\mathbb{Z}_{\geq 0}$	Variable of conductor $\mathfrak{p}^m$	Variable	$\mathbb{1}_{\mathfrak{p}^{m-n}}$

Table 2.2.A: Standard additive characters, measures and functions on each local component.

For each class of characters χ, our standard function $f \in \mathscr{S}(F)$ was constructed so that $Z(f, \chi)$ was a holomorphic function not identically zero. This makes the proof of part (b) valid even in the case where the zeta function is zero. We also record here the ϵ-factors corresponding each choice of χ, ψ and $\mathrm{d}x$, all of which are visibly holomorphic as functions of s. Using the fact that our local L-factors (defined in Eqns. 2.1-1, 2.1-3 and 2.1-2) are by construction holomorphic and non-vanishing on the strip $\mathfrak{Re}(s) \in (0,1)$, it follows that

$$\gamma(\chi, \psi, \mathrm{d}x) = \epsilon(\chi, \psi, \mathrm{d}x) \frac{L(\chi^\vee)}{L(\chi^\vee)}$$

is holomorphic on said strip, and thus admits a meromorphic continuation to all of $\mathscr{X}(F^\times)$.

	Multiplicative character χ	Zeta function $Z(f,\chi)$	$\epsilon(\chi, \psi, \mathrm{d}x)$
$F = \mathbb{R}$	$\chi = \lvert\cdot\rvert^s$	$\Gamma_{\mathbb{R}}(s)$	1
	$\chi = \mathrm{sgn}\lvert\cdot\rvert^s$	$\Gamma_{\mathbb{R}}(s+1)$	i
$F = \mathbb{C}$	$\chi_{s,n}$	$\Gamma_{\mathbb{C}}\left(s + \frac{\lvert n\rvert}{2}\right)$	$i^{\lvert n\rvert}$
F non-Arch	$\chi_{s,n},\ n = 0$	$\frac{q^{-ms}}{1-q^{-s}} \mathrm{Vol}(\mathcal{O}_F^\times, \mathrm{d}^\times x)$	$q^{ms}\,\mathrm{Vol}(\mathfrak{p}^m, \mathrm{d}x)$
	$\chi_{s,n}, n > 0$	$q^{(n-m)s} c\,\mathrm{Vol}(\mathcal{O}_F, \mathrm{d}x)\,\mathrm{Vol}(1+\mathfrak{p}^n, \mathrm{d}^\times x)\,\overline{g(\omega, \psi_{\pi^{m-n}})}^{-1}$	$\frac{1}{c} q^{(m-n)(s-1)} g(\overline{\omega}, \psi_{\pi^{m-n}})$

Table 2.2.B: The zeta functions and epsilon factors with inputs from Table 2.2.A.

We will now see how the γ-factor behaves as ψ and $\mathrm{d}x$ vary. In particular, we should hope it remains meromorphic as a function on $\mathscr{X}(F^\times)$ so that our computations in the previous subsection easily imply the result for arbitrary choices of ψ and $\mathrm{d}x$. Fortunately, this is indeed the case.

Proposition 2.2.1. Let F be a local field, χ a character of $F^\times$, ψ a character of F and $\mathrm{d}x$ a Haar measure. Then

(a) For every $t \in \mathbb{R}_{>0}$,

$$\gamma(\chi, \psi, t \cdot \mathrm{d}x) = t \cdot \gamma(\chi, \psi, \mathrm{d}x).$$

(b) For any $a \in F^\times$, if $\psi_a : x \mapsto \psi(ax)$ then

$$\gamma(\chi, \psi_a, \mathrm{d}x) = \chi(a)\,\lvert a\rvert_F^{-1}\, \gamma(\chi, \psi, \mathrm{d}x).$$

Proof. Since the Fourier transform of any $f \in \mathscr{S}(F)$ is dependent on choice of ψ, $\mathrm{d}x$, we will write $\widehat{f}(y, \psi, \mathrm{d}x)$ to denote this dependency (note if $(\psi, \mathrm{d}x)$ is not a self-dual pair then results like Fourier inversion need not hold, but this is no trouble for us presently). We thus have

$$\widehat{f}(y, \psi, t \cdot \mathrm{d}x) := \int_F f(x)\psi(xy) t\,\mathrm{d}x = t\widehat{f}(y, \psi, \mathrm{d}x).$$

We also note that $Z(f, \chi)$ has a dependence on $\mathrm{d}x$, if we define $\mathrm{d}^\times x := \mathrm{d}x / \lvert x\rvert_F$. In particular the scaling of $\mathrm{d}^\times x$ is precisely the amount we scale $\mathrm{d}x$ by. It is also clear from definition that $Z(f, \chi, t \cdot \mathrm{d}x) = tZ(f, \chi, \mathrm{d}x)$. Thus, we have

$$\frac{Z(\widehat{f}(y, \psi, \mathrm{d}x), \chi^\vee, t \cdot \mathrm{d}x)}{Z(f, \chi, t \cdot \mathrm{d}x)} = \frac{t^2}{t} \frac{Z(\widehat{f}(y, \psi, \mathrm{d}x), \chi)}{Z(f, \chi, \mathrm{d}x)}$$
$$\implies \gamma(\chi, \psi, t \cdot \mathrm{d}x) = t\gamma(\chi, \psi, \mathrm{d}x),$$

thus proving part (a).

Similarly, we have

$$\widehat{f}(y, \psi_a, \mathrm{d}x) := \int_F f(x)\psi(axy)\,\mathrm{d}x = \widehat{f}(ay, \psi, \mathrm{d}x).$$

Thus, we have

$$\begin{aligned} Z(\widehat{f}(x, \psi_a, \mathrm{d}x), \chi^\vee) &= \int_{F^\times} \widehat{f}(ax, \psi, \mathrm{d}x)\chi^\vee(x)\,\mathrm{d}^\times x \\ &= \int_{F^\times} \widehat{f}(x, \psi, \mathrm{d}x)\chi^\vee(a^{-1}x)\,\mathrm{d}^\times x \\ &= \chi^\vee(a^{-1}) Z(\widehat{f}(x, \psi, \mathrm{d}x), \chi^\vee) \\ &= \chi(a)\,|a|_F^{-1}\, Z(\widehat{f}(x, \psi, \mathrm{d}x), \chi^\vee), \end{aligned}$$

from which it easily follows that

$$\gamma(\chi, \psi_a, \mathrm{d}x) = \chi(a)\,|a|_F^{-1}\,\gamma(\chi, \psi, \mathrm{d}x),$$

proving part (b). □

In the case that the pair $(\psi, \mathrm{d}x)$ *is* self-dual, we also have nice results about the interaction of the γ-factors with the multiplicative character χ.

> **Proposition 2.2.2.** Let F be a local field, Suppose the pair $(\psi, \mathrm{d}x)$ is self-dual on F [a]. Then the gamma factors satisfy:
>
> (a) $\gamma(\chi, \psi, \mathrm{d}x)\gamma(\chi^\vee, \psi, \mathrm{d}x) = \chi(-1)$.
>
> (b) $\gamma(\overline{\chi}, \psi, \mathrm{d}x) = \chi(-1)\overline{\gamma(\chi, \psi, \mathrm{d}x)}$.
>
> [a] i.e the Fourier inversion formula with respect to the additive character ψ and measure $\mathrm{d}x$ holds on F.

Proof. Part (a): Since the Fourier inversion formula holds, we recall from classical Fourier theory that $\widehat{\widehat{f}}(x) = f(-x)$. Moreover, we make the observation that taking the shifted dual is an involution on the space of multiplicative characters, i.e. $(\chi^\vee)^\vee = \chi$. Putting these two facts together, we get:

$$Z(\widehat{\widehat{f}}, (\chi^\vee)^\vee) = \int_{F^\times} f(-x)\chi(x)\,\mathrm{d}^\times x = \chi(-1)\int_{F^\times} f(-x)\chi(-x)\,\mathrm{d}^\times x = \chi(-1)Z(f, \chi).$$

Next, recalling the main functional equation in Theorem 2.1.5 from the local theory and applying it twice, with the above equality, we get:

$$\begin{aligned} \chi(-1)Z(f, \chi) &= Z(\widehat{\widehat{f}}, (\chi^\vee)^\vee) \\ &= \gamma(\chi^\vee, \psi, \mathrm{d}x) Z(\widehat{f}, \chi^\vee) \\ &= \gamma(\chi^\vee, \psi, \mathrm{d}x)\gamma(\chi, \psi, \mathrm{d}x) Z(f, \chi). \end{aligned}$$

For any quasi-character χ we may look in Tables 2.2.A to find a function $f \in \mathscr{S}(F)$ such that Table 2.2.B will assure us that $Z(f, \chi)$ is nonvanishing. Thus, we may cancel the $Z(f, \chi)$ without worry to obtain the desired equality.

Part (b): First, using the fact that $\psi(xy) = \overline{\psi(-xy)}$, since ψ is additive, for any $f \in \mathscr{S}(F)$ we have:

$$\widehat{\overline{f}}(y) := \int_F \overline{f(x)}\psi(xy)\,\mathrm{d}x$$

$$= \int_F \overline{f(x)\psi(-xy)}\,\mathrm{d}x \ = \overline{\overline{f}}(-y).$$

This gives:

$$\begin{aligned} Z(\widehat{\widehat{\overline{f}}},\chi) &:= \int_{F^\times} \widehat{\widehat{\overline{f}}}(x)\chi(x)\,\mathrm{d}^\times x \\ &= \int_{F^\times} \overline{\overline{f}}(-x)\chi(x)\,\mathrm{d}^\times x \\ &= \chi(-1)Z(\overline{\overline{f}},\chi). \end{aligned}$$

Next we make the observation that complex conjugation commutes with taking the shifted dual, i.e. $\overline{\chi}^\vee = \overline{\chi^\vee}$. Applying this fact, along with the above deduced identity and the functional equation of Theorem 2.1.5, we get:

$$\begin{aligned} \overline{Z(\widehat{f},\chi^\vee)} = Z(\widehat{\overline{f}},\overline{\chi^\vee}) &= \chi(-1)Z(\widehat{\widehat{\overline{f}}},\overline{\chi}^\vee) \\ &= \chi(-1)\gamma(\overline{\chi},\psi,\,\mathrm{d}x\,)Z(\overline{f},\overline{\chi}). \end{aligned} \tag{2.2-1}$$

However, we also have:

$$\overline{Z(\widehat{f},\chi^\vee)} = \overline{\gamma(\chi,\psi,\,\mathrm{d}x\,)Z(f,\chi)} = \overline{\gamma(\chi,\psi,\,\mathrm{d}x\,)}Z(\overline{f},\overline{\chi}). \tag{2.2-2}$$

Combining Eqns. 2.2-1 and 2.2-2, and once again looking in our tables to pick f so that $Z(\overline{f},\overline{\chi})$ is nonvanishing, we get desired equality. □

Corollary 2.2.3. The statements of Propositions 2.2.1 and 2.2.2 hold true if we replace $\gamma(\chi,\psi,\,\mathrm{d}x\,)$ with $\epsilon(\chi,\psi,\,\mathrm{d}x\,)$.

Proof. This is immediate from the relationship between the epsilon factor and the gamma factor as defined in part (c) of Theorem 2.1.5, which we repeat below for convenience:

$$\epsilon(\chi,\psi,\,\mathrm{d}x\,) = \gamma(\chi,\psi,\,\mathrm{d}x\,)\frac{L(\chi)}{L(\chi^\vee)}.$$

Here we also implicitly use the observation (which the reader may choose to rigorously verify should they desire) that $\overline{L(\chi)} = L(\overline{\chi})$ for all the local L-factors and all inputs χ. □

The following definition was motivated by several classical computations in analytic number theory, in which L-functions have certain complex factors of norm 1 attached to them known as root numbers. Our local zeta functions give us a nice characterizations of this factor.

Definition 2.2.4. Let χ be a multiplicative character (i.e S^1-valued) of a local field F, and let $(\psi,\,\mathrm{d}x\,)$ be a self-dual pair of an additive character and Haar measure on F. Then define the **root number** $W(\chi)$ by

$$W(\chi) := \epsilon(\chi|\cdot|^{1/2},\psi,\,\mathrm{d}x\,).$$

Remark. At first glance this definition, in particular the choice of appending the factor of $|\cdot|^{1/2}$ to χ, may seem rather arbitrary. However, we will see that ensuring the input of the ϵ-factor is always a quasi-character of exponent 1/2 is crucial to make the root number well-defined.

> COROLLARY 2.2.5. For any multiplicative character χ of a local field, the root number $W(\chi)$ is well-defined independent of choice of ψ and $\mathrm{d}x$, and moreover $|W(\chi)| = 1$.

Proof. First we analyze dependence (or lack-thereof) on choice of additive character and Haar measure; let $(\psi, \mathrm{d}x)$ be a self-dual pair. By Theorem 1.2.9, any other additive character looks like $\psi_a = \psi(ax)$, for some $a \in F^\times$. One then knows (from classical Fourier theory in the Archimedean case and from examination of our proof of 1.2.9 in the non-Archimedean case) that $|a|^{1/2}\,\mathrm{d}x$ is the Haar measure that gives rise to a self-dual pair with ψ_a. Applying Proposition 2.2.1 (using Corollary 2.2.3 to use it for ϵ-factors), we get

$$\epsilon(\chi|\cdot|^{1/2}, \psi_a, |a|^{1/2}\,\mathrm{d}x) = |a|^{1/2}(\chi(a)|a|^{1/2})|a|^{-1}\epsilon(\chi|\cdot|^{1/2}, \psi, \mathrm{d}x) = \epsilon(\chi|\cdot|^{1/2}, \psi, \mathrm{d}x).$$

Thus $W(\chi)$ is well-defined.

For the second assertion, all this is really saying is that any quasi-character of exponent 1/2 has epsilon factor with norm 1. Note for an arbitrary quasi-character χ' we always have $\chi'(x)(\chi')^\vee(x) = |x|$. If we impose an exponent of 1/2 then we also have $\chi'(x)\overline{\chi'}(x) = |x|$. Combining these identities together and cancelling out $\chi'(x)$ we get $\overline{\chi'} = (\chi')^\vee$.
Using this fact and combining statements (a) and (b) of Corollary 2.2.3 (really Proposition 2.2.2), we obtain

$$\epsilon(\chi', \psi, \mathrm{d}x)^{-1} = \overline{\epsilon(\chi', \psi, \mathrm{d}x)},$$

and thus the epsilon factor of any exponent 1/2 quasicharacter has norm 1, proving our desired result. $\square$

Section 3

The Global Theory

§3.1 Adelic Fourier Theory (with some Riemann-Roch)

We first explore the theory of integration and Fourier transforms on general restricted products. Essentially we just want to confirm that all the results and identities one expects to hold actually do. Continuing with the notation of §1.1, we let $\{G_v\}$ be a collection of locally compact Hausdorff groups, cofinitely many of which are equipped with a respective compact subgroup H_v, a Haar measure $\mathrm{d}g_v$ with the property that $\mathrm{Vol}(H_v, \mathrm{d}g_v) = 1$, and let G denote their restricted product. We use $\mathrm{d}g$ to denote the induced Haar measure on the restricted product, as in 1.2.14.

Observation. For f an integrable function on G, we have the following identity:

$$\int_G f(g)\,\mathrm{d}g = \lim_S \int_{G_S} f(g_S)\,\mathrm{d}g_S\,,$$

where the limit is naturally taken over an increasing directed union of places. This is a direct consequence of the fact that we may compute and integral via a limit of any compact exhaustion, and the fact that any compact is contained in some G_S. Conversely, the dominated convergence theorem tells us that if this limit exists for any Borel function f, then it must be integrable.

Let S_0 be a finite set of indices containing J_∞ and the (finitely many!) finite v for which $\mathrm{Vol}(H_v, \mathrm{d}g_v) \neq 1$. Suppose for each v we have a continuous integrable f_v on G_v such that $f_v\,|_{H_v} = 1$ for all $v \notin S_0$. For $g = (g_v) \in G$, define

$$f(g) = \prod_v f_v(g_v)$$

Proposition 3.1.1. With the notation above, the function f is well-defined and continuous. For any $S \supseteq S_0$, we have

$$\int_{G_S} f(g_S)\,\mathrm{d}g_S = \prod_{v\in S} \int_{G_v} f_v(g_v)\,\mathrm{d}g_v$$

Proof. Since $g = (g_v)$ is in the restricted direct product, then $g_v \in H_v$ for all but finitely many v. Combining this with the fact that $f_v\,|_{H_v} = 1$ for almost all v, we deduce $\prod f_v(g_v)$ is a finite product, and thus well-defined.
To show f is continuous, consider a base for G consisting of sets of the form

$$\prod U_v \times \prod H_v$$

where the first product contains all the (finitely many!) finite v such that $f|_{H_v}$ is nontrivial. Thus f can be computed locally as a finite product of continuous functions, and is thus locally continuous, and thus continuous.
The fact that the integral behaves well with products is a consequence of definitions and Fubini's theorem. □

Corollary 3.1.2. With the notation as in the above result, if f is integrable then

$$\int_G f(g)\,\mathrm{d}g = \prod_v \int_{G_v} f_v(g_v)\,\mathrm{d}g_v$$

and $f \in L^1(G)$, when the limit exists for $|f|$.

Proof. By our earlier observation, we know f is integrable if and only if $\lim_S \int_{G_S} f(g_S)\,\mathrm{d}g_S$ exists, which is true if and only if the limit

$$\lim_S \prod_{v\in S} \int_{G_v} f_v(g_v)\,\mathrm{d}g_v = \prod_v \int_{G_v} f_v(g_v)\,\mathrm{d}g_v < \infty,$$

exists, by Proposition ??. □

Proposition 3.1.3. Suppose all the G_v are abelian, and suppose the continuous functions $\{f_v\}$ on G_v are the characteristic functions for H_v for all but finitely many v. Then f is integrable, with Fourier transform $\widehat{f}$ continuous and given by

$$\widehat{f}(\chi) = \prod_v \widehat{f_v}(\chi_v),$$

for $\chi = (\chi_v)_v \in \widehat{G} = \prod_v \widehat{G_v}$.

Proof. Firstly note that we have

$$\widehat{f}(\chi) = \int_G f(g)\chi(g)\,\mathrm{d}g = \prod_v \int_{G_v} f_v(g_v)\chi_v(g_v)\,\mathrm{d}g_v = \prod_{v\in S} \int_{G_v} f_v(g_v)\chi_v(g_v)\,\mathrm{d}g_v = \prod_{v\in S} \widehat{f_v}(\chi_v)$$

for some finite set S, since f_v is the characteristic function of H_v and $\chi_v|_{H_v} = 1$ for cofinitely many v. Thus we have a finite product of continuous functions of χ, as desired. □

From now on assume all groups are abelian. We want to build a Haar measure on the Pontryagin dual $\widehat{G}$ that is dual to $\mathrm{d}g$ in the sense of the Fourier inversion theorem (A.3.3). For each v, let

$$\mathrm{d}\chi_v = \widehat{\mathrm{d}g_v}$$

denote the dual measure to $\mathrm{d}g_v$ on $\widehat{G_v}$. Note for any $f_v \in L^1(G_v)$, we have that

$$\widehat{f_v}(\chi_v) = \int_{G_v} f_v(g_v)\overline{\chi_v}(g_v)\,\mathrm{d}g_v$$

by definition. If f_v is the characteristic function of H_v (which is visibly integrable and nonnegative) then

$$\widehat{f_v}(\chi_v) = \int_{H_v} \chi_v(g_v)\,\mathrm{d}g_v = \begin{cases} \mathrm{Vol}(H_v, \mathrm{d}g_v), & \chi_v|_{H_v} = 1 \\ 0, & \text{otherwise}, \end{cases}$$

since the integral of a non-trivial character over a compact group is zero. In other words, if $H_v^\perp$ is the subgroup of $\widehat{G_v}$ of characters trivial on H_v, then $\widehat{f_v}(\chi_v)$ is the characteristic function of H_v^* scaled by $\mathrm{Vol}(H_v, \mathrm{d}g_v)$. Thus, by Fourier inversion, for $y \in H_v$ we have:

$$\begin{aligned} 1n = \widehat{f_v}(y) &= \int_{\widehat{G_v}} \widehat{f_v}(\chi_v)\chi_v(y)\,\mathrm{d}\chi_v \\ &= \mathrm{Vol}(H_v, \mathrm{d}g_v) \int_{H_v^*} \mathrm{d}\chi_v \end{aligned}$$

$$= \mathrm{Vol}(H_v, \mathrm{d}g_v)\,\mathrm{Vol}(H_v^*, \mathrm{d}\chi_v)$$

where the first equality follows from our choice of f_v as the characteristic function of H_v. Recall in our construction of $\mathrm{d}g$ that $\mathrm{d}g_v$ assigns volume 1 to H_v for all but finitely many v. The above equation then tells us $\mathrm{Vol}(H_v, \mathrm{d}\chi_v) = 1$ almost everywhere as well, and so we can define the restricted product Haar measure

$$\mathrm{d}\chi := \prod_v \mathrm{d}\chi_v \tag{3.1-1}$$

as in 1.2-5, and this is also well-defined.

Proposition 3.1.4. The measure $\mathrm{d}\chi$, as defined in 3.1-1, is dual to $\mathrm{d}g$, defined in 1.2-5, in the sense that:

$$f(g) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(g)\,\mathrm{d}\chi$$

for all continuous $f \in L^1(G)$ such that $\widehat{f} \in L^1(\widehat{G})$.

Proof. For our purposes, since we know a dual measure to $\mathrm{d}g$ exists and is unique, it suffices to check on any nonzero continuous $f \in L^1(G)$, which we may choose to be a product function of the form

$$f = \prod f_v,$$

where f_v is the characteristic function of H_v for almost all v. For any input g, the left side above is a locally finite product $\prod f_v(g_v)$ by definition, and using the fact that each $\mathrm{d}\chi_v$ is dual to $\mathrm{d}g_v$, we get:

$$f_v(g_v) = \int_{\widehat{G_v}} \widehat{f_v}(\chi_v)\chi_v(g_v)\,\mathrm{d}\chi_v\,.$$

Multiplying them together (while applying Fubini) gives us our desired result. □

With the relevant desired properties verified, we are now ready to proceed to some actual adelic harmonic analysis.

Definition 3.1.5. For K a global field, if for every place v we have some $f_v \in \mathscr{S}(K_v)$ such that $f_v|_{\mathcal{O}_v} = 1$ for almost every v, then the "tensor product"

$$\bigotimes_v f_v =: f : \mathbb{A}_k \to \mathbb{C}$$
$$(x_v) \mapsto \prod f_v(x_v)$$

is locally a finite product, and we call such an f an **adelic Schwartz-Bruhat function**. We denote by $\mathscr{S}(\mathbb{A}_K)$ the set of all such functions.

Observation. It may not be immediately obvious at a glance, but $\mathscr{S}(\mathbb{A}_K)$ *is* a complex vector space. Clearly any scalar can be absorbed into a single component, and since any two adelic Schwartz-Bruhat functions are trivial on $\mathcal{O}_v$ at almost all v, their sum component-wise only affects the f_v on which they are mutually "non-trivial" (i.e only finitely many).

Similar to classical Fourier analysis, it is an exercise in simple-function approximation to see that $\mathscr{S}(\mathbb{A}_K)$ is a complex vector space dense in $L^2(\mathbb{A}_K)$ and $L^1(\mathbb{A}_K)$. Our goal will be to derive an adelic version of the Poisson summation formula of classical Fourier analysis.

First off, to define our Fourier transform we fix ψ to be the standard additive character[2] of $\mathbb{A}_K$ defined in the proof of Proposition 1.2.8 (i.e the "product" of all the standard characters on its local components). Moreover we let $\mathrm{d}x$ be the (self-dual!) Tamagawa measure on $\mathbb{A}_K$, defined in 1.2.16. Then for any $f \in \mathscr{S}(\mathbb{A}_K)$, the Fourier transform

$$\widehat{f}(x) := \int_{\mathbb{A}_K} f(y)\psi(xy)\,\mathrm{d}y \tag{3.1-2}$$

makes sense.

Proposition 3.1.6. The Fourier transform $f \mapsto \widehat{f}$ is an L^2-isometry of $\mathscr{S}(\mathbb{A}_K)$.

Proof. This follows immediately from the fact that on each component it is an isometry (from classical Fourier analysis at the Archimedean places, and from our proof of 1.2.9 at the non-Archimedean places), and from the multiplicative definition of the adelic norm in Def 1.3.9. We leave it to the reader to verify the (rather tedious) details should they want to do so. □

We will also particularly be interested in continuous integrable functions on $\mathbb{A}_K$ that are K-*periodic*, i.e $f(x) = f(x+\alpha)$ for every $\alpha \in K$. In these cases we may view f as a function on the quotient $\mathbb{A}_K/K$. In particular, we may consider its Fourier transform on $\widehat{\mathbb{A}_K/K}$ (canonically identified with K by 1.2.11)

$$\widehat{f}(\alpha) := \int_{\mathbb{A}_K/K} f(y)\psi(\alpha y)\,\mathrm{d}y\,, \tag{3.1-3}$$

for $\alpha \in K$.

Warning (Notational Abuse). There are a few slight abuses of notation here; we have already mentioned the one with our notation of f. Note also that ψ here is the character on $\mathbb{A}_K/K$ induced by our standard adelic character, which makes sense since ψ is trivial on K by construction. We also abuse notation for the measure $\mathrm{d}y$ in Eqn 3.1-3 is unique measure induced on the quotient by the self-dual $\mathrm{d}x$ on $\mathbb{A}_K$ and the counting measure on K. It is important to use this measure since, by Theorem 1.2.17, it gives $\mathbb{A}_K/K$ volume 1 as this is exactly the dual of the counting measure on K, and thus we avoid having to multiply by any normalizing constant in the Fourier inversion formula.

Fourier inversion then tells us that if $\widehat{f} \in L^1(K)$ then

$$f(x) = \sum_{\alpha\in K} \widehat{f}(\alpha)\overline{\psi}(x\alpha)$$

for $x \in \mathbb{A}_K/K$, since a convergent integral with respect to a counting measure is just an absolutely convergent sum.

Definition 3.1.7. A $\mathbb{C}$-valued continuous integrable function f on $\mathbb{A}_K$ is **admissible** if both

$$\sum_{\gamma\in K} f(x+\gamma), \quad \sum_{\gamma\in K} \widehat{f}(x+\gamma)$$

are absolutely and uniformly convergent on compact sets.

[2] which we recall rests on some *choice* of separating transcendence basis $\{t\}$ in the positive characteristic case.

Admissibility will be a vital property for us, as we will need to use it in the proofs of the upcoming results of Poissoin Summation and Riemann-Roch for the adeles, so we had better be sure everything we are working with is indeed admissible.

LEMMA 3.1.8. Any $f \in \mathscr{S}(\mathbb{A}_K)$ is admissible.

Proof. Since the Fourier transform $f \mapsto \widehat{f}$ carries $\mathscr{S}(\mathbb{A}_K)$ into itself, it suffices to just show the necessary convergence for an arbitrary adelic Schwartz-Bruhat function. By part (b) of 1.1.3, any compact $C \subseteq \mathbb{A}_K$ is contained in some set of the form

$$C := \prod_{v \in S} C_v \times \prod_{v \notin S} \mathcal{O}_v$$

for S a finite index set and $C_v \subseteq K_v$ compact. Thus it suffices to show (absolute and uniform) convergence on a set of the form C. Recall that, as a $\mathbb{C}$-vector space, $\mathscr{S}(\mathbb{A}_K)$ is generated by product functions built from characteristic functions of fractional ideals at all the non-Archimedean places v, and some Schwartz function at all the Archimedean completions. Showing convergence of any such generating function suffices, as it will imply convergence of any finite $\mathbb{C}$-linear combination of them.
For any such "product function" f we note that at any non-Archimedean place v there exists some $\delta_v \in \mathbb{R}_{>0}$ such that

$$|\gamma|_v > \delta_v \implies f_v(x_v + \gamma) = 0 \; \forall x_v \in C_v.$$

We note that $\delta_v = 1$ for almost all v, and so there is some maximum δ_v across all finite places.
In the positive characteristic case, since our K is some function field over a finite field and every place is non-Archimedean, this tells us the sum $\sum_{\gamma \in K} f(x + \gamma)$ is just a finite sum on C, and thus naturally absolutely and uniformly convergent.
In the number field case, our summation is restricted to some fractional ideal J of $\mathcal{O}_K$ whose prime factorization corresponds exactly to all the finite places where $\delta_v > 1$. However note that this is a discrete subset of $K_\infty := \prod_{v|\infty} K_v$, which we can treat as a subset of a lattice in $\mathbb{R}^n$. Note that, as we take an exhaustion of closed balls, the number of elements in the lattice contained in a shell of radius R and thicknes Δ (relatively small compared to R) grows like a fixed polynomial in R. However, Since $f_v \in \mathscr{S}(K_v)$ decreases faster than any polynomial, it will go to 0 much faster than the number of γ in the sum can grow. This gives us our desired result. $\square$

Now that we know our summation does give us a well-defined value, the statement of the following theorem makes sensse.

THEOREM 3.1.9 (Adelic Poisson Summation). For $f \in \mathscr{S}(\mathbb{A}_K)$,

$$\sum_{\gamma \in K} f(\gamma) = \sum_{\gamma \in K} \widehat{f}(\gamma)$$

Proof. For notational convenience, define

$$F(x) := \sum_{\gamma \in K} f(x + \gamma).$$

Note that since $F(x)$ is K-periodic we can view it as a continuous function on $\mathbb{A}_K / K$. By Pontryagin duality considerations we then know its Fourier transform $\widehat{F}$ can be viewed as a function $K \to \mathbb{C}$. For any $\alpha \in K$, we have:

$$\begin{aligned}
\widehat{F}(\alpha) &= \int_{\mathbb{A}_K/K} \sum_{\gamma\in K} f(x+\gamma)\psi(\alpha x)\,\mathrm{d}x \\
&= \int_{\mathbb{A}_K/K} \sum_{\gamma\in K} f(x+\gamma)\psi(\alpha(x+\gamma))\,\mathrm{d}x \\
&= \int_{\mathbb{A}_K} f(x)\psi(\alpha x)\,\mathrm{d}x \\
&= \widehat{f}(\alpha),
\end{aligned}$$

where the second equality follows from the fact that ψ is trivial on K and the third follows from the compatability of our measure on $\mathbb{A}_K/K$ with the counting measure on K. Since $\widehat{f} \in \mathscr{S}(\mathbb{A}_K)$, the above equality also tells us that $\widehat{F} \in L^1(K)$, so we may apply Fourier inversion to get

$$\begin{aligned}
F(x) := \sum_{\gamma\in K} f(x+\gamma) &= \sum_{\gamma\in K} \widehat{F}(\gamma)\overline{\psi(x\gamma)} \\
&= \sum_{\gamma\in K} \widehat{f}(\gamma)\overline{\psi(x\gamma)}.
\end{aligned}$$

Setting $x = 0$ gives us our desired result. □

We will also need a multiplicative version of the above result, relating to the average

$$\sum_{\gamma\in K} f(x\gamma),$$

for $x \in \mathbb{I}_K$.

Theorem 3.1.10 (Adelic Riemann-Roch). For any $x \in \mathbb{I}_K$ and $f \in \mathscr{S}(\mathbb{A}_K)$,

$$\sum_{\gamma\in K} f(\gamma x) = \frac{1}{|x|_{\mathbb{I}_K}} \sum_{\gamma\in K} \widehat{f}(\gamma x^{-1})$$

Remark. Note there is no confusion in the absolute value taken of x, since it is an element of K its normal absolute value and idelic absolute value coincide.

Proof. Fix some $x \in \mathbb{I}_K$, and set $g(z) := f(xz)$, for any $z \in \mathbb{A}_K$. Note that $g \in \mathscr{S}(\mathbb{A}_K)$ as well, and thus Poisson summation (3.1.9) applies to give us:

$$\begin{aligned}
\sum_{\gamma\in K} g(\gamma) &= \sum_{\gamma\in K} \widehat{g}(\gamma) \\
&= \sum_{\gamma\in K} \int_{\mathbb{A}_K} f(xz)\psi(\gamma z)\,\mathrm{d}z \\
&= \sum_{\gamma\in K} \int_{\mathbb{A}_K} \frac{1}{|x|_{\mathbb{I}_K}} f(z)\psi(\gamma z x^{-1})\,\mathrm{d}z \\
&= \frac{1}{|x|_{\mathbb{I}_K}} \sum_{\gamma\in K} \widehat{f}(\gamma x^{-1}),
\end{aligned}$$

completing the proof. □

At first glance, one might wonder why the above Theorem is named Riemann-Roch; indeed, there seem to be no compact Riemann surfaces anywhere in the picture. Even in Tate's exposition he does not expound upon this connection apart from declaring that it is a "number-theoretic analogue" of Riemann-Roch. In general, the theory of local and global fields in positive characteristic cannot be fully fleshed out and understood without the language of algebraic geometry, and to that end we indulge in a short interlude for the remainder of this subsection to explain this connection. Due the limits of space and time however we will not be able to describe nearly as many connections to broader concepts as we should; for a more detailed exposition we refer the reader to the second chapter of [Ser12], or the more number-theory-oriented (yet denser) sixth chapter of [Wei74].

The goal here will be to view K as the function field of some algebraic variety, and reformulate Theorem 3.1.10 in geometric terms that will correspond to the classical theorem of Riemann-Roch.

DEFINITION 3.1.11. Let K be a global field of positive characteristic. A **divisor** on K is a formal finite linear combination

$$D = \sum_{v \in \mathscr{P}_K} n_v \cdot v, \quad n_v \in \mathbb{Z}.$$

We denote by $\operatorname{Div}(K)$ the group of divisors on K (which can be thought of as the free abelian group generated by the places of K).
The **degree** of a place $\deg(v)$ is the degree of the residue field corresponding to v over the field of constants $\mathbb{F}_q$. For a divisor D we define its degree by

$$\deg(D) := \sum_{v \in \mathscr{P}_K} n_v \deg(v).$$

We denote by $\operatorname{Div}^0(K)$ the subgroup of degree-zero divisors.

For any $f \in K^\times$, we note the formal linear combination

$$\operatorname{div}(f) := \sum_{v \in \mathscr{P}_K} \operatorname{ord}_v(f) v \tag{3.1-4}$$

is a finite linear combination, by unique prime factorization of (nonzero) ideals in Dedekind domains. Hence this is a divisor, and moreover we note that

$$\operatorname{div}(f) + \operatorname{div}(g) = \operatorname{div}(fg)$$

Furthermore, recall the product formula (1.3.10) tells us for any $f \in K^\times$ that

$$\begin{aligned} 1 = \prod_{v \in \mathscr{P}_K} |f|_v &= \prod_{v \in \mathscr{P}_K} q_v^{\operatorname{ord}_v(f)} \\ &= q^{\sum_{v \in \mathscr{P}_k} \deg(v) \operatorname{ord}_v(f)} \\ \implies \deg(\operatorname{div}(f)) &= \sum_{v \in \mathscr{P}_k} \deg(v) \operatorname{ord}_v(f) = 0. \end{aligned}$$

Thus any such divisor induced by an element of $K^\times$ lies in $\operatorname{Div}^0(K)$, and moreover the set of all such divisors forms a subgroup (which is proper so long as $\mathcal{O}_K$ is not a PID). This motivates the following definitions.

DEFINITION 3.1.12. We denote by $\operatorname{div}(K^\times)$ the group of all **principal divisors** of the form $d(f)$ for $f \in K^\times$. The quotient $\operatorname{Div}(K)/\operatorname{div}(K^\times)$ is denoted $\operatorname{Pic}(K)$ and called the **Picard group** of K, whose elements are called **divisor classes**. We also define the **Picard group of degree zero**, denoted $\mathrm{Pic}^0(K)$, by the quotient $\mathrm{Div}^0(K)/\operatorname{div}(K^\times)$.

We note that if $\operatorname{div}(f) = \operatorname{div}(g)$ then $\operatorname{div}(f/g) = 0$, which means $f/g \in \bigcap_{v\in\mathscr{P}_K} \mathcal{O}_v \implies f/g \in \mathbb{F}_q^\times$. All of this tells us we have the following exact sequence:

$$1 \to \mathbb{F}_q^\times \to K^\times \to \mathrm{Div}^0(K) \to \mathrm{Pic}^0(K) \to 0.$$

Definition 3.1.13. We say a divisor $D = \sum_{v\in\mathscr{P}_K} n_v v$ is **effective** if $n_v \geq 0$ for all v.
The **linear system** $L(D)$ of a divisor D is the union of $\{0\}$ and set of all $\operatorname{div}(f) \in \operatorname{div}(K^\times)$ such that $D + \operatorname{div}(f)$ is effective.

Geometrically, the notion of an effective divisor corresponds to some (codimension-one) subvariety cut out by regular functions, not just rational ones. Analytically one could also think of it as the vanishing locus of a holomorphic function. This concept also allows us to "compare" divisors to one another. In particular, we say

$$D \geq D' \iff D - D' \text{ is effective} \iff n_v \geq n'_v \ \forall v \in \mathscr{P}_K.$$

Observation. The "functions" $f \in K^\times$ that are in $L(D)$, together with 0, form an $\mathbb{F}_q$-vector space, since $f \in L(D)$ if and only if $\operatorname{ord}_v(f) \geq -n_v$ for all $v \in \mathscr{P}_K$. We make the observation that if if $\deg(D) < 0$ then $\dim(L(D)) = 0$. In the more general case, we have the following proposition.

Proposition 3.1.14. For any divisor D, the vector space $L(D)$ is finite-dimensional.

Proof. The proof here uses the idelic formulation. First we define a *divisor map* on $\mathbb{I}_K$ by

$$\begin{aligned} \mathscr{D} : \mathbb{I}_K &\to \mathrm{Div}(K) \\ (x_v)_v &\mapsto \sum_v \operatorname{ord}_v(x_v)\cdot v. \end{aligned} \tag{3.1-5}$$

This map is visibly a surjective homomorphism, so to each divisor $D \in \mathrm{Div}(K)$ we may denote an idele $x(D) = (x(D)_v)_v$, i.e $\operatorname{ord}_v(x(D)_v) = \operatorname{ord}_v(D)$. Letting f_v be the characteristic function of $\mathcal{O}_v$ for each v, we may define $f := \bigotimes_v f_v \in \mathscr{S}(\mathbb{A}_K)$. Then by construction, for any $\gamma \in K^\times$, we have:

$$f(\gamma x(D)) = \begin{cases} 1, & \text{if } \operatorname{ord}_v(\gamma x(D)_v) \geq 0 \ \forall v, \\ 0, & \text{otherwise.} \end{cases}$$

In other words, $f(\gamma x(D)) = 1$ if and only if $\mathscr{D}(\gamma x(D))$ is an effective divisor, which in turn is true if and only if $\gamma \in L(D)$. Thus

$$\sum_{\gamma\in K} f(\gamma x(D)) = |L(D)|.$$

Note f is admissible by construction and so the adelic Riemann-Roch theorem applies to tell us the above sum converges, and thus $L(D)$ has finite cardinality, hence is finite dimensional. ☐

It will also be of later use to us to note some basic properties of the divisor map, all of which follow directly from definitions. We leave the reader to verify the following equalities, should they wish.

Proposition 3.1.15. If $\mathscr{D} : \mathbb{I}_K \to \mathrm{Div}(K)$ denotes the divisor map, then the following properties hold:

- $\ker(\mathscr{D}) = \prod_v \mathcal{O}_v^\times = I_{K,\varnothing}$.
- $\mathscr{D}(\mathbb{I}_K^1) = \mathrm{Div}^0(K)$.
- $\mathbb{I}_K/(K^\times \cdot \mathbb{I}_{K,\varnothing}) = \mathrm{Pic}(K)$.
- $\mathbb{I}_K^1/(K^\times \cdot \mathbb{I}_{K,\varnothing}) = \mathrm{Pic}^0(K)$.

Remark. The nontrivial map $\deg : \mathrm{Pic}(K) \to \mathbb{Z}$ not only has finite index, but is in fact *surjective*.

We are now ready for the formulation of the Riemann-Roch theorem.

Theorem 3.1.16 (Geometric Riemann-Roch). Let K be a global field of positive characteristic. Then there exists an integer $g \geq 0$, called the **genus** of K, and a divisor $\mathscr{K}$ of degree $2g-2$, called the **canonical divisor** of K, such that

$$\dim(L(D)) - \dim(L(\mathscr{K} - D)) = \deg(D) - g + 1$$

for all $D \in \mathrm{Div}(K)$.

The proof of this theorem relies on the extension of the notion of a divisor to that of characters as well. Let ψ be a non-trivial quasi-character of $\mathbb{A}_K$ trivial on K; we know this exists by Proposition 1.2.8. Moreover, the Breaking Down Lemma (1.2.5) tells us that the induced characters ψ_v on each are trivial on $\mathcal{O}_v$ at cofinitely many v. Thus, if we define $\mathrm{ord}_v(\psi_v) := \mathrm{ord}_{\mathfrak{p}_v}(\text{Conductor of } \psi_v)$, we get that the divisor

$$\mathrm{div}(\psi) := -\sum_v \mathrm{ord}_v(\psi_v) \cdot v \tag{3.1-6}$$

is well-defined. Fixing any such ψ, Proposition 1.2.11 tells us that any other such character of $\mathbb{A}_K$ trivial on K is obtained by $x \mapsto \psi(\alpha x)$ for some $\alpha \in K^\times \subseteq \mathbb{I}_K$. In other words, the set of all divisors induced by quasicharacters of $\mathbb{A}_K/K$ is given by a single divisor class modulo the group of principal divisors, i.e some element of $\Psi \in \mathrm{Pic}(K)$. This is the key to the proof.

Proof of 3.1.16. Pick any $\mathscr{K} \in \Psi$ defined above corresponding to some quasi-character ψ of $\mathbb{A}_K$ trivial on K, and define

$$g := 1 + \frac{\deg(\mathscr{K})}{2} \in \frac{1}{2}\mathbb{Z}.$$

Letting $f := \otimes_v f_v \in \mathscr{S}(\mathbb{A}_K)$ be defined as the tensor product of all the characteristic functions of $\mathcal{O}_v$ at each place, we have seen from the proof of 3.1.14 that

$$q^{\dim(L(D))} = \sum_{\gamma \in K} f(\gamma x(D))$$

for any divisor D. Recalling the adelic Riemann Roch theorem, we therefore get that

$$q^{\dim(L(D))} = \frac{1}{|x(D)|_{\mathbb{A}_K}} \sum_{\gamma \in K} \widehat{f}(\gamma x(D)^{-1}).$$

Note that

$$|x(D)|_{\mathbb{A}_K} = \prod_v q_v^{-\operatorname{ord}_v(D)} = q^{-\deg(D)},$$

so it only remains to show that

$$\sum_{\gamma \in K} \widehat{f}(\gamma x(D)^{-1}) = q^{1-g+\dim(L(\mathscr{K}-D))}$$

Using virtually the same computation as in Proposition 1.2.15, where we showed self-duality of the measures defined with respect to the standard characters, with respect to our arbitrary quasi-character ψ, we get that

$$\widehat{f_v} = N(\text{Conductor of } \psi_v)^{1/2}\mathbb{1}_{\text{Conductor of } \psi_v} = q^{\deg(v)\operatorname{ord}_v(\mathscr{K})/2}\mathbb{1}_{\text{Conductor of } \psi_v},$$

from which it follows that

$$\prod_v \widehat{f_v} = q^{-\deg(\mathscr{K})/2}\mathbb{1}_{\text{Conductor of } \psi_v} = q^{1-g}\mathbb{1}_{\text{Conductor of } \psi_v}$$

Thus, for any $\gamma \in K$ we have

$$\widehat{f}(\gamma x(D)^{-1}) = \begin{cases} q^{1-g}, & \operatorname{ord}_v(\gamma) \geq \operatorname{ord}_v(D) + \operatorname{ord}_v(\psi_v)\ \forall v \\ 0, & \text{otherwise.} \end{cases}$$

In particular, $\widehat{f}(\gamma x(D)^{-1})$ is nonzero if and only if $\operatorname{div}(\gamma) \in L(\mathscr{K} - D))$, whence the desired equality follows. This also shows that g must be equal to $\dim(L(\mathscr{K}))$, and hence must be a non-negative integer. □

§3.2 The Global Functional Equation

Definition 3.2.1. An **idele-class quasi-character** is a continuous homomorphism $C_K \to \mathbb{C}^\times$. Equivalently, it is a quasi-character of $\mathbb{I}_K$ trivial on $K^\times$. We will frequently abuse notation between these two.

Remark. Some texts also use the term **Hecke character** or **Großëncharaktere**, though we stick to the more explicitly descriptive term in this text.

Example 3.2.2 (Idelic Lifts of Dirichlet Characters). Recall a **Dirichlet character** of modulus m is a homomorphism $\chi : (\mathbb{Z}/m\mathbb{Z})^\times \to \mathbb{C}^\times$. We claim this can be lifted to an idele-class character. Recalling our decomposition of the rational idele group from Example 1.3.11, one sees immediately that

$$\mathbb{I}_\mathbb{Q}/\mathbb{Q}^\times \cong \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^\times \twoheadrightarrow \widehat{\mathbb{Z}}^\times \twoheadrightarrow (\mathbb{Z}/m\mathbb{Z})^\times \xrightarrow{\chi} \mathbb{C}^\times$$

gives us a character of $C_\mathbb{Q}$. A perhaps more non-obvious observation is that *every* quasi-character of $C_\mathbb{Q}$ arises in this fashion, up to some complex power of the idelic absolute value. To see this, note we have the exact sequence

$$0 \to C_\mathbb{Q}^1 \to C_\mathbb{Q} \to |\mathbb{I}_\mathbb{Q}|_{\mathbb{A}_\mathbb{Q}} \to 0,$$

and recall by Theorem 1.3.14 that $C_\mathbb{Q}^1$ is compact. We can decompose any idele-class quasi-character χ into $\mu\eta$, with $\mu : C_\mathbb{Q} \to S^1$ and $\eta : C_\mathbb{Q} \to \mathbb{R}_{>0}$. Since $\widehat{\mathbb{Z}}^\times$ is compact and totally disconnected, any complex quasi-character of it must have finite image, by a standard no-small-subgroups argument.

In the case of η which has its image in the positive reals, this means it must be trivial on $\widehat{\mathbb{Z}}^\times$ and therefore factors through the idelic norm $|\cdot|_{\mathbb{I}_\mathbb{Q}} : \mathbb{I}_\mathbb{Q} \twoheadrightarrow \mathbb{R}_{>0}$. Since any continuous endomorphism of $\mathbb{R}_{>0}$ is an exponential, we deduce η must be of the form

$$\eta(x) = \left(|x|_{\mathbb{I}_\mathbb{Q}}\right)^a,$$

for some $a \in \mathbb{R}$.
In the case of μ whose values are in S^1, it has finite image, and so on $\widehat{\mathbb{Z}}^\times$ it must be a lifted Dirichlet character $\widetilde{\gamma}$. Moreover its value on $\mathbb{R}_{>0}$ must be some imaginary power of the idelic norm, by a similar argument as with η, using the fact that any character of $\mathbb{R}_{>0}$ corresponds to imaginary exponentiation. Thus we deduce

$$\mu(x) = \left(|x|_{\mathbb{I}_\mathbb{Q}}\right)^{bi}\widetilde{\gamma}(x),$$

for $b \in \mathbb{R}$ and γ a Dirichlet character. Multiplying them together we get that

$$\chi(x) = \left(|x|_{\mathbb{I}_K}\right)^s\widetilde{\gamma},$$

for $s = a + bi \in \mathbb{C}$.

By a similar argument, we can generalize our work of the above example to deduce any idele class quasi-character, for any global field K, is of the form

$$\chi = \mu|\cdot|^s_{\mathbb{A}_K} \tag{3.2-1}$$

for μ a character and $s \in \mathbb{C}$. As in §2.1, we denote by σ its exponent (which is intrinsic to χ even though $s \in \mathbb{C}$ is not) and $\chi^\vee$ the shifted dual (see Definitions 2.1.2 and 2.1.3).
Also recall from Eqn 2.1-4, before our analysis on the local theory that, for each completion K_v, our Haar measure $\mathrm{d}^\times x_v$ on the multiplicative group was $\mathrm{d}x_v / |x_v|$ scaled by some constant c_v. We will now see the reason we gave ourselves this flexibility. If we simply chose the easy route setting $c_v = 1$ for all v, then the induced measure $\mathrm{d}^\times x = \prod_v \mathrm{d}^\times x_v$ on the ideles (in the sense of Eqn 1.2-5) is non-ideal in the following sense: the measure of any basic open set $\prod_{v\in S} U_v \times \prod_{v\notin S} \mathcal{O}_v^\times$ would be

$$\left(\prod_{v\in S}\int_{U_v}\mathrm{d}^\times x_v\right)\cdot\left(\prod_{v\notin S}\int_{\mathcal{O}_v^\times}\mathrm{d}^\times x_v\right) = \left(\prod_{v\in S}\mathrm{Vol}(U_v, \mathrm{d}^\times x_v)\right)\cdot\left(\prod_{v\notin S}(1 - N(\mathfrak{p}_v)^{-1})\right),$$

and the second term in the product converges to 0. To amend this, we set our scaling constant to

$$c_v = N(\mathfrak{p}_v)/(N(\mathfrak{p}_v) - 1). \tag{3.2-2}$$

Since our standard measure $\mathrm{d}x_v$ was normalized so that $\mathrm{Vol}(\mathcal{O}_v, \mathrm{d}x_v) = N(\mathfrak{D}_v)^{1/2}$ and $\mathrm{Vol}(\mathcal{O}_v, \mathrm{d}x_v/|x_v|) = (N(\mathfrak{p}_v) - 1)/N(\mathfrak{p}_v)$, it follows that our scaled measure $\mathcal{O}_v^\times$ the measure $N(\mathfrak{D}_v)^{1/2}$. We may then define $\mathrm{d}^\times x$ on $\mathbb{I}_K$ as the restricted product of all the appropriately scaled local measures, which makes sense since $\mathcal{O}_v$ is given measure 1 at all the finite unramified places v.

Definition 3.2.3. Given $f \in \mathscr{S}(\mathbb{A}_K)$ and χ an idele-class quasi-character, we define the **global zeta function** to be

$$Z(f,\chi) := \int_{\mathbb{I}_K} f(x)\chi(x)\,\mathrm{d}^\times x$$

By our characterization of idele-class quasi-characters in Eqn 3.2-1, we may, as we did similarly in the local theory, view the zeta function as having inputs on the Riemann surface $\mathscr{X}(C_K)$ of idele-class quasi-characters, with complex inputs s. We now are ready for the crux of Tate's thesis.

Theorem 3.2.4 (The Global Functional Equation). Let K be a global field. Then for any $f \in \mathscr{S}(\mathbb{A}_K)$ and idele-class quasi-character χ with exponent $\sigma = \mathfrak{Re}(\chi)$ the following hold:

(a) $Z(f,\chi)$ absolutely converges when $\sigma > 1$, and is holomorphic in χ.

(b) $Z(f,\chi)$ extends to a meromorphic function on all of $\mathscr{X}(C_K)$, with simple poles when $\chi = 1$ and $\chi = |\cdot|_{\mathbb{I}_K}$, with corresponding residues given by $-\operatorname{Vol}(C_K^1)f(0)$ and $\operatorname{Vol}(C_K^1)\widehat{f}(0)$.

(c) The functional equation $Z(f,\chi) = Z(\widehat{f},\chi^\vee)$ is satisfied.

Remark. In item (b) of the above Theorem, we have not explicitly what measure we are taking the volume of C_K^1 with respect to. The answer is what one might expect: first we induce the measure on $\mathbb{I}_K^1$ by the exact sequence

$$1 \to \mathbb{I}_K^1 \to \mathbb{I}_K \to |\mathbb{I}_K|_{\mathbb{A}_K} \to 1,$$

where $|\mathbb{I}_K|_{\mathbb{A}_K}$ is given the standard measure $|x|^{-1}\,dx$ if it is $\mathbb{R}_{>0}$ and the counting measure if it is (some finite-index subgroup of) $q^{\mathbb{Z}}$, and $\mathbb{I}_K$ is given the measure defined above. The reason for this will become clear in the computations involved in the proof of the above theorem.
Then, using that measure on $\mathbb{I}_K^1$ and the counting measure on $K^\times$, and the exact sequence

$$1 \to K^\times \to \mathbb{I}_K^1 \to C_K^1 \to 1,$$

we get the measure with respect to which we should compute the volume of C_K^1.
Weil, in [Wei74] normalizes the measure to make $\operatorname{Vol}(C_K^1) = 1$; this is essentially akin to the old question of "where to put the factor of 2π?" that one encounters in classical Fourier analysis. We elect not to do this and to compute its volume later in terms of the measure scaled locally by the factors described in Eqn. 3.2-2.

Proof of (a) of 3.2.4. Since $f \in \mathscr{S}(\mathbb{A}_K)$, recall from our Definition 3.1.5 that we can write $f = \prod_v f_v$ where f_v is the characteristic function of $\mathcal{O}_v$ for almost all v. Similarly we write χ_v for the restriction of an unramified representative $\chi = |\cdot|_{\mathbb{A}_K}^s$ to the v-th component. Thus, by 3.1.2 we can decompose our zeta function into a product of the form:

$$Z(f,\chi) = \prod_{v \in \mathscr{P}_K} Z(f_v,\chi_v).$$

Thus it suffices to show convergence of the product of $Z(\mathbb{1}_{\mathcal{O}_v^\times}, |\cdot|_{\mathbb{A}_K}^\sigma) = \left(1 - |\pi_v|^\sigma\right)^{-1}$ over all the finite places v. When K is a number field of degree d over $\mathbb{Q}$, we have (when $\sigma > 1$):

$$1 \le \prod_{v \nmid \infty} \frac{1}{1-|\pi_v|^\sigma} \le \prod_{p \text{ prime}} \left(\frac{1}{1-1/p^\sigma}\right)^d = \left(\sum_{n\ge1} \frac{1}{n^\sigma}\right)^d,$$

so we set absolutely convergence in this case.
When K has positive characteristic p, the relevant product to check for convergence is

$$\prod_{n\ge1}\left(\frac{1}{1-1/q^{n\sigma}}\right)^{N_n}$$

where q is the size of the algebraic closure of $\mathbb{F}_p$ in K and N_n is the number of monic irreducibles of degree n in $\mathbb{F}_q[t]$. The reasoning behind this is that for every monic irreducible of $\mathbb{F}_q(t)$ there are a finite number of finite places of K over that prime. We know there are $\leq q^n$ monic irreducibles of degree n, so it suffices to check convergence of

$$\prod_{n\geq 1}\left(\frac{1}{1-1/q^{n\sigma}}\right)^{q^n}.$$

Taking the formal logarithm gives us

$$\log\left(\prod_{n\geq 1}\left(\frac{1}{1-1/q^{n\sigma}}\right)^{q^n}\right) = \sum_{n\geq 1} q^n \log\left(\frac{1}{1-q^{-n\sigma}}\right).$$

A classical Taylor series expansion tells us $\log(1/(1-x)) \leq x$ (for $x \in (0,1/2)$, at least, which is what we need for our purposes), so the above converges if the following expression does:

$$\sum_{n\geq 1} q^n q^{-n\sigma} = \sum_{n\geq 1} q^{n(1-\sigma)}.$$

This is visibly convergent when $\sigma > 1$, so thus our original expression is also convergent in this domain, as desired. We will show holomorphicity in both cases in the proof of parts (b) and (c) of the theorem. $\square$

To prove the other parts of the theorem, we employ a classic integration trick: splitting up our integral by norm. We thus write our zeta function as the double integral

$$Z(f,\chi) = \int_{\mathrm{Im}\,|\cdot|_{\mathbb{A}_K}} \int_{\mathbb{I}_K^1} f(tx)\chi(tx)\,\mathrm{d}^\times x\,\frac{\mathrm{d}t}{t},$$

where by "tx" we mean multiplying x by the idele which is 1 everywhere except a single infinite place (recall in positive characteristic this depends on our choice of separating basis!) , where it is an element of norm t. We are being a little reductive via notation here since this integral notation is more valid for the number field case. More explicitly, the outer integral is over the entirety of $\mathbb{R}_{>0}$ when K is a number field and is a countable sum over powers of q when K is a function field.

Warning. When $K \cong \mathbb{F}_q(t)$, we have not yet shown that the image of the idelic norm is in fact all of $q^{\mathbb{Z}}$; this will arrive as a consequence of the proof of our main theorem on the global zeta function! However, we know it is some finite index subgroup of this, so for the purposes of the remainder of the proof we may abuse notation a little and take q to be a *generator* of this subgroup.

The path to proving our desired statements will be via proving a functional equation for the inner integrand

$$Z_t(f,\chi) := \int_{\mathbb{I}_K^1} f(tx)\chi(tx)\,\mathrm{d}^\times x \;=\; \int_{\mathbb{I}_K^t} f(x)\chi(x)\,\mathrm{d}^\times x\,, \tag{3.2-3}$$

where by $\mathbb{I}_K^t$ we mean the ideles of norm t. The key lemma is as follows.

Lemma 3.2.5. For any idele-class quasi-character χ and adelic Schwartz-Bruhat function f, the t-local zeta function $Z_t(f,\chi)$ defined in Eqn 3.2-3 satisfies

$$Z_t(f,\chi) + f(0)\int_{C_K^1} \chi(tx)\,\mathrm{d}^\times x \;=\; Z_{t^{-1}}(\widehat{f},\chi^\vee) + \widehat{f}(0)\int_{C_K^1} \chi^\vee(xt^{-1})\,\mathrm{d}^\times x\,.$$

WARNING (Notational Abuse). There are two slight abuses of notation in the above lemma statement. First we are integrating χ over C_K^1, when really we should be writing the projection of χ onto the quotient, and secondly we are writing $\mathrm{d}^\times x$ when are actually integrating with respect to the natural quotient measure on C_K^1 discussed in the remark following the Theorem statement of 3.2.4.

Proof. Since $\mathbb{I}_K^1/K^\times = C_K^1$ and χ is trivial on $K^\times$, we have that

$$Z_t(f,\chi) = \int_{C_K^1} \left(\sum_{\alpha \in K^\times} f(\alpha t x) \right) \chi(tx)\, \mathrm{d}^\times x\,.$$

Here we use discreteness of $K^\times$ in $\mathbb{I}_K$ (Corollary 1.3.6) so that a Haar measure on $K^\times$ is given by the counting measure, then using Fubini to separate the uniform summation from the integral. By Lemma 3.1.8, we know any $f \in \mathscr{S}(\mathbb{A}_K)$ is admissible, so the sum within the integral is (absolutely and uniformly!) convergent. Since we are integrating over a compact set, it follows $Z_t(f,\chi)$ is in fact convergent for any choices of f,χ. This naturally leads us to the following identity of the left side of the lemma:

$$Z_t(f,\chi) + f(0) \int_{C_K^1} \chi(tx)\, \mathrm{d}^\times x \; = \int_{C_K^1} \sum_{\alpha \in K} f(\alpha t x) \chi(tx)\, \mathrm{d}^\times x\,.$$

The vital step now is to apply Theorem 3.1.10, or the adelic Riemann-Roch to the summation over K. More precisely it tells us that $\sum_{\alpha \in K} f(\alpha t x) = 1/\left|tx\right|_{\mathbb{A}_K} \sum_{\alpha\in K} \widehat{f}(\alpha (xt)^{-1})$. Substituting this in, we get

$$\begin{aligned}
Z_t(f,\chi) + f(0) \int_{C_K^1} \chi(tx)\, \mathrm{d}^\times x \; &= \int_{C_K^1} \chi(tx) \frac{\chi(tx)}{\left|tx\right|_{\mathbb{A}_K}} \sum_{\alpha \in K} \widehat{f}(\alpha (xt)^{-1})\, \mathrm{d}^\times x \\
&= \int_{C_K^1} \left| t^{-1} x \right|_{\mathbb{A}_K} \chi(tx^{-1}) \sum_{\alpha \in K} \widehat{f}(\alpha x t^{-1})\, \mathrm{d}^\times x \\
&= \int_{\mathbb{I}_K^1} \chi^{-1}(xt^{-1}) \left| xt^{-1} \right|_{\mathbb{A}_K} \widehat{f}(xt^{-1})\, \mathrm{d}^\times x \; + \widehat{f}(0) \int_{C_K^1} \chi^{-1}(xt^{-1}) \left| xt^{-1} \right|_{\mathbb{A}_K} \mathrm{d}^\times x \\
&= Z_{t^{-1}}(\widehat{f}, \chi^\vee) + \widehat{f}(0) \int_{C_K^1} \chi^\vee(xt^{-1})\, \mathrm{d}^\times x\,,
\end{aligned}$$

where the second line follows from a change of variables x to x^{-1}. This completes the proof of the functional equation. □

Proof of parts (b) and (c) of 3.2.4. The first step is yet another classic integration trick: splitting up the integral. In the number field case, we write

$$Z(f,x) = \int_0^1 Z_t(f,\chi) \frac{\mathrm{d}t}{t} + \int_1^\infty Z_t(f,\chi) \frac{\mathrm{d}t}{t} \tag{3.2-4}$$

We note that we can write the second integral as

$$\int_1^\infty Z_t(f,\chi) \frac{\mathrm{d}t}{t} = \int_{x \in \mathbb{I}_K : |x|_{\mathbb{A}_K} \geq 1} f(x)\chi(x)\, \mathrm{d}^\times x\,. \tag{3.2-5}$$

By the settled part (a) of our theorem, we know that this converges when $\sigma > 1$. However since all the inputs of this integral have norm ≥ 1 convergence actually improves when our exponent σ shrinks. Thus we deduce that 3.2-5 is convergent everywhere.

It remains to analyse the first half of Eqn 3.2-4. By our "t-local" functional equation from Lemma 3.2.5, we note that

$$\int_0^1 Z_t(f,\chi) \frac{\mathrm{d}t}{t} = \int_0^1 Z_{t^{-1}}(\widehat{f}, \chi^\vee) \frac{\mathrm{d}t}{t} + \mathscr{R}(f,\chi), \tag{3.2-6}$$

where $\mathscr{R}$ is the "remainder" given by the functional equation of the lemma, i.e

$$\mathscr{R}(f,\chi) := \int_0^1 \left(\widehat{f}(0) \int_{C_K^1} \chi^\vee(xt^{-1})\, \mathrm{d}^\times x \, - f(0) \int_{C_K^1} \chi(tx)\, \mathrm{d}^\times x \right) \frac{\mathrm{d}t}{t}.$$

We note that by replacing t with t^{-1}, we get that

$$\int_0^1 Z_{t^{-1}}(\widehat{f},\chi^\vee)\frac{\mathrm{d}t}{t} = \int_1^\infty Z_t(\widehat{f},\chi^\vee)\frac{\mathrm{d}t}{t}.$$

By Proposition 3.1.6, $\widehat{f} \in \mathscr{S}(\mathbb{A}_K)$ and $\chi^\vee$ is by construction also an idele-class quasi-character, so the same argument as for Eqn 3.2-5 tells us that this is also convergent everywhere. Thus we have to analyse $\mathscr{R}(f,\chi)$.
If χ is nontrivial on the norm-1 ideles, then integrating either χ or $\chi^\vee$ over C_K^1 (a compact group!) yields zero, that is to say, $\mathscr{R}(f,\chi) = 0$ in this case for any f. Combined with our work done so far, we deduce that in this case $Z(f,\chi)$ is in fact an entire function!
Thus in subsequent analysis we assume that χ is trivial on the norm-1 ideles, or "unramified", in an analogous sense to Definition 2.1.2. That is to say, $\chi = |\cdot|_{\mathbb{A}_K}^s$, for some $s \in \mathbb{C}$. Thus, we get:

$$\int_{C_K^1} \chi(tx)\, \mathrm{d}^\times x \, = \int_{C_K^1} |xt|_{\mathbb{A}_K}^s \, \mathrm{d}^\times x \, = \mathrm{Vol}(C_K^1)t^s.$$

Substituting the above into our remainder factor we get:

$$\begin{aligned}\mathscr{R}(f,\chi) &= \int_0^1 \left(\widehat{f}(0)\,\mathrm{Vol}(C_K^1)(t^{-1})^{1-s} - f(0)\,\mathrm{Vol}(C_K^1)t^s\right)\frac{\mathrm{d}t}{t}.\\ &= \mathrm{Vol}(C_K^1)\int_0^1 \left(\widehat{f}(0)t^{s-1} - f(0)t^s\right)\frac{\mathrm{d}t}{t}.\end{aligned}$$

Applying the standard power rule of integration, we get that

$$\mathscr{R}(f,\chi) = \mathrm{Vol}(C_K^1)\left(\frac{\widehat{f}(0)}{s-1} - \frac{f(0)}{s}\right). \tag{3.2-7}$$

This is visibly meromorphic on $\mathbb{C}$ with the appropriate residues at the appropriate simple poles as mentioned in part (b) of the theorem in the number field case.
When K is instead a function field, let t_n denote a representative of a coset of $\mathbb{I}_K^1$ of norm q^{-n}. Recall from our discussion after $3.2-4$ the convention of splitting up t_0 across both integrals, but we ignore this term since it visibly does not contribute to convergence (or lack thereof) nor any poles.

$$\begin{aligned}\mathscr{R}(f,\chi) &= \sum_{n=1}^\infty \left(\widehat{f}(0)\chi^\vee(t_n^{-1})\int_{C_K^1}\chi^\vee(x)\,\mathrm{d}^\times x \, - f(0)\chi(t_n)\int_{C_K^1}\chi(x)\,\mathrm{d}^\times x\right)\\ &= \sum_{n=1}^\infty \mathrm{Vol}(C_K^1)\left(\widehat{f}(0)\left(q^{-n}\right)^{s-1} - f(0)\left(q^{-n}\right)^s\right)\\ &= \mathrm{Vol}(C_K^1)\left(\frac{\widehat{f}(0)}{1-q^{1-s}} - \frac{f(0)}{1-q^s}\right),\end{aligned} \tag{3.2-8}$$

where the final line follows from the standard geometric series identity. Here we also see we have the appropriate residues at the relevant (simple!) poles, thus proving part (b) of the theorem in the function field case.
Finally we need to show the functional equation. The two key identities needed for this are

$$\widehat{\widehat{f}}(x) = f(x) \;\text{ and }\; (\chi^\vee)^\vee = \chi,$$

where the second follows by definition and the first follows by self duality of $\mathbb{A}_K$ with respect to the Tamagawa measure (Corollary 1.2.10). Using this observation and combining Eqns 3.2-4 and 3.2-6, we get

$$\begin{aligned} Z(f,\chi) &= \int_1^\infty Z_t(f,\chi)\frac{\mathrm{d}t}{t} + \int_1^\infty Z_t(\widehat{f},\chi^\vee)\frac{\mathrm{d}t}{t} + \mathscr{R}(f,\chi) \\ &= \int_1^\infty Z_t(\widehat{\widehat{f}},(\chi^\vee)^\vee)\frac{\mathrm{d}t}{t} + \int_1^\infty Z_t(\widehat{f},\chi^\vee)\frac{\mathrm{d}t}{t} + \mathscr{R}(\widehat{f},\chi^\vee) \\ &= Z(\widehat{f},\chi^\vee), \end{aligned}$$

where in the penultimate line we use the observation from our explicit computations in Eqns 3.2-7 and 3.2-8 that $\mathscr{R}(f,\chi)$ is invariant to replacing f with $\widehat{f}$ and χ with $\chi^\vee$. This completes the proof of the functional equation, and in the words of Tate's thesis itself, *"our Main Theorem is proved!"* □

Section 4

Examples and Applications

§4.1 Examples and Corollaries of the Main Theorem

Here we go through a number of immediate results that follow from the main result of the prior section. Throughout, K denotes a global field.

Proposition 4.1.1. Let S be a finite set of places of K containing the infinite ones. Then the product

$$Z_S(s) := \prod_{v \notin S} \frac{1}{1 - N(\mathfrak{p}_v)^{-2}}$$

is absolutely convergent for $\mathfrak{Re}\,(s) > 1$, and moreover $(s-1)Z_S(s)$ tends to a finite positive limit as $s \to 1$.

Proof. □

Proposition 4.1.2. When K has positive characteristic and field of constants q, the image of $\mathbb{I}_K$ under the idelic norm is exactly $q^{\mathbb{Z}}$

Proof. It is clearly some finite index subgroup; it remains to show that we can generate an idele of norm q. We will be able to do this so long as the residue fields of all the completions give us enough freedom to do so. More precisely, if κ_v is the residue field of K_v (up to choice of isomorphism), we want to show that

$$\gcd(\{[\kappa_v : \mathbb{F}_q] : v \in \mathscr{P}_K\}) = 1.$$

Let this gcd be denoted d. Then each κ_v contains a copy of $\mathbb{F}' := \mathbb{F}_{q^d}$. Let X be a geometrically connected smooth projective curve with function field K, then consider

$$X' := X \otimes_{\mathbb{F}_q} \mathbb{F}',$$

which is a finite etale connected cover of X that has d points $v_1, \dots v_d$ over each points $v \in X$. Thus, we have

$$Z(f, \chi)$$

□

§4.2 Hecke L-functions and Global Epsilon Factors

The general theory of L-functions on a global field K is of great interest to the number theorist. One of the primary merits in Tate's approach is the insight into gamma factors and global root numbers of generalized ideal class characters that becomes available via the adelic language. To explore this we will first need to "globalize" the ϵ-factors discussed in Section 2.

DEFINITION 4.2.1. Let $(\psi = \prod \psi_v, \mathrm{d}x = \prod \mathrm{d}x_v)$ be a self-dual pair on $\mathbb{A}_K$. Then we define the **global epsilon factor** to be

$$\epsilon(\chi) := \prod_{v \in \mathscr{P}_K} \epsilon(\chi_v, \psi_v, \mathrm{d}x_v).$$

PROPOSITION 4.2.2. The global ϵ-factor in 4.2.1 is a nonvanishing and holomorphic function on $\mathscr{X}(C_K)$, independent of choice of self-dual pair $(\psi, \mathrm{d}x)$.

Proof. We first discuss independence of choice of character and measure. Pick an adelic character and Haar measure

$$\psi = \prod\nolimits_{v \in \mathscr{P}_K} \psi_v, \quad \mathrm{d}x = \prod\nolimits_{v \in \mathscr{P}_K} \mathrm{d}x_v,$$

where we recall from §1.2 that ψ_v is trivial on $\mathcal{O}_v$ and $\mathrm{d}^\times x_v$ assigns $\mathcal{O}_v$ measure 1 for almost all places v. Recall Proposition 1.2.11 tells us $\widehat{\mathbb{A}_K/K} \cong K$, so any other additive character has the form $\psi_b = \psi(bx)$, for some $b \in K^\times$. At each local component, viewing b as an element of $K_v^\times$ under the canonical inclusion, we have seen that the self-dual local Haar measure relative to $\psi_{v,b}$ is given by $|b|_v^{1/2}\,\mathrm{d}x_v$. By Proposition 2.2.1 (using Corollary 2.2.3 to apply it to the ϵ-factors), we get:

$$\epsilon(\chi_v, \psi_{v,b}, |b|_v^{1/2}\,\mathrm{d}x_v) = \chi_v(b)\,|b|_v^{-1/2}\,\epsilon(\chi_v, \psi_v, \mathrm{d}x_v).$$

Applying this over the product (which also verifies that the restricted product of these modified local measures is a valid measure globally) gives us:

$$\begin{aligned}\prod_{v \in \mathscr{P}_K} \epsilon(\chi_v, \psi_{v,b}, |b|_v^{1/2}\,\mathrm{d}x_v) &= \prod_{v \in \mathscr{P}_K} \chi_v(b)\,|b|_v^{-1/2}\,\epsilon(\chi_v, \psi_v, \mathrm{d}x_v) \\ &= \chi(b)\,|b|_{\mathbb{I}_K}^{-1/2} \prod_{v \in \mathscr{P}_K} \epsilon(\chi_v, \psi_v, \mathrm{d}x_v) \\ &= \prod_{v \in \mathscr{P}_K} \epsilon(\chi_v, \psi_v, \mathrm{d}x_v),\end{aligned}$$

where the final equality follows from the fact that χ is trivial on $K^\times$ and the product formula (1.3.10) telling us $|b|_{\mathbb{I}_K} = 1$. Thus, $\epsilon(\chi)$ is independent of choice of self-dual pair.

Firstly, examining our record of all local ϵ-factors in Table 2.2.B, (combined with Proposition 2.2.1) we see that they are non-vanishing and holomorphic for any choice of local $\chi_v, \psi_v, \mathrm{d}x_v$, so it remains only to check convergence.

Any $\chi \in \mathscr{X}(C_K)$ is unramified at almost all places, so examining the fourth row of Table 2.2.B tells us that at cofinitely many v the local epsilon factor is of the form

$$q^{|m_v|s}\,\mathrm{Vol}(\mathfrak{p}_v^{m_v}, \mathrm{d}x_v),$$

where $\mathfrak{p}_v^{m_v}$ is the conductor of ψ_v. However, we also recall that any adelic character must have trivial conductor (i.e $m_v = 0$) at almost all places, and that any adelic measure must assign $\mathcal{O}_v$ volume 1 at almost all places. Thus, for any χ, the product $\epsilon(\chi)$ reduces (locally, on $\mathscr{X}(C_K)$) to a finite product of non-vanishing holomorphic functions, thereby proving our claim. □

With this behind us, we may proceed to globalize the L-functions. Take some idele-class quasicharacter $\chi \in \widehat{C_K}$, and – viewing it as a character on $\mathbb{I}_K$ – recall we can consider the local restrictions χ_v for each place v.

Restricting ourself to the finite places, we have the **local** L-factors defined in 2.1-1, which we restate here for convenience:

$$L(\chi_v) := \begin{cases} (1-(\chi_v(\pi_v)))^{-1}, & \chi \text{ unramified,} \\ 1, & \text{otherwise.} \end{cases}$$

We can then define the global L-function by taking the product of all the local factors, i.e.

$$L(\chi) := \prod_{v \in \mathscr{P}_K} L(\chi_v). \tag{4.2-1}$$

We note this is well-defined since any idelic quasicharacter is trivial on $\mathcal{O}_v$ for almost all v, hence unramified on that component.

In the previous proposition and the following one, as with the Global and Local theory of §??, we consider the global ϵ-factor and L-function as having inputs on the Riemann surface $\mathscr{X}(C_K)$ of equivalence classes of idele-class quasi-characters. Moreover we also recall the **different ideal** $\mathfrak{d}_K$ of a global field K (which in the function field case depends on a choice of separating transcendence basis $\{t\}$), whose norm, in the number field case, recovers the discriminant of K.

> Proposition 4.2.3. For an idele-class quasicharacter χ the global L-function $L(\chi)$ converges when $\mathfrak{Re}(\chi) > 1$, and has a meromorphic continuation to all of $\mathscr{X}(C_K)$, holomorphic outside of $\chi = \mathbb{1}$ and $\chi = |\cdot|_{\mathbb{I}_K}$, where it has respective residues $-\operatorname{Vol}(C_K^1)$ and $(N\mathfrak{d}_K)^{-1/2}\operatorname{Vol}(C_K^1)$. Moreover, it satisfies the following functional equation:
>
> $$L(\chi) = \epsilon(\chi)L(\chi^\vee).$$

Proof. Let $(\psi, \mathrm{d}x)$ be the self-dual pair of the standard character and Tamagawa measure with respect to which we defined the Fourier transform, all the way back in Eqn. 3.1-2. If we can show the desired properties of the global L-functions with respect to this pair, For each place v, we choose some $f_v \in \mathscr{S}(K_v)$ such that $f_v = \mathbb{1}_{\mathcal{O}_v}$ for cofinitely many places, and consider the adelic Schwartz-Bruhat function $f := \bigotimes_v f_v \in \mathscr{S}(\mathbb{A}_K)$. We also know that $\chi_v|_{\mathcal{O}_v} = 1$ for cofinitely many places as well, and thus $f_v\chi_v$ and $\widehat{f_v}\chi_v^\vee$ are the characteristic functions of $\mathcal{O}_v$ for cofinitely many v.

Thus the conditions for Proposition 3.1.1 apply (i.e the integral over the restricted product is the product of the component-wise integrals, so we get) to give us:

$$\begin{aligned} Z(f,\chi) &:= \int_{\mathbb{I}_K} f(x)\chi(x)\,\mathrm{d}^\times x \\ &= \prod_v \int_{K_v^\times} f_v(x)\chi_v(x)\,\mathrm{d}^\times x_v \\ &= \prod_v Z(f_v,\chi_v) \end{aligned} \tag{4.2-2}$$

for $\mathfrak{Re}(\chi) > 1$. Similarly, using Proposition 3.1.3 to assure us that the Fourier transform behaves the way we expect it to, we also get that

$$Z(\widehat{f},\chi^\vee) = \prod_v Z(\widehat{f_v},\chi_v^\vee), \tag{4.2-3}$$

on the domain $\mathfrak{Re}(\chi < 0)$. Thus the domains on which the product decompositions are valid are disjoint. However, recall part (b) of Theorem 3.2.4 tells us $Z(f,\chi)$ may be meromorphically continued to the domain $\mathfrak{Re}(\chi) < 0$.

Thus, we may write:

$$\begin{aligned} Z(f,\chi) &= Z(\widehat{f},\chi^\vee) \\ &= \prod_{v\in\mathscr{P}_K} Z(\widehat{f_v},\chi_v^\vee) \\ &= \prod_{v\in\mathscr{P}_K} \epsilon(\chi_v,\psi_v,\mathrm{d}x_v)\frac{L(\chi_v^\vee)}{L(\chi_v)}Z(f_v,\chi_v) \\ &= \epsilon(\chi)\frac{L(\chi^\vee)}{L(\chi)}\prod_{v\in\mathscr{P}_K} Z(f_v,\chi_v) \end{aligned}$$

Using the global functional equation in part (c) of Theorem 3.2.4, we may equate Eqns. 4.2-2 and 4.2-3 to deduce that:

$$1 = \frac{\prod_v Z(\widehat{f_v},\chi_v^\vee)}{\prod_v Z(f_v,\chi_v)}.$$

Substituting in the local functional equation on each component as described in part (b) of Theorem 2.1.5, we get:

$$1 = \prod_v \gamma(\chi_v) = \prod_v \epsilon(\chi_v)\frac{L(\chi_v^\vee)}{L(\chi_v)}.$$

Multiplying out the denominator we get

$$L(\chi) = \epsilon(\chi)L(\chi^\vee),$$

proving the functional equation and the meromorphicity of $L(\chi)$ on the given domain, which follows as a consequence of the meromorphicity of the global zeta function $Z(f,\chi)$.

It remains to compute the residues. Since the L-function is independent of f, we may proceed with any nonzero $f\in\mathscr{S}(\mathbb{A}_K)$, and we choose one by letting $f_v = \mathbb{1}_{\mathcal{O}_v}$ for all components v. Then $f(0)=1$, so part (b) of the main theorem on the global functional equation (3.2.4) tells us the residue at 0 is $-\operatorname{Vol}(C_K^1)$.

For the other pole at 1, we recall the proof of Proposition 1.2.15, in which we computed

$$\widehat{f_v} = (N\mathfrak{D}_v)^{-1/2}\mathbb{1}_{\mathfrak{D}_v^{-1}}.$$

Thus we deduce

$$\widehat{f}(0) = \prod_v (N\mathfrak{D}_v)^{-1/2} = |\mathfrak{d}_K|,$$

giving us a residue of $|\mathfrak{d}_K|\operatorname{Vol}(C_K^1)$, as desired. □

Classically, the L-function is written in the manner given in the following definition.

Definition 4.2.4. For χ an idele-class quasi-character, the **Hecke L-function** is

$$L(s,\chi) := L(\chi|\cdot|^s).$$

In the above notation, the functional equation proved in 4.2.3 is given by

$$L(s,\chi) = \epsilon(s,\chi)L(1-s,\chi^{-1})$$

This L-function in itself is a generalization of something that is a much more natural extension of the classical Riemann-zeta function, and the results we have proven thus far may be applied to these.

§4.3 Dedekind Zeta Functions

Definition 4.3.1. For K a global field, the global L-function with $\chi = 1$ is called the **Dedekind zeta function**, and denoted

$$\zeta_K(s) = \prod_{v \text{ finite}} \frac{1}{1 - |\pi_v|_v^s}.$$

We note that unique prime factorization of ideals in a Dedekind domains tells us, via a computation similar to classical Euler expansion,

$$\zeta_K(s) = \prod_{v \text{ finite}} \frac{1}{1-(N(\mathfrak{p}_v))^{-s}} = \prod_{v \text{ finite}} \left(\sum_{n \in \mathbb{N}} \frac{1}{N(\mathfrak{p}_v)^{ns}} \right) = \sum_{\mathfrak{a} \trianglelefteq \mathcal{O}_K} \frac{1}{N(\mathfrak{a})^s}. \tag{4.3-1}$$

Example 4.3.2. When $K = \mathbb{Q}$ then Eqn. 4.3-1 in the above observation tells us

$$\zeta_{\mathbb{Q}}(s) = \sum_{I \trianglelefteq \mathbb{Z}} \frac{1}{|\mathbb{Z}/I|} = \sum_{n \geq 1} \frac{1}{n^s},$$

i.e. we recover the classical Riemann zeta function.

Joke. An underclassman once asked me "what is a zeta function?" I gave a recursive definition: first, we declare the Riemann zeta function to be a zeta function. Inductively, a zeta function is anything that looks like a zeta function.

Our goal will be to give explicit characterizations of the Dedekind zeta function for both the number field and function field case. We begin with the case of the number fields. Before we do so, however, we need to recall a few objects from algebraic number theory.

Suppose K has r_1 real embeddings and r_2 complex embeddings, each corresponding to an infinite place $v \in S_\infty := \mathscr{P}_{K,\infty}$. By Dirichlet's unit theorem (Example 1.4.3, which itself is a consequence of the generalized S-unit theorem), the unit group $\mathcal{O}_K^\times$ has rank $r-1$, where $r := r_1 + r_2$.

Definition 4.3.3. For K a number field, let $u_1, \dots u_{r-1}$ be a set of generators for $\mathcal{O}_K^\times/\mu(K)$, and for each $v \in S_\infty$, let N_v be 1 if v is real and 2 if it is complex. Consider the $(r-1) \times r$ matrix $M := [N_v \log(|u_i|_v)]_{v \in S_\infty, i \in [r-1]}$, and let M' be the $(r-1) \times (r-1)$ matrix obtained by deleting any one column of M. Then the **regulator** of K is defined as $\det(M)$.

Essentially, the regulator is a measure of the "relative size" of the group of units of a number field. The larger the regulator, the "fewer" units there are. That the regulator is well-defined is a consequence of the fact that the sum of any row is zero, which itself is a consequence of Lemma 4.3.5 which is used in the proof of the main theorem in the number field case. Geometrically, what is happening is we are taking embedding the group of units as a lattice in a hyperplane in $\mathbb{R}^{r_1+r_2}$, and then taking its covolume (equivalently, the volume of the hyerplane quotiented out by the lattice).

Finally before the theorem statement, we recall the local L-factors in the Archimedean cases, which we write here as:

$$\Gamma_{\mathbb{C}}(s) := (2\pi)^{-s}\Gamma(s), \quad \Gamma_{\mathbb{R}}(s) := \pi^{-s/2}\Gamma(s/2),$$

where Γ is the classical analytic extension of the factorial function.

THEOREM 4.3.4. Let K be a number field with r_1 real places, r_2 imaginary places, root of unity group $\mu(K)$, regulator R_K, class number h_K, and discriminant $\mathfrak{d}_K$. Then the function $\zeta_K(s)$ is holomorphic except for simple poles at $s = 0$ and 1 with respective residues $-c_K$ and $|\mathfrak{d}_k|^{-1/2} c_K$, where

$$c_K := \frac{2^{r_1}(2\pi)^{r_2} h_k R_k}{|\mu(K)|}.$$

By Proposition 4.2.3 detailing the properties of the global L-function, we see that all we have left to do is show that $\mathrm{Vol}(C_K^1)$ (with respect to the measure induced appropriately scaled $\mathrm{d}^\times x$ on $\mathbb{I}_K$ and the counting measure on K) is equal to c_K. Thus our work is reduced to computing this volume.

Let S denote any finite set of places of K, and recall from 1.4.1 the S-ideles $\mathbb{I}_{K,S}$. We define the *S-idelic logarithm map* by

$$\begin{aligned} \mathscr{L}_S : \mathbb{I}^1_{K,S} &\to \mathbb{R}^{|S|} \\ (x_v)_v &\mapsto (\log |x_v|_v)_{v \in S}. \end{aligned} \tag{4.3-2}$$

LEMMA 4.3.5. When $S = S_\infty$, define the hyperplane

$$H_S := \left\{ (t_v)_{v \in S} : \sum_{v \text{ real}} t_v + 2 \sum_{v \text{ complex}} t_v = 0 \right\}.$$

Then $\mathrm{Im}(\mathscr{L}_S) = H_S$ and $\ker(\mathscr{L}_S) = \mathbb{I}_{K,\varnothing}$.

Proof. By definition, $\prod_{v \in S} |x_v|_v = 1$, so thus $\sum_{v \in S} \log(|x_v|_v) = 0$, which is precisely the polynomial defining H (recall our normalized absolute value on $\mathbb{C}$ defined in 1.3.8 is the square of the standard one). Moreover given any $(t_v)_v \in H$ we can simply construct an element of $\mathbb{I}^1_{K,S}$ by setting x_v of absolute value e^{t_v} for $v \in S$ and picking some $x \in \mathcal{O}_v^\times$ for all the $v \notin S$. This proves the image of the logarithm is exactly H.
For the kernel, note $(x_v)_v \in \ker(\mathscr{L}_S)$ if and only if $|x_v|_v = 1$ (which is true if and only if $x_v \in \mathcal{O}_v^\times$) for all $v \in S$. Thus, $(x_v)_v \in \prod_v \mathcal{O}_v^\times = \mathbb{I}^1_{K,\varnothing} = \mathbb{I}_{K,\varnothing}$. □

REMARK. For the reasons discussed following Definition 4.3.3, Ramakrishnan [RV98] called the restrict of $\mathscr{L}_{S_\infty}$ to $\mathcal{O}_K^\times$ the *regulator map*, though this terminology is nonstandard.

We are now ready for the proof of the zeta function.

Proof of 4.3.4. Let $S = S_\infty$ be the (nonempty, since we are in characteristic zero) set of Archimedean places of K. Recall the S-class group $C_{K,S}$, which, from the observation following Definiton 1.4.6, using the fact that S is nonempty we may write

$$C_{K,S} = \mathbb{I}^1_K / (K^\times \cdot \mathbb{I}^1_{K,S}).$$

We make the observation that the kernel of the projection $C_K^1 \twoheadrightarrow C_{K,S}$ is precisely $\mathbb{I}^1_{K,S}/(K^* \cap \mathbb{I}^1_{K,S})$. By Proposition, 1.4.7, since S is nonempty we deduce $C_{K,S}$ is a finite group, hence we have that

$$\mathrm{Vol}(C_K^1) = |C_{K,S}| \cdot \mathrm{Vol}(\mathbb{I}^1_{K,S}/(K^* \cap \mathbb{I}^1_{K,S})). \tag{4.3-3}$$

Since $S = S_\infty$, we may recall Example 1.4.8, where we showed that $|C_{K,S}| = h_K$, where h_K is the class number of K (i.e the size of the ideal class group). It remains to compute the volume of the second factor.

Note that $\mathcal{O}_K^\times$ can be viewed as the diagonal embedding of $K^\times$ into $\mathbb{I}_{K,S}^1$. By Lemma 4.3.5, we see that

$$\ker(\mathscr{L}_S|_{\mathcal{O}_K^\times}) = \mathcal{O}_{K,\varnothing}^\times = \mu(K),$$

where the second equality follows from the generalized S-unit theorem (Theorem 1.4.2). This gives us the exact sequence

$$1 \to \mu(K) \to \mathcal{O}_K^\times \to \mathscr{L}_S(\mathcal{O}_K^\times) \to 1. \tag{4.3-4}$$

Since K embeds discretely into any $\mathbb{I}_{K,S}^1$, it follows that $\mathscr{L}_S(\mathcal{O}_K^\times)$ is a discrete subgroup of $\mathscr{L}_S(\mathbb{I}_{K,S}^1) = H_S$, as defined in 4.3.5. Moreover recall from 1.4.5 that $\mathbb{I}_{K,S}^1/\mathcal{O}_{K,S}$ is compact, and since $H_S, \mathscr{L}_S(\mathcal{O}_K^\times)$ are the continuous images of $\mathbb{I}_{K,S}^1$ and $\mathcal{O}_{K,S}(= \mathcal{O}_K^\times)$ respectively, we deduce the quotient $H_S/\mathscr{L}_S(\mathcal{O}_K^\times)$ is also compact. This gives us the exact sequence

$$0 \to \mathscr{L}_S(\mathcal{O}_K^\times) \to H_S \to H_S/\mathscr{L}_S(\mathcal{O}_K^\times) \to 0. \tag{4.3-5}$$

Combining Eqns. 4.3-4 and 4.3-5 together along with all of our prior observations, we get the following grid of exact sequences:

$$\begin{tikzcd}
 & 1 \arrow[d] & 1 \arrow[d] & 0 \arrow[d] & \\
1 \arrow[r] & \mu(K) \arrow[r] \arrow[d] & \mathcal{O}_K^\times(= K \cap \mathbb{I}_{K,S}^1) \arrow[r] \arrow[d] & \mathscr{L}_S(\mathcal{O}_K^\times) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & \mathbb{I}_{K,\varnothing}^1 \arrow[r] \arrow[d] & \mathbb{I}_{K,S}^1 \arrow[r] \arrow[d] & H_S \arrow[r] \arrow[d] & 0 \\
1 \arrow[r] & \mathbb{I}_{K,\varnothing}^1/\mu(K) \arrow[r] \arrow[d] & \mathbb{I}_{K,S}^1/(K \cap \mathbb{I}_{K,S}^1) \arrow[r] \arrow[d] & H_S/\mathscr{L}_S(\mathcal{O}_K^\times) \arrow[r] \arrow[d] & 0 \\
 & 1 & 1 & 0 &
\end{tikzcd}$$

It follows that the desired volume we want is given by:

$$\mathrm{Vol}(\mathbb{I}_{K,S}^1/(K \cap \mathbb{I}_{K,S}^1)) = \mathrm{Vol}(H_S/\mathscr{L}_S(\mathcal{O}_K^\times)) \cdot \mathrm{Vol}(\mathbb{I}_{K,\varnothing}^1/\mu(K))$$

Examining the definition of the regulator and the S-idelic logarithm map, we see that $\mathscr{L}_S(\mathcal{O}_K^\times)$ is precisely the embedding of the unit group as a lattice into H_S, and that thus the determinant of the matrix in Definition 4.3.3 is equivalent to the covolume of this lattice. Hence,

$$\mathrm{Vol}(H_S/\mathscr{L}_S(\mathcal{O}_K^\times)) = R_K. \tag{4.3-6}$$

To compute the measure of $\mathbb{I}_{K,\varnothing}^1/\mu(K)$, we use the fact that $\mu(K)$ is a finite subgroup to see that the measure of the quotient is simply given by $\mathrm{Vol}(\mathbb{I}_{K,\varnothing}^1)/|\mu(K)|$. It thus remains to compute this volume. For $v \in S_\infty$, define $U_v := \{\pm 1\}$ if v is real and S^1 if v is complex. Note then that we have

$$\mathbb{I}_{K,\varnothing}^1 = \prod_{v \in S_\infty} U_v \times \prod_{v < \infty} \mathcal{O}_v^\times \subseteq \mathbb{I}_{K,\infty}^1 \subseteq \mathbb{I}_{K,\infty}.$$

First we determine the measure on $\mathbb{I}_{K,\infty}$ (not just norm-1)!. Recall that our measure we placed on the finite components of the ideles using the scaling constant of Eqn. 3.2-2 that gives the unit group volume 1 at all the unramified finite places. Thus if we let T be the set of all infinite and ramified finite places, if $\mathrm{d}^\times x$ denotes the standard restricted product measure on the ideles, it follows that:

$$\mathrm{d}^\times x\,|_{\mathbb{I}_{K,\infty}} = \mathrm{d}^\times x_T = \prod_{v\in T} \mathrm{d}^\times x_v\,.$$

In other words, the Haar measure on $\mathbb{I}_{K,\infty}$ is literally just the product of the normalized measure at all the components in T. At all the ramified finite places, we have

$$\prod_{v<\infty \text{ ramified}} \mathrm{Vol}(\mathcal{O}_v^\times, \mathrm{d}^\times x_v) = \prod_{v<\infty \text{ ramified}} N(\mathfrak{p}_v)^{-1/2} = |\mathfrak{d}_K|^{-1/2}.$$

For the infinite places, The Haar measure on U_v should be precisely the one that recovers the measure on $\mathbb{I}_{K,\infty}$ when integrating along $|\mathbb{I}_K|_{\mathbb{I}_K} = \mathbb{R}_{>0}$, in the sense of Fubini. Since the measure on $\mathbb{I}_{K,\infty}$ was seen to be the product of the normalized measures on each component, picking an infinite place $v_0 \in S_\infty$, and noting that $K_v^\times = U_v \times \mathbb{R}_{>0}$, we want to impose the measure on U_v such that

$$\int_{K_v^\times} f(x_v)\,\mathrm{d}^\times x_v = \int_{U_v}\int_{\mathbb{R}_{>0}} f(x_v)\frac{\mathrm{d}t}{t}\,\mathrm{d}u = \int_{\mathbb{R}_{>0}}\int_{U_v} f(x_v)\,\mathrm{d}u\,\frac{\mathrm{d}t}{t}$$

for any $f \in L^1(K_v^\times)$. Since this is the standard Lebesgue measure when $K_v^\times = \mathbb{R}^\times$, we have to give $U_v = \{\pm 1\}$ the counting measure at the real places (we equally add up the volume along the positive and negative half of the real line), and since this is twice the Lebesgue measure when $K_v^\times = \mathbb{C}^\times$, we have to give $U_v = S^1$ measure 2π (the standard Lebesgue measure gives a circle area πr^2). Thus $\mathrm{Vol}(U_v) = 2$ if v is real and 2π if v is complex. Combining these we get

$$\mathrm{Vol}(\mathbb{I}^1_{K,\varnothing}) = \mathrm{Vol}\left(\prod_{v\in S_\infty} U_v \times \prod_{v<\infty \text{ ramified}}\right) = (2)^{r_1}(2\pi)^{r_2}|\mathfrak{d}|^{-1/2}. \tag{4.3-7}$$

Putting together our observations and computations in Eqns 4.3-3, 4.3-6 and 4.3-7, we get

$$\mathrm{Vol}(C_K^1) = \mathrm{Vol}(\mathbb{I}^1_{K,S}/(K\cap \mathbb{I}^1_{K,S})) = \frac{2^{r_1}\cdot(2\pi)^{r_2} h R_K}{|\mu(K)|\,|\mathfrak{d}_K|^{1/2}},$$

as desired. □

We now proceed to prove the corresponding result in the positive characteristic case.

Theorem 4.3.6. Let K be a global field of positive characteristic with field of constants $\mathbb{F}_q$ and genus g. Then

$$\zeta_K(s) = \frac{P(q^{-s})}{(1-q^{-s})(1-q^{1-s})},$$

where $P \in \mathbb{Z}[t]$ of degree $2g$, with $P(0) = 1$ and $P(1) = |\mathrm{Pic}^0(K)|$.

Before we proceed to the proof we will need a notion of a differential idele; we recall the language of divisors used in §3.1. In particular, we recall from Eqn. 3.1-5 the *divisor map* on the ideles:

$$\mathscr{D} : (x_v)_v \mapsto \sum_v \mathrm{ord}_v(x_v)\cdot v.$$

DEFINITION 4.3.7. We say an idele $x = (x_v)_v \in \mathbb{I}_K$ is a **differential idele** if $\mathscr{D}(x)$ is a canonical divisor (i.e if $\mathrm{ord}_v(x) = \mathrm{ord}_v(\psi)$ for all v, for some quasicharacter ψ of $\mathbb{A}_K$ trivial on K).

Recall that the geometric Riemann-Roch theorem (3.1.16) tells us that any differential idele has degree $2g-2$ as a divisor, and therefore $\|x\|_{\mathbb{A}_K} = q^{2g-2}$.

Proof. Since $|\mathbb{I}_K|_{\mathbb{A}_K} = q^{\mathbb{Z}}$ and the global L-function in this case takes $|\cdot|^s_{\mathbb{A}_K}$ as an input, we immediately see from Proposition 4.2.3 that $\zeta_K(s)$ may be written as a meromorphic function $f(q^{-s})$, with simple poles at 1 and q^{-1}, corresponding to $s = 0$ and 1 in the proposition statement respectively.

Recall here that by convention we have our single infinite (but still non-Archimedean!) place of K which we omit from our sum. By mimicking our Euler expansion in Example 4.3.2, we see when $K = \mathbb{F}_p(t)$ that we have

$$\zeta_K(s) = \sum_{\text{monic } x \in \mathbb{F}_p[t]} p^{-\deg(x)s} = \sum_{d \geq 0} p^{d(1-s)} = \frac{1}{1-p^{1-s}}.$$

This is clearly greater than 1 for any s, and for general K of degree n over $\mathbb{F}_p(t)$ we may bound $\zeta_k(s)$ in absolute value from above by $\left|\zeta_{\mathbb{F}_p(t)}(s)\right|^n$. In any event, the expression makes it clear that $\zeta_{\mathbb{F}_p(t)}(s) \to 1$ as $\mathfrak{Re}(s) \to \infty$, and the same must therefore hold for any K.

Thus, our meromorphic (away from 1 and q^{-1}) function $f(q^{-s})$ tends to 1 as $q^{-s} \to 0$, and so it must be holomorphic at 0, and be equal to 1 there. From all these and the standard characterization of complex analytic functions we deduce we must have

$$f(x) = \frac{P(x)}{(1-x)(1-qx)},$$

for P an holomorphic function on the complex plane.

Recall that the functional equation of the global L-function (4.2.3) gives us in this case:

$$\zeta_K(1-s) = \epsilon(s,1)\zeta_K(s).$$

Letting $x = q^{-s}$, this tells us that

$$P(1/(qx)) = \epsilon(s,1)P(x)$$

Noting that in this case our multiplicative character $\chi = 1$ is trivial and thus $\mathrm{ord}_v(\chi_v) = 0$ at all v, we recall from the proof of the local functional equation the expression for the local ϵ-factor $\epsilon(1_v)$ (from Eqn 2.1-14), given by

$$\epsilon(s, 1_v) = q^{\mathrm{ord}_v(\psi_v)s}\, \mathrm{Vol}(\mathfrak{p}_v^{\mathrm{ord}_v(\psi_v)}, \, dx_v\,).$$

Here the Haar measure dx_v must be chosen as to be self-dual with respect to our character ψ_v. This is exactly the one that sets the volume of $\mathcal{O}_v$ to be $N(\mathfrak{p}_v^{\mathrm{ord}_v(\psi_v)})^{-1/2} = q^{-\mathrm{ord}_v(\psi_v)/2}$. It thus follows that

$$\epsilon(s, 1_v) = q^{\mathrm{ord}_v(\psi_v)(s-1/2)},$$

and so the global ϵ-factor is given by

$$\epsilon(s,1) := \prod_v \epsilon(s, 1_v) = q^{(s-1/2)(-\sum_v \mathrm{ord}_v(\psi_v))} = q^{(s-1/2)(2g-2)},$$

using the fact that $\mathrm{div}(\psi)$ is by definition a canonical divisor, and thus has genus $2g-2$ by our geometric Riemann-Roch theorem. Thus we have:

$$\frac{P(1/(xq))}{(1-q^{s-1})(1-q^s)} = x^{-2g}q^{-g}q^{1-2s}\frac{P(x)}{(1-q^{-s})(1-q^{1-s})}.$$

Multiplying both sides by $1/(q^{1-2s})$ gives us:

$$\begin{aligned} x^{-2g}q^{-g}\frac{P(x)}{(1-q^{-s})(1-q^{1-s})} &= \frac{1}{q^{1-s}\cdot q^{-s}}\frac{P(1/(xq))}{(1-q^{s-1})(1-q^{s})} \\ &= \frac{P(1/xq)}{(q^{1-s}-1)(q^{-s}-1)}. \end{aligned}$$

We can cancel out the denominators, since each factor on the right side is off by a factor of -1, giving us:

$$P(1/(xq)) = x^{-2g}q^{-g}P(x).$$

Note as $|x| \to \infty$, we have $P(1/xq) \to P(0) = 1$, as already shown. Thus we have $|P(x)| = O(|x^{2g}|)$ for x or large enough norm. For any $a \in \mathbb{C}$ of large enough norm, we therefore have $|P(a)| \leq B|a|^{2g}$ for some $B \in \mathbb{R}$. Pick some $R > |a|$, and let C_R denote the circle of radius R in the complex plane. Cauchy differentiation tells us

$$\begin{aligned} \left|P^{(2g)}(a)\right| &= \left|\frac{(2g)!}{2\pi i}\int_{C_R}\frac{P(z)}{(z-a)^{2g+1}}\,\mathrm{d}z\right| \\ &\leq \frac{(2g)!}{2\pi}\left|\int_0^1 \frac{BR^{2g}}{R^{2g+1}}\,\mathrm{d}t\right| \\ &\leq CB, \end{aligned}$$

for some constant C independent of R. In other words, we have that the $2g$-th derivative of P is a bounded entire function, hence constant by Liouville. It follows that P must be a polynomial of degree $2g$, as desired.

□

Remark. Theorem 4.3.6, in particular the observation that zeta function of a smooth projective curve is rational with specific degree properties related to its genus and reciprocal roots that are powers of q is the beginning of a deep and dark forest that is the Weil Conjectures, the postulation of which lead to the development of modern algebraic geometry as we know it today. Our theorem can be seen as a proof of the Weil conjectures in the special case of a smooth projective curve.

Chapter III

Automorphic Representations (a.k.a GL_2)

[TO BE WRITTEN]

Section 5

Adelic Algebraic Groups

§5.1 Two (Equivalent) Definitions

Throughout this section we want to work with so-called "adelic algebraic groups". There are two reasonable ways to define what this type of object should look like, both with their advantages in various scenarios. The goal of this subsection, in addition to providing a swift review of the core definitions and concepts of linear algebraic groups, is to compare these two approaches to defining adelic algebraic groups and show that they coincide in the way that we want them to.

Definition 5.1.1. Let K be any field. A **linear algebraic K-group** G is a smooth affine K-scheme of finite type, equipped with K-morphisms $m : G \times G \to G$ (i.e "multiplication"), $i : G \to G$ (i.e. "inversion"), and a dsitinguished rational point $e \in G(K)$ satisfying the "group axioms", in the sense that the following diagrams should commute:

Associativity:
$$\begin{tikzcd} G \times G \times G \arrow[r, "\mathbb{1}_G \times m"] \arrow[d, "m \times \mathbb{1}_G \times"'] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"] & G \end{tikzcd}$$

Inversion:
$$\begin{tikzcd} G \arrow[rr, "\mathbb{1}_G \times i"] \arrow[dr] \arrow[dd, "i \times \mathbb{1}_G"'] & & G \times G \arrow[dd, "m"] \\ & \mathrm{Spec}(k) \arrow[dr, "e"] & \\ G \times G \arrow[rr, "m"] & & G \end{tikzcd}$$

Identity:
$$\begin{tikzcd} G \arrow[r, "(\mathbb{1}_G, e)"] \arrow[d, "(e, \mathbb{1}_G)"'] \arrow[dr, equal] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"] & G \end{tikzcd}$$

Remark. Although it is canonical to use a lowercase "k" to denote the fields over which the above objects are defined, here we remain consisten with our use of K to denote our field of interest.

Example 5.1.2 (GL_n). Consider the classical group of invertible matrices GL_n, which we may represent as an affine K-scheme by the following closed subset:

$$\mathrm{GL}_n := V(y \det(x_{ij}) - 1) \subseteq \mathbf{A}_K^{n^2+1}. \tag{5.1-1}$$

Via the classical Rabinovitsch correspondence we see that we can realize this as an open subset of $\mathbb{A}_K^{n^2}$, hence it is immediately smooth and connected. It is not too hard of an exercise to show that any connected algebraic K-group is also *geometrically* connected and irreducible (i.e remains connected and irreducible when base-changing over any finite extension of K).

Another reason that GL_n is of particular interest is the following (highly non-obvious!) theorem, which we omit the proof of.

THEOREM 5.1.3. Any linear algebraic K-group arises as a K-subgroup scheme of GL_n for some n.

Here a K-subgroup means what one intuitively should imagine it to mean: a closed group subscheme H such that the relevant morphisms commute with the inclusion map $H \hookrightarrow G$.

Similar to the above example one can construct all the classical matrix groups ($\mathrm{SL}_n, \mathrm{SO}(n)$, pick your favorite) as linear algebraic groups, as they are all defined in terms of polynomial relations over a ground field. There are a couple of examples that are so special, however, they ge their own customized notation in the setting of algebraic groups.

EXAMPLE 5.1.4 ($\mathbb{G}_m$ and $\mathbb{G}_a$). As a special case of Example 5.1.2, we denote $\mathbb{G}_m := \mathrm{GL}_1$. One can directly verify then that

$$\mathbb{G}_m(\mathbb{A}_K) = \mathbb{I}_K.$$

This is what we meant by the title of Ch. II.
Another key elementary example is $\mathbb{G}_a := \mathbb{A}^1$, which we think of as an additive group. We note we can realize this as a matrix group in the following sense:

$$\mathbb{G}_a(R) = \left\{ \begin{bmatrix} 1 & r \\ 0 & 1 \end{bmatrix} : r \in R \right\} \subseteq \mathrm{GL}_2(R),$$

which is in line with our expectations keeping Theorem 5.1.3 in mind.

Our discussion will be focused with the adelic points of these linear algebraic groups, i.e $G(\mathbb{A}_K)$ for some global field K. One could imagine, however, defining the notion of an "adelic group scheme" in a different but also seemingly natural way; one could consider the rational points $G(K_v)$ over all the places $v \in \mathscr{P}_K$ and then consider the scheme-theoretic restricted product

$$\coprod G(K_v) = \varinjlim_S \prod_{v \in S} G(K_v) \times \prod_{v \notin S} G(\mathcal{O}_v),$$

i.e the directed union over all finite sets of places S, as defined all the way back in Eqn 1.1-3. It is intuitive and expected but rigorously non-obvious that these constructions actually coincide, in the sense that they are topologically isomorphic. This will be convenient as we will be free to alter which definition we're using in any scenario without concern. Proving this fact will be the primary goal of this subsection.

One notes that when G is affine then $\Gamma(G, \mathcal{O}_G)$ can be represented as $K[t_1, \ldots, t_n]/I$, and so as a literal set we can represent $X(\mathbb{A}_K)$ as the vanishing locus

$$V(I) := \{\vec{x} : f(\vec{x}) = 0 \, \forall f \in I\} \subseteq \mathbf{A}^n_{\mathbb{A}_K} \cong \prod_{i \in [n]} \mathbb{A}_K.$$

WARNING (Notational Overload). It is unfortunately standard for the adeles and affine n-space to be typset with black-board capital A. To avoid confusion in this text we have opted for the bolded $\mathbf{A}$ for affine n-space and the standard blackboard $\mathbb{A}$ for the adeles. Moreover it is also standard to denote by $\mathcal{O}_X$ the structure sheaf of a scheme X, which is frustratingly similar to the ring of integers $\mathcal{O}_K$ notation, but discernable enough from context that we allow this overlap.

We note that $V(I)$ is naturally given a locally compact subspace topology from $\mathbf{A}^n_{\mathbb{A}_K}$. However, we want to verify that this topology is independent of our choice of presentation, i.e if we had some different n and some other ideal I such that the constructed quotient was K-isomorphic to our coordinate ring then we'd have the same topology. This is the purpose of the following lemma.

LEMMA 5.1.5. Let R be a topological K-algebra and G a linear algebraic K-group, for K a global field. Then there is a unique way to topologize $G(R)$ that is functorial in X, compatible with the formation of fiber products, carries closed immersions of schemes into topological embeddings of rings, and gives $G_{\mathbb{A}}(R) = R$ its usual topology.

Proof. Let $A := \Gamma(G, \mathcal{O}_G) \xrightarrow{\sim} R[t_1, \ldots, t_n]/I$ be a choice of (K-linear) isomorphism, and identify $G(R)$ with the elements of R^n on which all functions vanish, and imbue it with the subspace topology. □

THEOREM 5.1.6. Let G_S be a finite-type affine $\mathcal{O}_{K,S}$-scheme. Using projections from $\mathbb{A}_{K,S}$ to K_v and $\mathcal{O}_v$ for $v \in S$ and $v \notin S$ respectively, the natural map

$$G_S(\mathbb{A}_{K,S}) \to \prod_{v\in S} G_v(K_v) \times \prod_{v\notin S} G_{S,v}(\mathcal{O}_v)$$

is a topological isomorphism.

Proving the above will be the main goal of this subsection.

§5.2 Weil's Restriction of Scalars

Section 6

The Representation Theory of $\mathrm{SL}_2(\mathbb{A}_\mathbb{Q})$

Appendices

The following sections are here primarily for reference on some of the background for the main content of this text. The material here does follow a narrative flow in each section but the proofs are either sketched or omitted entirely in most places, as this content is more used as a tool for this thesis as opposed to being the main focus of our exposition.

Appendix A

Fourier Theory of Locally Compact Groups

§A.1 Introduction: Dual Groups and Haar Measure

In this section we build some of the main theory of Fourier transforms on a LCA group G, via the *dual* $\widehat{G}$, the group of continuous homomorphisms $G \to S^1$. We assume the reader has some familiarity with the classical Fourier transform (i.e the one over $\mathbb{R}$), and some elementary (i.e undergraduate-level) functional analysis and measure theory. The primary purpose of this appendix is to discuss the main ideas involved in proving the generalization of some of these classical results, including Fourier inversion and self-duality. We refer the reader to the Section on

Definition A.1.1. The **Pontryagin** [a] **Dual** of G, denoted $\widehat{G}$, is the group of all continuous complex homomorphisms $G \to S^1$ (where S^1 is given its standard subspace topology from $\mathbb{C}$), with multiplication as the group operation. Elements of $\widehat{G}$ are often referred to as **characters** of G.

[a] A Russian name, sometimes also anglicized as Pontrjagin or Pontriagin

Example A.1.2. The dual group of the circle is the integers, i.e

$$\widehat{S^1} \cong \mathbb{Z}.$$

Proof. Recall the continuous homomorphism $\exp : \mathbb{R} \to S^1$ given by $x \mapsto e^{2\pi i x}$, the so-called *universal covering map*. For any continuous homomorphism $\phi : S^1 \to S^1$, we have the following diagram:

$$\begin{tikzcd} \mathbb{R} \arrow[d, two heads, "\exp"'] \arrow[dr, "\psi"] & \\ S^1 \arrow[r, "\phi"'] & S^1 \end{tikzcd}$$

from which we see that any such ϕ must induce a continuous homomorphism $\psi = \phi \circ \exp : \mathbb{R} \to S^1$. Moreover, since the kernl of the universing covering map is exactly $\mathbb{Z}$, it follows ψ must also have kernel containing $\mathbb{Z}$.

A standard vector-space argument shows any continuous homomorphism $\mathbb{R} \to \mathbb{R}$ is of the form $x \mapsto \lambda x$ for $\lambda \in \mathbb{R}$. using the universal lift we deduce any continuous homomorphism $\mathbb{R} \to S^1$ is of the form $\psi_\lambda(x) = e^{2\pi i \lambda x}$, for $\lambda \in \mathbb{R}$. Since we want $\mathbb{Z}$ to be in the kernel we deduce $\lambda \in \mathbb{Z}$.

Thus we have:

$$\psi = e^{2\pi i n x} = \phi(e^{2\pi i x}) \implies \phi(y) = y^n$$

which was what we wanted to show. □

A similar argument to A.1.2 also in fact tells us that $\widehat{\mathbb{Z}} \cong S^1$. If you are not an algebraic topologist and this is new to you, it's a worthwhile exercise to attempt. Putting this two together we get the wonderful identities:

$$\widehat{\widehat{S^1}} \cong S^1 \text{ and } \widehat{\widehat{\mathbb{Z}}} \cong \mathbb{Z}.$$

It turns out this duality holds true for ANY topological group that is locally compact and abelian.

Theorem A.1.3 (Pontryagin Duality). The map

$$\begin{aligned} \alpha : G &\to \widehat{\widehat{G}} \\ g &\mapsto (\chi \mapsto \chi(g)) \end{aligned}$$

(i.e evalution on the dual $\widehat{G}$) is an algebraic and topological isomorphism.

Remark (Historical Aside). Lev Pontryagin's foundational work on this duality result initially imposed some auxiliary conditions of second-countabilitiy and compactness; it was later generalized to locally compact groups by van Kampen and Weil. Pontryagin's other contributions were largely in topology (he is often referred to as one of the co-founders of cobordism theory), which is especially surprising considering he was blind since the age of 14, i.e. was making huge strides in the study of shapes and spaces without being able to draw or see!

Proving Theorem A.1.3 will be the main goal of this section. The primary tool used here is Fourier analysis over these topological groups. To this end we recall some of the basic definitions and objects used to build a theory of integration over arbitrary groups.

Definition A.1.4. Any LCA (locally compact abelian) group G admits a positive regular measure μ (over the Borel σ-algebra $\mathcal{B}_G$) that is translation invariant, i.e

$$\mu(E+g) = \mu(E) \; \forall g \in G, E \in \mathcal{B}_G.$$

This is known as the **Haar measure**.

The reader will notice that the above is a proposition disguised as a definition. The reason for this is the proof of existence of this measure isn't particularly enlightening, and we leave the reader to check other resources (e.g section 1.2 of [RV98]) for the details. The key idea is the same one trick any measure theorist uses to construct a measure; construct some nontrivial translation-invariant linear functional

$$T : \mathscr{C}_c(G) \to \mathbb{R},$$

and the local compactness and Hausdorff hypotheses on G allow us to use the Riesz-representation theorem to obtain a regular measure μ on G.

The more astute reader will also have noticed the phrasing of *the* Haar measure, hinting at some sort of canonicity. The reason this is justified is the following proposition.

Proposition A.1.5. If μ, μ' are two Haar measures on a LCA group G, then $\mu = \lambda\mu'$ for some $\lambda \in \mathbb{R}_{\geq 0}$.

Proof. We refer the reader to 1.1.3 of [Rud91], but the essential idea is to use the construction via Riesz representation and apply Fubini on $G \times G$. □

§A.2 Topologizing the dual

We want to put a topology on $\widehat{G}$ that tells us if two functions are "close together". For a function $f \in \widehat{G}$, suppose we take a compact set $K \subseteq G$, and an open neighborhood $V \supseteq f(K)$ of the image of this compact set. Since S^1 is Hausdorff, $f(K)$ is a closed set. As such, we might expect that if we wiggle $f(K)$ around a little, it shoud still be inside V. Thus we want to characterize a function g as being "close" to f if $g(K) \subseteq V$ as well. This motivates the following construction.

Definition A.2.1. The **compact-open topology** on $\widehat{G}$ is the one generated by the following neighborhood base of the trivial character:

$$\left\{W(K,V) := \left\{\chi \in \widehat{G} : \chi(K) \subseteq V\right\}\right\}_{K,V}$$

where K ranges over all compact sets in G and V ranges over all neighborhoods of the identity in S^1.

Example A.2.2. If G is discrete, then the compact-open topology on $\widehat{G}$ is the same as the topology of pointwise convergence.

Proof. Recall the topology of pointwise convergence is generated by sets of the form

$$S(g,V) := \left\{\chi \in \widehat{G} : \chi(x) \in U\right\}_{g\in G;\ V\subseteq_{\text{open}} S^1}$$

For any G, this is finer than the compact-open topology, since for any compact $K \subseteq G$ we can write:

$$W(K,V) = \bigcup_{g\in K} S(g,V)$$

If G is discrete then every compact set is necessarily finite (and vice versa!), and thus $\{g\}$ is compact, so every subbasis element $S(g,U)$ is also a basis element of the compact-open topology. Thus the two topologies are equivalent. □

Proposition A.2.3 (Properties of Abelian Topological Groups). Suppose G is an abelian topological group, and let $\widehat{G}$ have the compact-open topology. Then:

(a) G discrete $\implies$ $\widehat{G}$ compact.

(b) G compact $\implies$ $\widehat{G}$ discrete.

(c) G locally compact $\implies$ $\widehat{G}$ locally compact.

Proof. This is largely an exercise in point-set topology working with the basis defined in A.2.1, and in particular also making use of the canonical universal cover projection map $\phi : \mathbb{R} \to S^1$ given by $r \mapsto e^{2\pi i r}$. Then for some $\epsilon \in (0,1]$, we can define

$$N(\epsilon) := \phi\left(\left(\frac{-\epsilon}{3}, \frac{\epsilon}{3}\right)\right)$$

i.e $N(\epsilon)$ is some small symmetric arc around 1 in S^1. In the notation of Definition A.2.1, we can set $V = N(1)$, and WLOG reduce to this case to do all our computations. We refer the interested reader to Prop 3-2 of [RV98] for a full rigorous proof. □

The above result is useful in all sorts of ways, but of prime importance to us is part (c). This allows us to use theorems and results involving local compactness (which most of them do, in the world of topological groups) on

$\widehat{G}$, which we couldn't do a-priori. In particular it will give us a Haar measure on $\widehat{G}$, so Fourier inversion will make sense.

§A.3 Fourier Inversion

Throughout this subsection we let G be a locally compact abelian group (we'll use the acronym LCAG for this) with bi-invariant Haar measure $\mathrm{d}x$, with continuous character group $\widehat{G}$. We'll Our main tool for establishing Pontryagin duality is the Fourier Inversion formula.

DEFINITION A.3.1. For $f \in L^1(G)$, we define its **Fourier transform** as:

$$\begin{aligned} \widehat{f} : \widehat{G} &\to \mathbb{C} \\ \chi &\mapsto \int_G f(y)\overline{\chi(y)}\,\mathrm{d}y \end{aligned}$$

SANITY CHECK. Verify that $\widehat{f}$ is well-defined, and that $\left|\widehat{f}(\chi)\right| \leq \|f\|_1$.

EXAMPLE A.3.2. In the case $G = \mathbb{R}$, with the addition operation, we can identify each $t \in \mathbb{R}$ with the character $s \mapsto e^{ist}$. The Fourier transform then gives us:

$$\widehat{f}(t) = \int_{\mathbb{R}} f(s)e^{-ist}\,\mathrm{d}s$$

which is the classical Fourier transform.

WARNING. Pedantically speaking, the Fourier transform in the above example is a function on $\widehat{\mathbb{R}}$ rather than $\mathbb{R}$.

Let $V(G)$ denote the complex span of the continuous functions of positive type on G, and define

$$V^1(G) := V(G) \cap L^1(G).$$

THEOREM A.3.3 (Fourier Inversion). There exists a Haar measure $\mathrm{d}\chi$ on $\widehat{G}$ such that, for all $f \in V^1(G)$,

$$f(y) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(y)\,\mathrm{d}\chi\,.$$

Moreover, the Fourier transform $f \mapsto \widehat{f}$ identifies $V^1(G)$ with $V^1(\widehat{G})$.

DEFINITION A.3.4. The measure $\mathrm{d}\chi$ is called the **dual measure** of $\mathrm{d}x$, the given Haar measure on G.

To prove the existence of the dual measure, we recall some more properties of convolution.

PROPOSITION A.3.5 (Properties of Convolution). Let f, g be $\mathbb{C}$-valued Borel functions on a LCA group G. Then:

(a) If the convolution $f * g(x)$ exists for some $x \in G$, then so does $g * f(x)$, and they are equal to one another.

(b) If $f, g \in L^1(G)$, then $f * g(x)$ exists for almost all $x \in G$. Moreover, $f * g \in L^1(G)$ and

$$\|f * g\|_1 \leq \|f\|_1 \|g\|_1 .$$

(c) If $f, g, h \in L^1(G)$ then $(f * g) * h = f * (g * h)$.

(d) Products of Fourier transforms correspond to Fourier transforms of convolutions, i.e $\widehat{f}\widehat{g} = \widehat{f * g}$.

Proof. These results are standard over $\mathbb{R}$ or $\mathbb{C}$, and the proof method in the general case is virtually identical. □

Observation. The above properties of convolution allow us to deduce that $L^1(G)$ is a Banach algebra with respect to convolution. If G is discrete, then the characteristic function of the identity also maps $L^1(G)$ unital. It is also a fact (that we do not prove here) that the converse is true: i.e if $L^1(G)$ is unital wrt convolution then G must be discrete.

Proposition A.3.6. Let B denote the Banach algebra $L^1(G)$, and let $\widehat{B} := \hom_{\mathbb{C}}(B, \mathbb{C}) \setminus \{\mathbf{0}\}$ be the non-zero complex characters of B. For any $\chi \in \widehat{G}$ and $f \in B$, define

$$v_\chi(f) = \widehat{f}(\chi) = \int f(y)\overline{\chi(y)}\,\mathrm{d}y .$$

Then $v_\chi \in \widehat{B}$ for all χ. Moreover, the map

$$\begin{aligned} \widehat{G} &\to \widehat{B} \\ \chi &\mapsto v_\chi \end{aligned}$$

is a bijection.

Proof. INsert reference here. □

§A.4 Pontryagin Duality

This is the main theorem of this appendix that will be used repeatedly throughout the thesis.

The first step to proving this is the observation that this mapping is indeeed injective. This follows from the following lemma.

Lemma A.4.1. For every non-identity $g \in G$, there exists some $\chi \in \widehat{G}$ such that $\chi(g) \neq 1$.

Proof. Assume FTSOC there was some $s \in G$ for which this was not the case. Then Recall our function shifting notation in **??**, we note that

$$\widehat{f} = \widehat{L_s f}$$

for all $f \in L^1(G)$ □

Appendix B

Some Useful Theorems on Local and Global Fields

[Remove some stuff here. Figure out what stuff should be assumed by default an what stuff is useful to include for reference.]

Recall an absolute value on a field F is a multiplicative function $|\cdot| : F \to \mathbb{R}_{\geq 0}$ that is nonzero on any non-zero element of F and satisfies the triangle inequality, i.e. $|x+y| \leq |x| + |y|$ for all $x, y, \in F$.

Definition B.0.1. A **local field** is a field with a non-trivial absolute value, under the induced topology of which is locally compact.

Remark. Some texts also require the definition of a local field to assume the absolute value by Archimedean, which occludes $\mathbb{R}$ and $\mathbb{C}$. This is a useful exclusion for class field theory, it will be more useful to consider them as local fields for this thesis.

Theorem B.0.2. Every local field arises as the completion of a finite separable extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$. In particular, every global field is isomorphic to one of the following:

1. (Archimedean) $\mathbb{R}$ or $\mathbb{C}$,
2. (Non-Archimedean, Characteristic Zero) A finite extension of $\mathbb{Q}_p$ for some any p,
3. (Non-Archimedean, Positive Characteristic) $\mathbb{F}_q((t))$, the field of laurent series over a finite field.

The above motivates the following definition.

Definition B.0.3. A **global field** is a finite separable extension of $\mathbb{Q}$ or $\mathbb{F}_q(t)$.

There are several ways one could define the ring of integers of a general global field; the following will be the most convenient for us.

Definition B.0.4. For a global field K, its **ring of integers** is the intersection

$$\mathcal{O}_K := \{x \in K : |x|_v \leq 1 \ \forall v \text{ finite}\}.$$

Theorem B.0.5. The ring of integers of a global field is always a Dedekind Domain.

Corollary B.0.6. The ring of integers of a local field is always a discrete valuation ring.

Proposition B.0.7. For x a non-zero element of the maximal ideal $\mathfrak{m}$ of a local field F, and let $n = \operatorname{ord}(x)$.

Then for every $z \in \mathbb{Z}$, every $y \in \mathfrak{m}^{nz}$ has a unique "Laurent Series" expansion

$$y = \sum_{i=z}^{\infty} a_i x^i,$$

where each $a_i \in \mathcal{O}_F \setminus \mathfrak{m}^n$.

Proof. See Theorem 6 of Chapter 1 Section 4 of [Wei74], along with the subsequent corollaries and discussion. □

LEMMA B.0.8. For any finite separable extension E/K of global fields and any non-zero $a \in \mathcal{O}_K$ to invert, we have $(\mathcal{O}_E)_a = \overline{(\mathcal{O}_K)_a}$. For any place v not dividing a, an A_a-basis of B_a is an $(A_a)_v$-basis of $(B_a)_v$.

§B.1 Modules and Quasicharacters of Locally Compact Fields

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